

Childhood Kidney Disease

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KEY POINTS

- Pediatric kidney disease is intrinsically different from adult kidney disease, with unique genetic, developmental, and congenital bases not commonly seen in adults requiring age-specific diagnostics and therapeutic approaches.
- Kidney disease in children requires special care and multimodal therapy because of the associated impairment of growth and development.
- Congenital anomalies of the kidney and urinary tract is the leading cause of kidney disease in children and requires a specialized approach to management that considers the congenital and genetic bases of the disease, which differ substantially from acquired renal diseases seen in adults.
- Pediatric nephrotic syndrome is mainly idiopathic and the most common glomerular disease. It often requires long-term immunosuppressive therapy, which has unique implications in children's immune and physical development, thus requiring careful balancing to minimize adverse effects.
- Emerging treatments for pediatric kidney diseases focus on targeting immune pathways and specific proteins involved in glomerular function.

KIDNEY AND URINARY TRACT DISORDERS IN CHILDREN

The spectrum of kidney and urinary tract disorders varies significantly between pediatric and adult populations. Congenital disorders, such as anomalies of the kidney and urinary tract (CAKUT) and non-CAKUT hereditary

nephropathies, account for the majority of causes, followed by glomerulonephritides (Fig. 71.1) and appear to be more common in higher-resource countries.¹ Acquired kidney diseases appear to prevail in lower-resource countries, but this may reflect bias due to limitations in access to diagnostics in these settings.¹ CAKUT predominates in younger patients ages newborn to 5, while the incidence of glomerulonephritis increases progressively with age, and

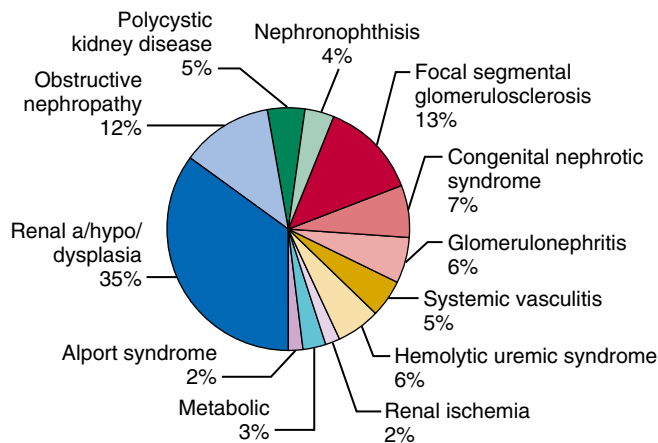


Fig. 71.1 Spectrum of underlying renal diagnoses in 1367 children with end-stage kidney disease. (From the International Pediatric Peritoneal Dialysis Network Registry.)

at ages 6 to 13 is nearly as common as CAKUT.² In Europe, the incidence of chronic kidney disease (CKD) in children ranges from 8 to 14 per million age-related population (pmarp) for CKD stages 2 to 5 and approximately 8 pmarp for CKD stages 4 to 5.³ The highest incidence rates for end-stage kidney failure (ESKF) in children were reported from the United States, New Zealand, and Austria, at 14.8, 13.6, and 12.4 per million population, respectively.⁴ The incidence and prevalence of CKD and ESKF in children are notably greater in males than females, reflecting the increased prevalence of congenital disorders, such as obstructive uropathy and renal dysplasia in boys.⁴ The causes of CKD differ among populations of different current or ancestral continental associations. For instance, focal segmental glomerulosclerosis (FSGS), a common cause of primary glomerular disease, has a threefold higher prevalence in populations of current or recent African compared with European or West Eurasian ancestry, with comparative rates of 19% versus 6%, and a notably higher prevalence (35%) among adolescents.⁵ Because of the predominantly genetic origin of pediatric nephropathies, CKD in children is frequently associated with a variety of extrarenal abnormalities. Impaired neurodevelopment and sensory dysfunctions are among the most severe general disabilities interfering with psychosocial adjustment and integration in children with CKD.

EVALUATION OF KIDNEY FUNCTION IN CHILDREN

Accurate assessment of glomerular filtration rate (GFR) is crucial for evaluation of kidney function (see [Chapter 23](#)). GFR is naturally higher in adults due to their larger filtration surface area and increased renal blood flow. To standardize measurements for children, GFR is adjusted on the basis of their body surface area (BSA), typically standardized at 1.73 m², equivalent to that of an average adult.⁶ Nephron development is complete by 32 to 36 weeks' gestation, with continued growth in size and function.⁷ Term infants have a GFR of approximately 20

to 30 mL/min/1.73 m², which increases rapidly during their first 2 years, correlating with BSA. Beyond age 2, the conventional Kidney Disease Improving Global Outcomes (KDIGO) CKD staging system, which uses estimated GFR (eGFR) multiples of 15 and 30 mL/min/1.73 m², is used, as in adults. The KDIGO CKD guidelines recommend using a validated equation to calculate eGFR.⁸ For decades, serum creatinine levels have been used to estimate GFR in children ages 1 to 16 using the modified "bedside" Schwartz equation (eGFR = Height [cm] × 0.413/serum creatinine [mg/dL]). This equation estimates GFR from the ratio of height to serum creatinine concentration multiplied by a uniform constant for both genders, K.⁹ [Table 71.1](#) outlines multiple equations used to estimate GFR. The CKiD U25 2021 eGFRcr equation is the currently recommended formula as it has been externally validated in multiracial cohorts with reduced and normal GFR. The performance of the CKiD U25 2021 eGFRcr equation is, however, uncertain in the very young, those with very low GFR, or in populations outside of Europe and North America.⁸ The European Kidney Function Consortium (EKFC) equation has been validated in both children and adults. Both the CKiD U25 and EKFC equations may be of use during the transition between adolescence and young adulthood.⁸ Reviews have highlighted the inaccuracy of solely using creatinine as a GFR estimation measurement parameter due to its high interindividual variability in its determinants such as muscle creatinine production, unreliability in the presence of reduced GFR, placental transfer, and technical difficulty in measurement.¹⁰⁻¹² Additional molecules that may be used to assess kidney function in conjunction with creatinine include, but are not limited to, cystatin C (CysC), β -2 microglobulin, and β -trace protein (BTP).⁶ All of these substances exhibit their highest serum concentrations during early infancy, with a subsequent decline during the first 2 years of life. CysC is a protein that distributes in the extracellular space, unlike creatinine, which is distributed throughout the total body water.¹³ This difference allows for greater sensitivity of CysC in detecting changes in GFR. The use of CysC is advantageous as it is not affected by muscle mass; thus it does not require anthropometric data and requires small blood sample sizes, which is an advantage in small children. Current KDIGO guidelines state there is insufficient externally validated data to determine the utility of combining CysC and creatinine in children (this is available in the CKiD U25 calculators); however, CysC-based equations are likely to be more appropriate in children with neurologic disorders, muscle wasting, and those with metabolic disorders on a very low-protein diet ([Table 71.2](#)).⁸ β -2 microglobulin is a protein that composes the β chain of the major histocompatibility complex (MHC). Serum levels reflect kidney function along with inflammation and infections.⁶ CysC and B2M do not cross the placental barrier and thus may be used to determine fetal kidney function using cord blood. BTP is a small glycoprotein synthesized primarily in the central nervous system and is almost exclusively eliminated by the kidney.¹⁴ The use of BTP is advantageous as it is not influenced by height, gender, age, or muscle mass. To enhance precision, assessment of kidney filtration function in children should involve a combination of measures.

Table 71.1 Serum Creatinine-Based Formulas for Glomerular Filtration Rate Estimation for Pediatric and Neonatal use

Equation	Calculation	Age Range
CKID U25 ⁸⁰⁸	$eGFR = k \times (\text{height}/sCr)$, with height in m and serum creatinine in mg/dL In males, k is calculated as: - For 1 to <12 years old: $39.0 \times 1.008^{(\text{age} - 12)}$ - For 12 to <18 years old: $39.0 \times 1.045^{(\text{age} - 12)}$ - For 18 to 25 years old: 50.8 In females, k is calculated as: - For 1 to <12 years old: $36.1 \times 1.008^{(\text{age} - 12)}$ - For 12 to <18 years old: $36.1 \times 1.023^{(\text{age} - 12)}$ - For 18 to 25 years old: 41.4	1-25 years old
CKID “bedside” ⁸⁰⁹	$eGFR = 0.413 \times (\text{height}/\text{serum creatinine})$, with height in m and serum creatinine in mg/dL or $eGFR = 36.5 \times (\text{height}/\text{serum creatinine})$, with height in m and serum creatinine in $\mu\text{mol/L}$	1-16 years old
Q(height) – eGFR ⁸¹⁰	$eGFR = 107.3 / (sCr/Q)$, with serum creatinine in mg/dL In males, Q is calculated as: $Q = 0.21 + 0.057 \times \text{Age} - 0.0075 \times \text{Age}^2 + 0.00064 \times \text{Age}^3 - 0.000016 \times \text{Age}^4$ In females, Q is calculated as: $Q = 0.23 + 0.034 \times \text{Age} - 0.0018 \times \text{Age}^2 + 0.00017 \times \text{Age}^3 + 0.0000051 \times \text{Age}^4$	1-25 years old
Flanders metadata ⁸¹¹	$eGFR = [0.0414 \times \ln(\text{age}) + 0.3018] \times (\text{height}/sCr)$, with age in years, height in cm, and sCr in mg/dL	1 month to 14 years old
Gao quadratic ⁸¹²	$eGFR = 0.68 \times (\text{height}/sCr) - 0.0008 \times (\text{height}/sCr)^2 + 0.48 \times \text{age} - (21.53 \text{ in males or } 25.68 \text{ in females})$, with age in years, height in cm, and sCr in mg/dL	2-18 years old
Brion ⁸¹³	$eGFR = k \times Ht/SCr$, where $k = 0.33$ [preterm], $k = 0.45$ [term infants]	25 weeks' gestation age to 8 weeks of life
Léger ⁸¹⁴	$eGFR = (56.7 \times \text{weight} + 0.142 \times \text{height}^2)/PCr$, with weight in kg, height in cm, and plasma creatinine in μM	0.8-18 years old

Adapted from Glomerular Filtration Rate Estimation Formulas for Pediatric and Neonatal Use, Edit Muhari-Stark et al and Age- and sex-dependent clinical equations to estimate glomerular filtration rates in children and young adults with chronic kidney disease, Pierce et al.⁸⁰⁸

Table 71.2 Cystatin C–Based Formulas for Glomerular Filtration Rate Estimation for Pediatric and Neonatal Use

Equation	Calculation	Age Range
Filler ⁸¹⁵	$eGFR = 91.62 \times (\text{CysC})^{-1.123}$	1-18 years old
Grubb ⁸¹⁶	$eGFR = 84.69 \times (\text{CysC})^{-1.680} \times 1.384^{\text{if } <14 \text{ years old}}$	0.3-18 years old
Zappitelli ⁸¹⁷	$eGFR = 74.94/(\text{CysC})^{1.17} \times 1.2^{\text{if Tx}}$ $eGFR = (43.82 \times e^{0.003 \times \text{height}})/(\text{CysC}^{0.635} \times \text{SCr}^{0.547})$	8-17 years old
Bouvet ⁸¹⁸	$eGFR (\text{mL/min}) = 63.2 \times (\text{SCr}/1.086)^{-0.35} \times (\text{CysC}/1.2)^{-0.56} \times (\text{Weight}/45)^{0.30} \times (\text{age}/14)^{0.40}$	1.4-22.8 years old
Schwartz ⁸¹⁹	$eGFR = 39.8 \times (\text{height}/\text{SCr})^{0.456} \times (1.8/\text{CysC})^{0.418} \times (30/\text{BUN})^{0.079} \times (1.076)^{\text{male}} \times (\text{height}/1.4)^{0.179}$	1-16 years old
CKID U25 ⁸⁰⁸	$eGFR = k \times (1/\text{CysC})$, with serum cystatin C in mg/L For males, k is calculated as: - For 1 to <15 years old: $k = 87.2 \times 1.011^{(\text{age}-15)}$ - For 15 to <18 years old: $k = 87.2 \times 0.960^{(\text{age}-15)}$ - For 18 to 25 years old: $k = 77.1$ For females, k is calculated as: - For 1 to <12 years old: $k = 79.9 \times 1.004^{(\text{age}-12)}$ - For 12 to <18 years old: $k = 79.9 \times 0.974^{(\text{age}-12)}$ - For 18 to 25 years old: $k = 68.3$	1-25 years old
Full spectrum ⁸²⁰	$eGFR = 107.3/(\text{CysC}/Q)$, with serum cystatin C in mg/L and $Q = 0.82$ for males and females <70 years old	<70 years old

Adapted from Glomerular Filtration Rate Estimation Formulas for Pediatric and Neonatal Use, Edit Muhari-Stark et al and Age- and sex-dependent clinical equations to estimate glomerular filtration rates in children and young adults with chronic kidney disease, Pierce et al.⁸⁰⁸

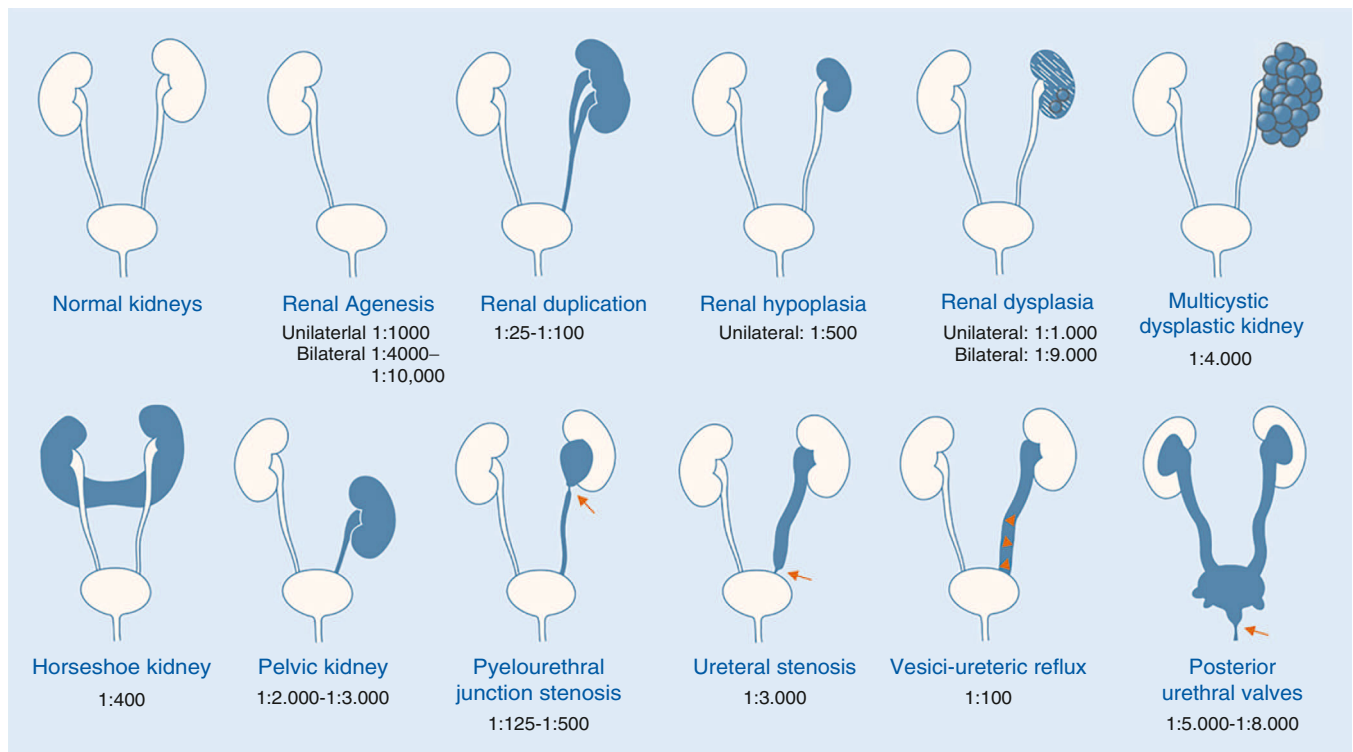


Fig. 71.2 Congenital abnormalities of the kidney and urinary tract. (Modified from Kosfeld A, Martens H, Hennies I, et al. *Kongenitale Anomalien der Nieren und ableitenden Harnwege (CAKUT)*. *Medgen*. 2018;30:448–460.)

CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

Congenital anomalies of the kidney and urinary tract (CAKUT) encompass a range of abnormalities (outlined in Fig. 71.2) from benign structural abnormalities to complete absence of kidney tissue.¹⁷ Kidney–urinary tract structural abnormalities are the most common cause of all birth defects, causing 30% to 50% of all cases of ESKF in children, and can predispose to adult-onset diseases such as hypertension.^{18,19} Roughly two-thirds of individuals with congenital upper urinary tract anomalies also present with concurrent anomalies affecting other organ systems, including the skeletal, cardiovascular, gastrointestinal, and central nervous systems.²⁰ This section is focused on the classification, epidemiology, pathogenesis, and clinical management of kidney malformations.

CLINICAL CLASSIFICATION OF KIDNEY MALFORMATIONS

The kidneys begin formation at week 3 of gestation and continue developing until week 36, making them vulnerable to environmental factors that may disrupt development throughout pregnancy. Multiple genes determine kidney and urinary tract development, and mutations in these genes are associated with CAKUT. Development of the kidney is discussed in detail in Chapter 1. The wide spectrum of CAKUT includes parenchymal defects of the kidney (e.g., agenesis, hypoplasia, dysplasia, and multicystic dysplastic kidney [MCDK]); upper urinary

tract defects (e.g., ureteropelvic junction obstruction, obstructive and/or refluxing megaureter, VUR); and lower urinary tract obstruction (such as posterior urethral valves).²¹ VUR is a risk factor for recurrent urinary tract infections (UTIs) and pyelonephritis. Kidney–urinary tract anomalies are classified within the CAKUT categories outlined in Table 71.3.

Renal agenesis is characterized by the total absence of one or both kidneys resulting from the failure of embryonic kidney development. Simple renal hypoplasia may be unilateral or bilateral, in which the kidneys maintain normal shape and architecture but are smaller in size and have a reduced number of nephrons.

Renal dysplasia is defined as an abnormality of tissue patterning (Fig. 71.3), and affected kidneys are often hypoplastic and contain disorganized elements of improperly differentiated tissue.²² Histopathologically, the dysplastic kidney is characterized by several primary features: 1. abnormal differentiation of mesenchymal and epithelial elements, 2. decreased nephron number, 3. loss of corticomedullary differentiation, and 4. metaplastic transformation of mesenchyme to cartilage and bone. Sonographically, renal dysplasia is characterized by reduced corticomedullary differentiation and/or diffuse cortical thinning. MCDK is an extreme form of renal dysplasia in which large polymorphic cysts dominate the kidney structure (Fig. 71.4).

Ectopic kidneys are characterized by anomalous positioning, resulting from the failure of the kidney to ascend to its typical retroperitoneal renal fossa during embryonic development. Ectopic kidneys can be categorized as either simple (on the same side as the ureter) or crossed (on the opposite side of the ureteric orifice), and they may

Table 71.3 Congenital Anomalies of the Kidneys and Urinary Tract Categories		
Types of Anomaly	CAKUT Disorder	Description
Kidney number Kidney size and morphology	Renal agenesis	Failure of formation of kidney/outflow system; unilateral or bilateral
	Renal hypoplasia	Small kidney size with reduced nephrons; normal shape; unilateral or bilateral
	Renal dysplasia	Abnormal kidney shape and tissue differentiation with reduced number of nephrons; unilateral or bilateral
Kidney position	Multicystic dysplastic kidney	Multiple cysts within a dysplastic kidney causing an abnormal shape
	Horseshoe kidney	Posterior fusion of kidneys forming a horseshoe shape
	Ectopic kidney	Kidney in an abnormal location, typically pelvic
Outflow abnormalities	Ureteropelvic junction obstruction	Blocked urine flow from kidney pelvis due to obstruction of junction between kidney and ureter; unilateral or bilateral
	Vesicoureteric reflux	Urine backflow from bladder due to defective junction between ureter and bladder; unilateral or bilateral
	Duplex collecting system	Duplication of ureter and kidney pelvis, duplicated kidneys may be present; reflux or obstruction of outflow system may result; unilateral or bilateral
	Megaureter	Distension of ureter impairs urine flow; unilateral or bilateral
	Posterior urethral valves	Membrane formation in urethra; prevents emptying of bladder; limited to males

Modified from Murugapoopathy V, Gupta IR. A primer on congenital anomalies of the kidneys and urinary tracts (CAKUT). *Clin J Am Soc Nephrol.* 2020;15(5):723–731.

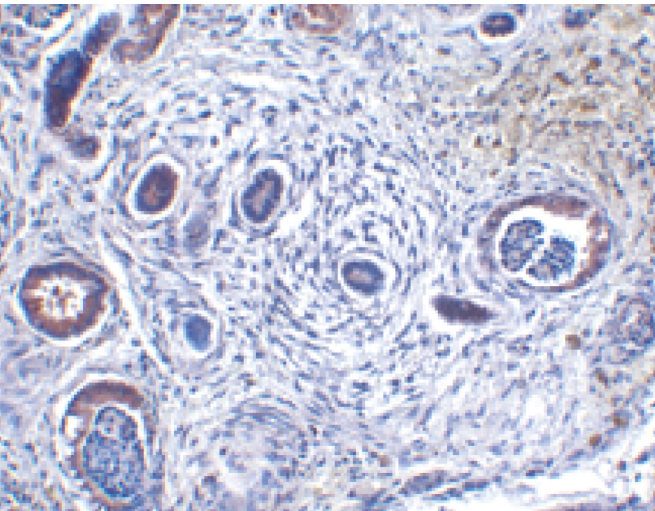


Fig. 71.3 Histologic section of renal dysplastic tissue. Dysplastic renal tissue demonstrates a paucity of glomerular and tubular elements, disorganization of tissue elements, an abundance of stroma, and thickened and dilated renal tubules.

be unilateral or bilateral (Fig. 71.5). In cases of simple renal ectopia, the kidney is most frequently situated in the pelvis and iliac region.²⁰

Renal fusion occurs when a portion of one kidney is fused to the other. The most common fusion anomaly is the horseshoe kidney, which involves abnormal migration of both kidneys (ectopy), resulting in fusion (see Fig. 71.2).

EPIDEMIOLOGY OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

CAKUT is the most frequent malformation detected in utero, occurring in 3% to 11% of the population and accounting

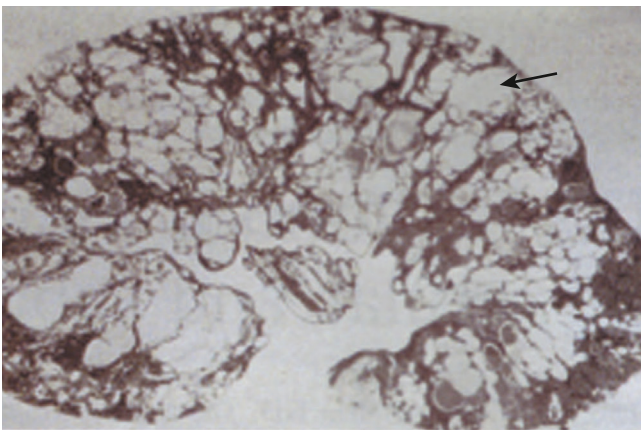


Fig. 71.4 Histologic section of a multicystic dysplastic kidney. The kidney consists in large part of multiple polymorphic cysts (arrow). Normal renal tissue elements cannot be identified.

for 50% of all congenital abnormalities (see Fig. 71.2).²³ The reported incidence of kidney–urinary tract malformations varies depending on the data collection method. Most rates are not derived from population-based pregnancy studies but rather from autopsy or selected live-born infant studies. Rates obtained through methods other than population-based ascertainment may underestimate the actual incidence because they may not account for fetal loss, and CAKUT can remain clinically silent in surviving fetuses.

Lower urinary tract anomalies account for roughly 50% of CAKUT cases, with 25% exhibiting vesicoureteral reflux (VUR), 11% presenting with ureteropelvic junction (UPJ) obstruction, and another 11% displaying ureterovesical junction (UVJ) obstruction.²⁴ Duplication of the renal collecting system, whether complete or partial, is the most

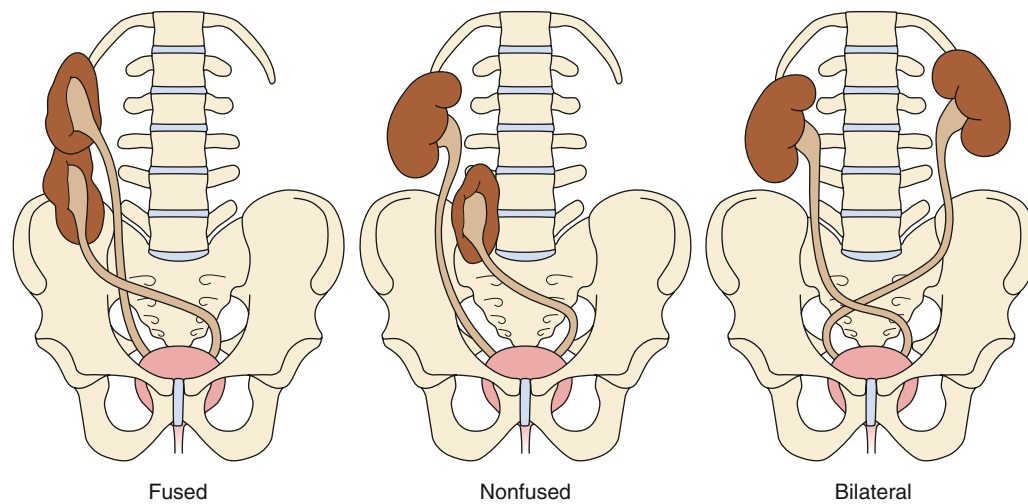


Fig. 71.5 Crossed fused ectopia. Renal ectopia is classified as simple (nonfused), fused, and bilateral. A fused kidney migrates across the midline and fuses to the lower pole of the normally positioned contralateral kidney. A simple crossed (nonfused) kidney migrates across the midline, does not fuse with the normally positioned contralateral kidney, and is usually positioned at the rim of the pelvis. In bilateral ectopia, both kidneys are ectopic and cross the midline with their native ureters, which are inserted normally into the bladder.

common congenital anomaly of the urinary tract with autopsy studies suggesting an incidence ranging from 0.8% to 5%.^{25,26}

Bilateral renal agenesis, characterized by the absence of both kidneys, is a rare congenital anomaly occurring in 1 of 3000 to 4000 births and is not compatible with life.²⁷ In contrast, unilateral renal agenesis has an estimated incidence of 1 in 2000 births, typically remaining asymptomatic and often incidentally diagnosed.²⁸ The incidence of unilateral dysplasia is approximately 1 in 3000 to 5000 births (1:3640 for the MCDK), with a male-to-female ratio of 1.92:1. Bilateral dysplasia is less common, affecting 1 in 10,000 births, with a male-to-female ratio of 1.32:1.^{29,30} A pelvic kidney is the most common form of simple renal ectopia, with an incidence ranging from 1 in 2200 to 1 in 3000 in autopsy studies. Bilateral cases occur in approximately 10% of affected individuals.³¹ Crossed renal ectopia, a rare condition, is observed in roughly 7.5 out of 10,000 newborns, with a male-to-female ratio of 3:2.³² Horseshoe kidney is the most common congenital anomaly of the upper urinary tract and can be found in 1 in 400 adults, with a male-to-female ratio of 2:1, and may be detected via ultrasound.^{33,34}

PATHOGENESIS OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

MECHANISMS OF INHERITANCE

The genetic inheritance of CAKUT is multifactorial and complex. CAKUT is responsible for roughly 40% of cases of ESKF that typically becomes evident during the initial 3 decades of life. Studies suggest that CAKUT frequently results from mutations in individual (monogenic) genes; approximately 40 mutations have been identified to date (25 dominant and 15 recessive), outlined in (Table 71.4). In approximately 30% of affected individuals, these malformations occur as part of a multiorgan genetic syndrome. Currently, more than 200 monogenic syndromes have been identified that involve the kidney and urinary

tract.³⁵ The genetics of kidney diseases are discussed in more detail in Chapter 44.

The development of CAKUT is grounded in the disturbance of typical nephrogenesis, attributed to either environmental or genetic factors. Most genetic causes do not follow a clear Mendelian pattern of inheritance. Although many cases of CAKUT appear to be sporadic, with no other affected member in the same family, familial clustering is common, suggesting that the pathogenesis is influenced by genetic factors.³⁶ Among affected family members carrying the same mutation, there is often variability in expressivity and penetrance, leading to a spectrum of renal phenotypes including agenesis, dysplasia, or isolated collecting system abnormalities such as hydronephrosis.³⁷ Approximately 10% of those with a congenital solitary kidney have a first-degree relative with renal or urinary tract disease, frequently presenting asymptotically. Multiple studies have confirmed an elevated recurrence risk for CAKUT among relatives, estimated to range from 4% to 20%.³⁸ This underscores the hereditary nature of the condition and highlights the crucial importance of screening first-degree relatives.

Numerous studies have sought to explore the genetic predisposition of CAKUT. Roodhooft and colleagues³⁹ conducted a study on individuals with bilateral renal agenesis or bilateral renal dysgenesis. Their findings revealed that 9% of first-degree relatives exhibited some form of kidney or urinary tract malformation. In a similar fashion, Bulum and colleagues³⁶ studied 180 families with known CAKUT and identified new anomalies in 23% of asymptomatic first-degree relatives. In a study involving 232 families, conducted by van der Ven and colleagues,⁴⁰ whole exome sequencing was employed to investigate CAKUT genotypes and found a potential single, novel monogenic CAKUT gene in 8% of families.

The predominant autosomal dominant single-gene defects leading to CAKUT are characterized by well-defined syndromic disorders, primarily arising from mutations in genes such as *PAX2*, *HNF1B*, *SALL1*, *WT1*, *SIX1*, and *EYA1*.⁴¹

Table 71.4 Human Gene Mutations Associated With Syndromic Congenital Anomalies of the Kidney and Urinary Tract

Primary Disease	Gene	Kidney Phenotype	References
Alagille syndrome	<i>JAGGED1</i> <i>NOTCH2</i>	Cystic dysplasia	821,822
Apert syndrome	<i>FGFR2</i>	Hydronephrosis	823
Atypical DiGeorge syndrome	<i>CRKL</i>	Hypodysplasia, vesicoureteral reflux (VUR), obstructive uropathy	824
Beckwith-Wiedemann syndrome	<i>p57^{KIP2}</i>	Medullary dysplasia	825
Branchio-oto-renal (BOR) syndrome	<i>EYA1, SIX1, SIX5</i>	Unilateral or bilateral agenesis/dysplasia, hypoplasia, collecting system anomalies	826
Campomelic dysplasia	<i>SOX9</i>	Dysplasia, hydronephrosis	827
Duane radial ray (Okihiro) syndrome	<i>SALL4</i>	UNL agenesis, VUR, malrotation, cross-fused ectopia, pelviectasis	828
Fraser syndrome	<i>FRAS1</i> <i>FREM1</i> <i>FREM2</i>	Agenesis, dysplasia	42,829
Isolated renal hypoplasia	<i>BMP4</i> <i>RET</i>	Hypoplasia, VUR	37,54
Hypoparathyroidism, sensorineural deafness, and renal anomalies syndrome	<i>GATA3</i> <i>KDM6A, KMT2D</i>	Dysplasia VUR, hypodysplasia, ectopia	830 831
Kabuki syndrome	<i>KAL1, FGFR1,</i>	Agenesis	832
Kallmann syndrome	<i>PROK2, PROK2R</i>	Hypodysplasia, VUR, ectopia, horseshoe kidney	833
Leukemia, acute B cell	<i>PBX1</i>		
Mammary-ulnar syndrome	<i>TBX3</i>	Dysplasia	834
Okihiro syndrome	<i>SALL4</i>	Ectopia, dysplasia	827
Pallister-Hall syndrome	<i>GLI3</i>	Agenesis, dysplasia, hydronephrosis	835,836
Renal-coloboma syndrome	<i>PAX2</i>	Hypoplasia, vesicoureteral reflux	61
Renal tubular dysgenesis	<i>RAS components</i>	Tubular dysplasia	82
Renal cysts and diabetes syndrome	<i>HNF1b</i>	Dysplasia, hypoplasia	837
Rubinstein-Taybi syndrome	<i>CREBBP</i>	Agenesis, hypoplasia	838
Simpson-Golabi-Behmel syndrome	<i>GPC3</i>	Medullary dysplasia	839
Smith-Lemli-Opitz syndrome	<i>7-hydroxy-cholesterol reductase</i>	Agenesis, dysplasia	840
Townes-Brock syndrome	<i>SALL1</i>	Hypoplasia, dysplasia, VUR	841
Ulnar-mammary syndrome	<i>TBX3</i>	Hypoplasia	834
Zellweger syndrome	<i>PEX1</i>	VUR, cystic dysplasia	842

In contrast, autosomal recessive defects are infrequent and often result in isolated CAKUT. The most common causes of autosomal recessive CAKUT are associated with mutations in *FRAS1*, *FREM1*, *FREM2*, and *GRIPI*.⁴²

Studies suggest that polymorphic variants in genes that control renal development and copy number variants (CNVs) contribute to the pathogenesis of CAKUT.⁴³ CNVs are the most common form of genetic changes in the human genome, involving the gain or loss of DNA segments. The most commonly observed genomic disorder was the Chr.17q12 deletion, indicating the presence of the renal cyst and diabetes syndrome (RCAD). This was followed by the Chr.22q11.2 deletion, indicative of the DiGeorge/velocardiofacial syndrome.⁴¹

MOLECULAR PATHOGENESIS

Normal development of the kidney and urinary tract begins during weeks 5 to 6 of gestation with invasion of the metanephric mesenchyme (MM) by the ureteric bud (UB), both of which are derived from the intermediate mesoderm.

Following this, the branching morphogenesis phase involves repetitive branching events, which ultimately form the collecting system, which is an intricate, treelike network of tubular structures including the ureter, calyces, and both medullary and cortical collecting ducts. The number of UB branches determines the final nephron count by influencing MM cells to initiate nephrogenesis. Kidney and urinary tract development continues until weeks 34 to 36 of gestation.^{44,45} Kidney development is discussed in detail in [Chapter 1](#).

The molecular pathogenesis of CAKUT has been the subject of many studies in human and mouse models. In this section, the functions of genes mutated in human CAKUT and for which a function has been explained are discussed as a framework for understanding the molecular pathogenesis of CAKUT.

URETERIC BUDDING, ROBO2, AND BMP4

Normal growth of the UB is dependent on TGF- β family member glial cell-line-derived neurotrophic factor (GDNF) signaling via the RET receptor and GPI-linked cell surface coreceptor (Gfra1). RET, a tyrosine kinase receptor, is

present on the surface of ureteric cells and is coexpressed with *Gfra1*, whereas GDNF is expressed by MM cells. Mouse studies have demonstrated that genetic disruptions in the *RET*, *GDNF*, and *Gfra1* genes can lead to impaired ureteric outgrowth and renal agenesis. Skinner and colleagues⁴⁵ investigated the genetic makeup of 33 stillborn fetuses with renal aplasia or severe renal dysplasia and found human *RET* mutations were found in 37% of fetuses with bilateral renal agenesis and in 20% of fetuses with unilateral renal agenesis. These findings underscore the critical role of the GDNF signal in promoting the growth of the UB and Wolffian duct (WD).^{46,47}

ROBO2, a cell surface receptor expressed in the nephrogenic mesenchyme, is linked to VUR and UTIs when disrupted. VUR is associated with reflux nephropathy due to the retrograde flow of urine from the bladder to the ureter.^{48,49} Normally involved in axon guidance and cell migration, mutations to ROBO2 result in the formation of additional UBs that remain inappropriately connected to the nephric duct.⁵⁰ It has been suggested that an ectopic position of the ureteral orifice in the bladder is associated with VUR, whereas ectopic positioning of the UB is associated with renal dysplasia.⁵¹

In the mesenchyme surrounding the ureteric stalk, bone morphogenetic protein-4 (BMP4) is present. Its roles include preventing the abnormal development of ureteric tips while facilitating the elongation of the ureteric stalk. Furthermore, BMP4 exerts an inhibitory effect on the branching of UB tips, whereas GDNF has a positive regulatory role.⁵² As a result, mutations can lead to the development of cystic kidneys, hypoplastic kidneys, and hydroureter. Human studies have found an association between BMP4 mutations with renal dysgenesis and VUR.⁵⁴

URETERIC BRANCHING, PAX2, AND RET

After the UB invades the MM, it undergoes numerous branching cycles to establish the collecting duct system, responsible for directing urine flow to the bladder. The branching phase involves approximately 19 to 21 generations of branching events and is a major determinant of final nephron number as each UB branch tip induces a discrete subset of MM cells to undergo nephrogenesis.⁵⁵

PAX2 is a paired-box transcription factor located in the collecting duct, nephron progenitors, and epithelial components of the developing nephron.⁵⁶ Deficiency of PAX2 is associated with autosomal dominant renal-coloboma syndrome (RCS), also known as papillorenal syndrome. RCS is characterized by optic nerve dysplasia (frequently described as a coloboma) and varied renal malformations, such as vesicoureteric reflux (VUR), renal hypoplasia, renal hypodysplasia, MCDK, oligomeganephronia, and horseshoe kidney. Studies by Chang and colleagues⁵⁷ and Porteous and colleagues⁵⁸ both show renal hypoplasia was the most prevalent congenital defect. Characteristic ocular features include an enlarged optic disk with blood vessels emanating from the periphery of the optic nerve head, rather than the center, and a wide spectrum of visual acuity, ranging from normal to severe impairment including blindness.⁵⁹ In contrast, PAX2 overexpression results in epithelial hyperproliferation and the occurrence of renal cysts.⁶⁰

The *PAX2* gene is located on chromosome 10q24-25. It encodes a transcription factor that belongs to the paired-box family of homeotic genes. During renal development, *PAX2* is expressed in the WD, UB, and metanephric mesenchyme. Mice studies have shown that the absence of the *PAX2* gene results in branching defects and significantly reduces the number of nephrons.

The GDNF-RET signaling pathway plays a vital role in not only ureteric outgrowth but also the essential process of ureteric branching. RET expressed on ureteric tip cells influences branch quantity and pattern by prompting surrounding mesenchymal cells to condense into epithelial vesicles, which then differentiate into nephron segments.⁶³ Clarke and colleagues⁶⁴ showed mice with a heterozygous mutation in *RET* resulted in a 22% reduction in glomerular density. This is supported by human studies that similarly showed infants with a common *RET* mutation (the *RET*[1476A] allele) have a 10% reduction in kidney volume.³⁷ Signal transduction through the glial-derived neurotrophic factor (GDNF)–GDNF receptor $\alpha 1$ (GFRA1)–RET pathway is necessary for the UB to protrude from the nephric duct and grow toward the MM.⁶⁵ Regulatory molecules, such as *PAX2*, *EYA1*, *SIX1*, *SIX4*, and *HOXA11*, which stimulate the GDNF–GFRA1–RET pathway, are essential for UB emergence and branching during the process of nephrogenesis. In addition, inhibitory molecules, such as ROBO2, *FOXC1*, *FOXF1*, and *FOXP1*, prevent ectopic or multiple UB formation.

CONTROL OF GDNF EXPRESSION IN THE METANEPHRIC MESENCHYME: *Sall1*, *Eya1*, AND *Six1*

In the MM, GDNF is essential for normal growth of the UB and branching morphogenesis. *Sall1*, *Eya1*, and *Six1* positively control GDNF expression.⁶⁶ *Sall1* is a member of the Spalt family of transcriptional factors, and inactivation causes decreased GDNF expression, resulting in Townes-Brock syndrome (TBS). This syndrome is inherited in an autosomal dominant manner and is characterized by imperforate anus, preaxial polydactyly or triphalangeal thumbs, or a combination of both, along with external ear abnormalities, sensorineural hearing loss, and a spectrum of kidney, urogenital, and heart malformations including renal agenesis.⁶⁷ The prevalence of TBS has been estimated to be 1:250,000, but overlap with VACTERL association (Vertebral defects, Anal atresia, Cardiac defects, Tracheo-esophageal fistula, Renal anomalies, and Limb abnormalities) may lead to an overascertainment of true prevalence.⁶⁸

EYA1, a DNA-binding transcription factor expressed in MM cells, together with *SIX1*, regulates the expression of GDNF, all of which play essential roles in the development of otic and branchial tissues. Mutations of *EYA1* and *SIX1* result in the autosomal dominant branchio-oto-renal (BOR) syndrome, which is characterized by kidney and urinary tract malformations and conductive, sensorineural, or combined hearing loss. Renal manifestations include unilateral or bilateral renal agenesis, hypodysplasia, as well as malformation of the lower urinary tract including VUR, pyelo-ureteral obstruction, and ureteral duplication. Branchiootic syndrome (BO) is a related disorder without renal anomalies, also caused by *EYA1* mutations. BOR/BO syndromes have a prevalence of 1:40,000 in the general

population and are responsible for 2% of profound deafness in children.^{67,69}

HEDGEHOG SIGNALING, GLI3 REPRESSOR, AND CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

Hedgehog (Hh) signaling plays a crucial role in tissue patterning, cell differentiation, and growth, and disturbances in this signaling pathway can lead to CAKUT. Binding of Hh ligand including Sonic Hedgehog (SHH) to its cognate cell surface receptor stimulates nuclear translocation of full-length GLI transcriptional activators and inhibits proteolytic processing of full length GLI3 to a shorter transcriptional repressor.⁷⁰ SHH is located on human chromosome 7q36 and is localized to the epithelium of the presumptive ureter and medullary collecting ducts. Mutations in the SHH pathway are linked to VACTERL syndrome in humans, whereas deletions are associated with renal anomalies such as hydroureter.⁷¹ Smith-Lemli-Opitz Syndrome is an autosomal recessive condition resulting from a Hh loss-of-function mutation in the DHCR7 gene. This gene encodes for sterol delta-7-reductase, which is required to convert 7-dehydrocholesterol into cholesterol. The renal manifestations of Smith-Lemli-Opitz Syndrome include renal agenesis and hypodysplasia.^{72,73} Truncating variations in the central third of the GLI3 gene in humans have been identified as responsible for development of the autosomal recessive Pallister-Hall Syndrome. These truncated variants are able to mimic the GLI3 repressor, which in SHH deficient mice, have resulted in various forms of CAKUT in 27%, with the most common being aplasia and hypoplasia followed by dysplasia, ectopic kidney, single kidney, and VUR.⁷⁴

THE MEDULLA AND GLYPICAN-3

Glypicans are a class of heparan sulfate proteoglycans that link to cells via a glycosyl-phosphatidylinositol anchor. Glypican-3 (GPC3) is essential for growth of the medullary kidney and regulates IGF-II activity. Mutations in GPC3 are associated with Simpson-Golabi-Behmel syndrome (SGBS). The pathogenesis of this X-linked disorder stems from disruption of GPC3's ability to regulate cell proliferation and apoptosis during development, leading to selective UB overgrowth followed by destruction of medullary collecting ducts due to apoptosis. The renal manifestations of SGBS include: cystic kidneys and medullary renal dysplasia.^{75,76} The cell signals regulated by GPC3 may be mediated by p57. In patients with Beckwith-Wiedemann syndrome, around 5% have mutations in p57. p57KIP2 is a cell cycle inhibitor at the G1/S phase and mutations are associated with medullary dysplasia.

TCF2, MODY5, AND SPORADIC FORMS OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

Human transcription factor 2 gene (TCF2) encodes for HNF-1B, which is required for the development of the pancreas, kidneys, liver, and intestine. TCF2 mutations are associated with the initially named maturity-onset diabetes of the young type 5 (MODY5), though the term renal cysts and diabetes syndrome (RCAD) has gained relevance to highlight the renal phenotypes of this condition. Nonketotic

diabetes mellitus is present in approximately 60% of all the cases reported, usually occurring before 25 years of age.⁷⁸ Cystic kidney diseases encompass renal cysts, cystic dysplastic kidney, and hypoplastic familial glomerulocystic kidney (FGCKD).⁷⁹ While TCF2 mutations were first identified in MODY5/RCAD, currently more than 40 renal abnormalities are known to be linked to TCF2 mutations/deletions. These include solitary functioning kidney, oligomeganephronia, atypical juvenile hyperuricemic nephropathy, autosomal dominant tubulointerstitial kidney disease (ADTKD), horseshoe kidney, bilateral hyperechogenic kidneys, renal agenesis, pelvicalyceal dilation, and MCDK. Decramer and colleagues⁸⁰ studied 62 newborns/fetuses with antenatally diagnosed bilateral hyperechogenic kidneys and revealed that large genomic TCF2 deletions were the most frequent cause (29%). Similarly, Ulinski and colleagues⁸¹ investigated 80 children with an ultrasound-confirmed structural renal abnormality, of whom 31% were carriers of a TCF2 anomaly and 88% of those carriers were noted to have a heterozygous anomaly. Among the heterozygous anomaly cohort, 64% were found to have the complete deletion of TCF2.

TUBULAR DYSGENESIS AND MUTATIONS OF RAS SYSTEM ELEMENTS

Renal tubular dysgenesis (RTD) is a severe fetal condition characterized by absence/poor maturation of proximal renal tubules and anuria, resulting in oligohydramnios and Potter sequence—altered physical appearance of the neonate with typical facies, pseudoepicanthus, recessed chin, posteriorly rotated flattened ears, flattened nose, pulmonary hypoplasia, hip dislocations, joint contractures, clubbed feet, reduced fetal movement, and ossification abnormalities of the skull. In most cases, death occurs early due to pulmonary hypoplasia or refractory hypotension.⁸² Acquired causes include the donor twin of severe twin-to-twin transfusion syndrome, congenital hemochromatosis, major cardiac conditions, or in fetuses exposed to RAS blockers.⁸³ Inherited cases are typically autosomal recessive and include mutations in the genes encoding renin (REN), angiotensinogen (AGT), angiotensin-converting enzyme (ACE) or angiotensin II receptor type 1, with the majority occurring in ACE and REN, 65.5% and 20% of cases, respectively.⁸⁴ Cases of RTD are typically diagnosed prenatally via ultrasound due to early oligohydramnios or anhydramnios, but it has been noted that some cases present with late-gestation oligohydramnios and normal second trimester ultrasound, highlighting the high level of suspicion required when treating oligohydramnios in spite of a normal second trimester renal ultrasound.⁸⁵

CHD1L, CHD7, AND CHARGE SYNDROME

Chromodomain helicase DNA-binding protein 1-like protein (CHD1L) and chromodomain helicase DNA-binding protein-7 (CHD7) are part of the Snf2 family of helicase-related ATP-hydrolyzing proteins.⁸⁶ This family of proteins plays a role in the formation of CAKUT via their effects on chromatin structure and accessibility.⁸⁷ CHD7 causes the autosomal dominant CHARGE syndrome, named for its symptoms: coloboma, heart defects, atresia of choanae, retardation, genital anomalies, and ear anomalies. Additionally, renal abnormalities are seen in 20% of cases including horseshoe kidneys, renal agenesis, renal

dysplasia, VUR, UVJ obstruction, posterior urethral valves, and renal cysts.⁸⁸ The prevalence of CHARGE syndrome is 1:10,000 with a wide variation in severity and symptom expression.⁸⁹ CHD1L is associated with hepatocellular carcinoma.⁹⁰ Brockschmidt and colleagues⁸⁶ identified CHD1L heterozygous missense mutations in 3 of 85 children affected with CAKUT.

THE ENVIRONMENT IN UTERO AND CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

In addition to genetic factors, there are many environmental factors that predispose to the development of CAKUT in utero. The renal system is particularly susceptible to the effects of teratogens and environmental toxins during fetal development, as nephron maturation continues until the 36th week of gestation, allowing for an extended period of vulnerability to these exposures.⁹⁶

In the late 1980s Barker and colleagues⁹⁷ established an association between low birth weight and increased risk of cardiovascular disease in adulthood, suggesting developmental programming of risk. Around the same time, Brenner and colleagues^{98,99} suggested a link between low birth weight and a potential nephron deficit which would predispose to elevated blood pressure and CKD in later life. As nephron number does not increase after around 36 weeks of gestational age, an individual's nephron complement at birth is a factor determining lifelong risk of kidney disease, especially in the face of superimposed kidney injury or stress. In children with CAKUT, the nephron number at birth may be an important modulator of the rate of loss of kidney function. Infants with simple renal hypoplasia or moderate to severe degree of hypodysplasia exhibit renal insufficiency. Low nephron endowment may predispose to many adult-onset conditions such as: hypertension, type 2 diabetes, and kidney disease. The impact of fetal development on kidney health across the life course is discussed in detail in [Chapter 20](#).

Factors such as uteroplacental insufficiency, maternal undernutrition, and use of teratogenic medications create a suboptimal intrauterine environment resulting in low birth weight or intrauterine growth restriction (IUGR) and have been associated with renal hypoplasia ([Table 71.5](#)). A number of human and animal studies clearly show the relationship between IUGR and reduced nephron number (discussed in detail in [Chapter 20](#)).¹⁰⁰⁻¹⁰²

Maternal dietary protein restriction results in decreased nephron number, reduced renal function, and hypertension though multiple mechanisms including increased maternal corticosterone levels, which hinder fetal intrarenal prostaglandin levels, as well as causing dysregulation of GDNF, WT1, Pax2, WNT4, BMP4, and FGF7.^{103,104} Results suggest maternal protein deficiency affects the renal function of offspring in a sex-specific manner with female rats exhibiting milder phenotypes.^{105,106}

Maternal vitamin A deficiency has been shown to predispose to reduced nephron number, with rat pups born to mothers with moderate vitamin A deficiency having 20% fewer nephrons at term.¹⁰⁷ It is hypothesized maternal retinol deficiency restricts the production of all-trans retinoic acid (atRA), impacting the branching morphogenesis of

Table 71.5 Factors Influencing in Utero Environment That Are Associated With Renal Hypoplasia

Fetal Exposure to	Renal Phenotype	References
Uteroplacental insufficiency	Hypoplasia	843
Vitamin A deficiency	Hypoplasia, hydronephrosis/ureter	844
Low-protein diet	Hypoplasia	845,846
Hyperglycemia	Agenesis, ectopic/horseshoe, cystic/dysplasia, hypoplasia, hydronephrosis/ureter	847
Cocaine	Agenesis, hypoplasia, hydronephrosis/ureter	113
Alcohol	Agenesis, ectopia/horseshoe, cystic	848,116
Angiotensin-converting enzyme inhibitors and angiotensin II receptor blockers	dysplasia, hypoplasia, hydronephrosis/ureter Renal dysgenesis	112

the adjacent UB and ultimately determining nephron number.^{108,109} In rodents, a single dose of retinoic acid administered at midgestation is able to normalize kidney size and nephron number in rat offspring exposed to maternal protein restriction, raising the possibility of preventative approaches in humans.¹¹⁰

Maternal diabetes and in utero exposure to drugs and alcohol are associated with renal hypoplasia in the absence of reduced birth weight. Fetal exposure to hyperglycemia is associated with kidney hypoplasia and nephron apoptosis via intrarenal RAS activation and NF-κB signaling.¹¹¹ Exposure of human fetuses to ACE inhibitors during the first trimester is associated with an increased risk of renal dysplasia, as well as cardiovascular and central nervous system malformations.¹¹² Human infants exposed to cocaine or alcohol in utero have an increased risk for a wide range of renal tract anomalies.¹¹³ Prenatal zinc deficiency acts as a coteratogen with alcohol, and alcohol is known to cause zinc metabolism dysregulation.^{114,115} Rat models of alcohol exposure during gestation noted reduced expression of Gdnf and Wnt11, both of which are required during ureteric branching, consistent with a decrease in nephrogenesis.¹¹⁶

FUNCTIONAL CONSEQUENCES OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

Nephrogenesis peaks between weeks 18 and 32 of gestation, reaching completion around week 35. Essential for the proper functioning of the renal system is the development of renal tubules and the expansion of the glomerular cross-sectional area, both during prenatal development and after birth. Disruptions in these pathways, as seen in renal

dysplasia and hypoplasia, contribute to reduced GFR and tubular function.

The human kidney does not exhibit a capacity to accelerate the rate of nephron formation in children born prematurely or to extend the period of nephrogenesis beyond the equivalent of 34 weeks' gestation.¹¹⁷ Thus the integrity of nephron formation in utero is absolutely critical to postnatal life. Growth of renal tubules and expansion of glomerular cross-sectional area in utero and after birth are critical to renal functional capacity. The general observation that tubule number, cross-sectional area, and cellular maturation are abnormal in renal dysgenesis is consistent with clinical observations that infants with moderate-to-severe renal hypoplasia or dysplasia demonstrate a limitation of GFR and tubular function.

A major increase in glomerular basement surface area after birth contributes to the maturational increase in GFR during infancy, childhood, and adolescence.¹¹⁸ GFR reaches adult levels at 1–2 years of age with adult level of urine concentrating capacity attained by 6 to 12 months of age. Low GFR at birth limits excretion of free water, increasing the susceptibility of newborns to hyponatremia in association with a hypotonic fluid challenge.¹¹⁹ Maximum urine concentrations achieved by preterm and term infants following fluid restriction are 600 and 800 mOsm/kg, respectively.¹²⁰ An adult level of urine concentrating capacity is attained by 6 to 12 months of age.¹²¹ Establishment of cortical and medullary domains during the 22nd to 24th week of gestation is critical to urine concentration.¹¹⁷ During embryogenesis, the renal cortex grows along a circumferential axis with a 10-fold increase in its volume. The renal medulla expands 4.5-fold in thickness along a longitudinal axis, an increase that is mainly due to elongation of the outer medullary collecting ducts.¹²² Longitudinal growth of the medulla contributes to lengthening of the loops of Henle such that they reach the inner renal medulla in the mature kidney. Elongation of the loops of Henle is important to the urine concentration mechanism because the magnitude of sodium and urea transport is greatest in longer loops, which generate steeper medullary tonicity gradients. The responsiveness of collecting duct cells to vasopressin is limited in newborns. This is thought to be due to high intrarenal levels of prostaglandins, which antagonize vasopressin.¹²³

Maturation of sodium transport in the fetus and infant is dependent on growth and differentiation within the proximal tubule, loop of Henle, and distal tubule. Normal newborn infants are limited in their capacity to respond to sodium restriction by reducing urinary sodium excretion. Interruption of tubule generation, differentiation, and growth, hallmark features of renal dysplasia, contribute to an exaggerated limitation in the capacity to absorb sodium in affected infants and children. The proximal tubule exhibits dramatic growth and maturation during renal development. The epithelium matures from a columnar to a cuboidal epithelium, microvilli are elaborated on the apical and basal domains, and expression of the Na⁺,K⁺-ATPase and the type 3 sodium-hydrogen exchanger (NHE3) increases.^{124–126} At birth, proximal tubule length is heterogeneous between the inner and outer cortex¹¹⁸; by 1 month of life, proximal tubule length becomes uniform and tubule length and

diameter have increased.¹²⁴ Maturation of tubule length is associated with the capacity to absorb sodium.¹²⁷ The loop of Henle is also characterized by increased spatial expression of transporters (NKCC2, NHE3, ROMK, Na⁺, and K⁺-ATPase) key to sodium transport.¹²⁸ Similarly, expression of NCCT and epithelial sodium channel (ENaC) is low in the neonatal kidney and increases thereafter.¹²⁹

Urinary potassium secretion is achieved predominantly by secretion of potassium in the cortical collecting duct via apical ROMK K⁺ channels. In neonates, K⁺ secretion is lower than in children due to the low secretory capacity of the cortical collecting duct. The postnatal increase in K⁺ secretion is thought to be due to a developmental increase in the number of ROMK channels,¹³⁰ as well as the BK potassium channel of the collecting duct.¹³¹ Malformation of tubules in disorders such as renal dysplasia is commonly associated with limited K⁺ secretion, particularly during infancy. Critical for urine concentrating abilities of the kidney is the longitudinal growth of the medulla, and thus the loops of Henle, which allows for creation of a greater concentration gradient allowing for increased transport of sodium and urea.^{132–134}

CLINICAL PRESENTATION OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

CLINICAL PRESENTATION IN THE FETUS

Antenatal ultrasound imaging is the primary method in which 60% to 85% of CAKUT is diagnosed with a sensitivity of approximately 80%. The fetal kidney can be visualized at 12 to 15 weeks of human gestation, and screening antenatal ultrasound is recommended between 16 and 20 weeks of gestation. Ultrasound allows for detection of structural abnormalities, as well as quantification of amniotic fluid volume, which is primarily composed of urine produced by the fetal kidneys.^{96,135} CAKUT may also become apparent during infancy or childhood due to UTIs, while in adolescence or adulthood, symptoms such as hypertension may arise.

Fetal ureters are not typically visualized on ultrasound. If they are observed, it may suggest potential concerns such as ureteral or bladder obstruction, or vesicoureteral reflux (VUR).¹³⁶ Lower urinary tract obstruction, such as posterior urethral valves (PUV) in males, may present on ultrasound as a distended fetal bladder in the first trimester. In the second trimester, findings indicating PUV include oligohydramnios, bilateral hydronephrosis, and obstructive renal dysplasia.¹³⁷

The development of the kidney in utero is monitored using fetal renal length, standardized for gestational age. The mean combined kidney length (MKL) of 20.87 ± 0.75 mm and 41.41 ± 0.07 mm at 20 and 41 weeks, respectively, have been reported.¹³⁸ Assessing renal function in utero is measuring amniotic fluid volume. Fetal urine production begins at 9 weeks of gestation. By 20 weeks of gestation and thereafter, fetal urine is the primary source of amniotic fluid.¹³⁹ Insufficient amniotic fluid, or oligohydramnios may be caused by bilateral renal dysplasia (or a critical defect in one kidney where a solitary kidney exists), bilateral ureteral obstruction, or obstruction of the bladder outlet. In the second trimester, severe oligohydramnios may cause

lung compression and increased fluid losses via the trachea ultimately resulting in lung hypoplasia. The most severe consequence of oligohydramnios and pulmonary hypoplasia is Potter sequence.^{140,141} In contrast, polyhydramnios reflects an increased amount of amniotic fluid and is associated with intestinal and esophageal atresias, rather than renal structural abnormalities.¹⁴² Congenital Bartter syndrome or other abnormalities associated with fetal polyuria may be associated with polyhydramnios, which may precipitate preterm birth.¹⁴³

The composition of fetal urine is also used as a marker of renal function. Increases in urea and creatinine in the second and third trimesters observed in amniotic fluid and fetal urine are due to an increase in the GFR of the fetal kidneys and signify a maturation of their tubular function.¹⁴⁴ Urine sodium and β -2-microglobulin decrease with increasing gestational age due to increasing reabsorption capacity of proximal tubular cells.¹⁴⁴ Fetuses with bilateral renal dysplasia or severe bilateral obstructive uropathy present with increased sodium and β -2 microglobulin and reduced osmolality due to impaired resorption and renal function.¹⁴⁵ In general, sodium and chloride concentration >90 mEq/L (90 mmol/L) and urinary osmolality <210 mOsmol/kg-H₂O (210 mmol/kg-H₂O) are indicative of fetal renal tubular impairment and poor renal prognosis. In addition, urinary β -2-microglobulin levels >6 mg/L are predictive of severe renal damage with a sensitivity and specificity of 80% and 71%, respectively.^{146,147}

CLINICAL PRESENTATION OF SPECIFIC FORMS OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

Renal agenesis

Unilateral renal agenesis (URA) is defined as the congenital absence of renal tissue resulting from failure of embryonic kidney formation on one side of the body. It has an incidence of approximately 1 in 2000 and is most often detected either during routine antenatal ultrasonography or during the assessment of an accompanying urinary tract abnormality.¹⁴⁸ URA is typically asymptomatic but may be associated with other congenital abnormalities. The most common of these is vesicoureteral reflux (VUR), found in 24% of cases, followed by pelviureteric junction obstruction (PUJO) in 6%. URA should be distinguished from spontaneous involution of a MCDK, contralateral nonfunctioning kidney, or an ectopic location of the second kidney, such as in the pelvis.^{149,150}

Renal dysplasia

Renal dysplasia is observed in around 0.1% among infants.¹⁵¹ It encompasses a range of disorders in which the kidneys begin to form, but nephrons and collecting ducts fail to progress into fully developed structures. This is due to a failure in metanephric differentiation.¹¹⁷ Microscopically renal dysplasia appears as primitive ducts with a fibromuscular collar and lobar disorganization and may be diffuse, segmental, or focal.¹¹⁸ Diagnosis is made via ultrasound, typically in infancy or childhood. Sonographic features include reduced corticomedullary differentiation and/or diffuse cortical thinning in the absence of urinary tract dilatation. Renal dysplasia frequently co-occurs with

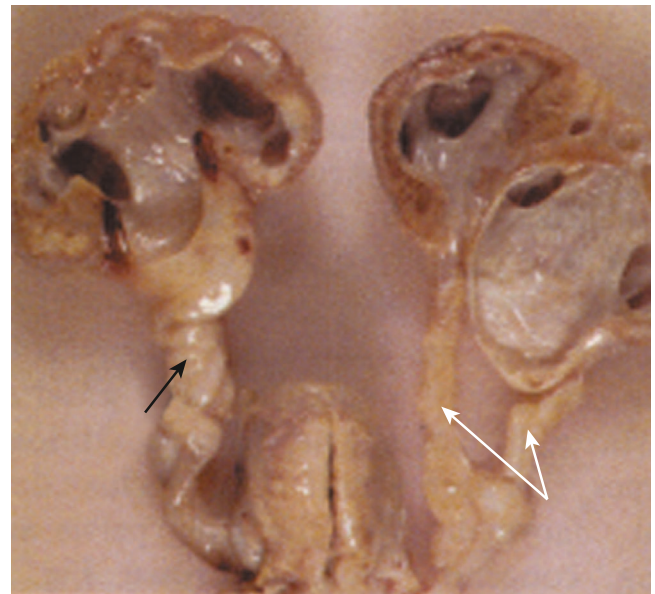


Fig. 71.6 Duplicated collecting system. Duplicated ureters (white arrows) are shown on the right. A single dilated ureter (black arrow) is shown on the left. Each ureter is dilated due to obstruction at the level of the bladder.

associated complications, such as UPJ obstruction, ureteral atresia, urethral obstruction, and vesicoureteral reflux.¹¹⁹ Large cystic elements can contribute to large kidney size. Kidney prognosis is worse in patients with bilateral dysplasia compared with unilateral dysplasia.¹²⁰ The MCDK is an extreme example of a large dysplastic kidney (see Fig. 71.4).

The MCDK is characterized by multiple large cysts and a lack of normal renal parenchyma, causing the kidney to be nonfunctional. It is usually unilateral. Bilateral MCDK is incompatible with life. The majority of cases, 60% to 80%, are diagnosed antenatally and 20% are diagnosed postnatally. Unilateral MCDK is typically asymptomatic but may be associated with contralateral abnormalities in 25% of cases, including VUR (7%–26%), UPJ obstruction (1.5%–5%), and UVJ obstruction (2%).¹²¹ While there have been suggestions of potential associations between MCDK and conditions such as undescended testicles (UDT), hypertension, and malignancies including Wilms tumor, it should be noted that hypertension occurs rarely (0.01%–0.1% of cases), and incidence of RCC and Wilms tumor in MCDK cases does not significantly deviate from that of the general population.^{122,123} The natural history of an MCDK is gradual reduction in size, with involution in up to 60% of affected kidneys at 2 years of age.

Double Collecting System

Complete or partial duplication of the renal collecting system is the most common congenital anomaly of the urinary tract (Fig. 71.6).¹²⁴ Premature division of the UB or double UBs results in a double collecting system. Completely duplicated systems are characterized by two separate pelvicalyceal systems and two ureters and are often associated with VUR.¹²⁴ Partially duplicated systems are more common and are characterized by two separate

pelvicalyceal systems with either a single ureter or two ureters that unite before insertion into the bladder. Partial duplication is associated with ureteroureteral reflux or UPJ obstruction of the lower pole of the kidney.¹²⁵ The most common presenting symptoms of an ectopic ureter are UTI and incontinence.

Renal Ectopy

Renal ectopy (Fig. 71.5) is defined as an abnormally located kidney due to failure of migration during embryologic development. During fetal development the kidneys develop in the pelvic region and migrate to their position in the retroperitoneal renal fossa by week 8 of gestation.¹²⁶ With ascension comes a 90-degree rotation from a horizontal to a vertical position with the renal hilum finally directed medially. Failure of the kidney to ascend results in simple congenital ectopy in which the kidney typically remains in the pelvis and is termed a “pelvic kidney,” which can be unilateral or bilateral. Less commonly, the kidney may lie on the contralateral side of the body, a state that is termed crossed ectopy without fusion. Renal ectopia is typically asymptomatic, though complications include calculi formation and infection.^{127,128} Rarely, an unusually high ascent of the metanephros can result in a diaphragmatic defect and a kidney located in the thorax, which is generally asymptomatic and incidentally diagnosed.¹²⁹

Renal Fusion

Renal fusion anomalies are defined as the congenital fusion of the kidneys. Partial fusion anomalies include horseshoe kidney and crossed fused renal ectopia, whereas complete fusion anomalies include “cake” kidney or fused pelvic kidney. Horseshoe kidney is the most common anomaly and is defined as a median fusion anomaly in which either the lower poles (most common) or the upper poles (rare) of the two kidneys are connected to each other by an isthmus.¹³⁰ It is hypothesized that the migration of the developing kidneys may be disrupted by an abnormal position of the umbilical artery, resulting in partial renal fusion.¹³¹ The isthmus may be composed of either parenchymal or fibrous tissue. Depending on the site of fusion, horseshoe kidneys may be classified as symmetric (midline fusion) and asymmetric (lateral fusion).¹⁵² The vascular supply of horseshoe kidneys is anomalous and is associated with accessory pathways. Though most patients remain asymptomatic, the anomalous collecting system position may result in urinary stasis, leading to infection and renal calculi occur in 20% of cases.¹⁵³ In addition, horseshoe kidney is associated with UPJ obstruction, collecting system duplication, and increased risk of Wilms tumor.^{154,155}

Crossed fused ectopy occurs when the ectopic kidney and ureter cross the midline to fuse with the contralateral kidney but the ureter of the ectopic kidney maintains its normal insertion into the bladder. The affected kidney is typically situated inferior to its contralateral partner. The majority of patients affected remain asymptomatic and are diagnosed incidentally, though complications such as obstructive uropathy and VUR may occur.¹⁵⁶

Cake kidney, or fused pelvic kidney, is a complete renal fusion anomaly. It is defined as “an anomaly in which the

entire renal substance is fused into one mass, lying in the pelvis and giving rise two separate ureters which enter the bladder in normal relationship.”¹⁵⁷ Affected patients are typically asymptomatic, but this condition is associated with accompanying congenital anomalies such as anomalies of testicular descent, anomalies of vas deferens, vaginal agenesis, bicornuate or unicornuate uterus, sacral agenesis, caudal regression syndrome, tetralogy of Fallot, and spina bifida.¹⁵⁶

CLINICAL MANAGEMENT OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

CAKUT is a major cause of ESKF in children, accounting for 30% to 50% of cases.¹⁵⁸ Therefore the prompt identification and commencement of treatment is imperative to minimize renal damage.

APPROACH TO MANAGEMENT AND IN THE IMMEDIATE POSTNATAL PERIOD

A coordinated multidisciplinary approach including teams from obstetrics, pediatric nephrology, pediatric urology, and neonatology is required. It is critical to counsel families appropriately and clearly communicate diagnosis and prognosis. A diagnosis of CAKUT is often made antenatally. Some cases allow for surgical intervention before symptomatic consequences. Interventions that aim to relieve lower urinary tract obstruction include serial ultrasound-directed vesicocentesis, vesicoamniotic shunting, fetal cystoscopy, and valve ablation.¹⁵⁹ These interventions aim to relieve urinary tract obstruction and promote pulmonary development. To date, little evidence exists that relief of urinary tract obstruction in utero prevents the development of associated renal dysplasia or renal scarring. In contrast, insertion of a bladder–amniotic cavity shunt in the fetus with an obstruction below the bladder neck can rescue oligohydramnios and pulmonary hypoplasia.^{160,161}

Following delivery, a thorough history and examination must be undertaken in newborns with antenatally diagnosed/suspected CAKUT. This includes evaluation of the respiratory system to assess for pulmonary insufficiency; the abdomen to assess for masses indicating renal enlargement or MCDK; the ears, as outer ear abnormalities are associated with an increased risk of CAKUT; and the umbilicus, as a single umbilical artery is associated with an increased risk of CAKUT.¹⁶² Examination should also assess for Potter sequence resulting from severe oligohydramnios.^{67,163}

An ultrasound within the first 24 hours post-birth is required for newborns with known/suspected bilateral renal malformations, a solitary malformed kidney, or a history of oligohydramnios to assess if bladder decompression is required. Newborns with unilateral renal malformations should be assessed via ultrasound between 48 hours and 1 week after birth as hydronephrosis may not be immediately evident. If severe hydronephrosis is detected on ultrasound, antibiotics (prophylactic dose, adjusted for age—amoxicillin, ampicillin, cotrimoxazole, trimethoprim, and nitrofurantoin) should be commenced as prophylaxis for pyelonephritis and a voiding cystourethrography (VCUG) should be performed to assess if reflux is the

underlying etiology and, if yes, the grade of the reflux. Serum creatinine concentration should be assessed in the newborn after the first 24 hours as prior levels reflect maternal creatinine. Creatinine should decline to normal values (serum creatinine 0.3 to 0.5 mg/dL [27–44 $\mu\text{mol/L}$]) within approximately 1 week in term infants and 2 to 3 weeks in preterm infants.⁶⁷ Pediatric nephrology support is required from birth for optimal management of fluid and electrolytes.

MANAGEMENT OF SPECIFIC TYPES OF CONGENITAL ANOMALIES OF THE KIDNEY AND URINARY TRACT

Vesicoureteral Reflux

An increased risk of VUR exists in renal dysplasia, but VUCG is not currently recommended unless a child develops recurrent febrile UTIs and/or has dilatation of the urinary tract.¹⁶⁴ In the case of recurrent UTI or suspected high-grade reflux on ultrasound, VUCG is recommended to assess for associated VUR.^{165–167} Magnetic resonance imaging (MRI) can provide improved visualization of ectopic ureters and more sensitive detection of renal scarring, which may be used as an indication for urologic intervention. The diagnostic workup for children should assess overall health and development, including height, weight, and blood pressure, with kidney and bladder ultrasound used as the primary postnatal evaluation tool.¹⁶⁸ Toilet-trained children with both lower urinary tract dysfunction and VUR face a higher risk of recurrent febrile UTIs compared with those with isolated VUR. Bladder and bowel dysfunction (BBD) is common in toilet-trained children with febrile UTIs, with or without primary VUR, and the coexistence of BBD and VUR doubles the risk of recurrent febrile UTIs. Therefore all children presenting with UTI should be assessed for BBD and appropriately managed before VUR treatment. UTIs in general are discussed in [Chapter 38](#). [Fig. 71.7](#) outlines the approach to infant with a suspected UTI. [Fig. 71.8](#) outlines the necessary workup in a child with recurrent pyelonephritis or a UTI with an uncommon pathogen.

The primary goal in managing VUR is to preserve renal function, with treatment generally divided into nonsurgical and surgical approaches. Nonsurgical management includes close monitoring, intermittent or continuous antibiotic prophylaxis at lower doses, and BBD rehabilitation. Follow-up involves imaging, height, weight, blood pressure, and serum creatinine monitoring to track VUR resolution and kidney health. Antibiotic prophylaxis is most beneficial in specific cases, such as infants with grade III–V VUR, though it may increase resistance to some pathogens.¹⁶⁹ Surgical options, such as endoscopic injection of bulking agents or ureteral reimplantation (open or minimally invasive), are reserved for patients with antibiotic noncompliance, breakthrough infections, or persistent symptomatic VUR. Although high-grade VUR in infants often resolves more successfully with injection therapy, it's not universally recommended, as spontaneous improvement can occur.

Ureteropelvic Junction Stenosis

Ureteropelvic junction stenosis (UPS) is a narrowing in the area of the pyeloureteral junction (see [Fig. 71.2](#)). Depending on the urodynamic relevance, there is a morphological dilatation of the renal pelvic caliceal system and functional impending loss of renal function. UPS is the most common

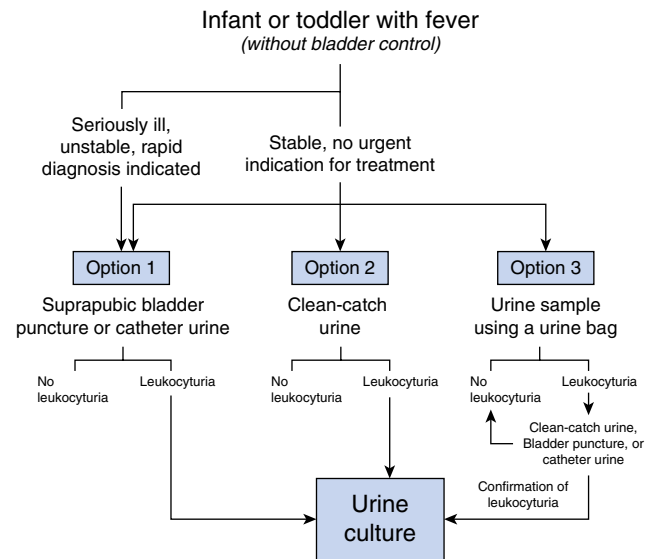


Fig. 71.7 Algorithm for infant or toddler with fever and concern for urinary tract infection. (From Pediatric Urinary Tract Infection—Diagnostics, therapy and prophylaxis. AWMF Register Nr. 2021;166–204.)

cause of sonographically detected hydronephrosis with an incidence of 1:1000 to 1:4000. In two thirds of cases, UPS is found unilaterally on the left side with a clear male predilection. Bilateral UPS was found in only 6% of operated patients. In one in four newborns there is an association with other urological anomalies. In addition to sonography, the most important investigation is dynamic $^{99\text{mTc}}$ -MAG3 renal scintigraphy. In compensated conditions (stable split renal function, functional obstructive outflow pattern), a conservative approach should be taken. Ureteropyeloplasty is performed in cases of decompensated outflow or decreasing split renal function. Postoperative results and prognosis are very good.

Aetiology. A distinction is made between intrinsic and extrinsic causes of ureteropelvic junction stenosis. The intrinsic cause is an alteration in the structure of the ureteral wall at the pyeloureteral junction in the form of fibromuscular dysplasia. The aetiology and interpretation of this change is still unclear. It accounts for about 75% of cases. In about 10% of cases there is a so-called high ureteral orifice with consequent obstruction of the outflow from the renal pelvis. The extrinsic causes are either an abnormal lower pole vessel or a kink of the ureter fixed by fibrotic cords. This is often associated with or caused by recurrent inflammation, e.g., as a result of vesicoureteral reflux. Patients with a lower pole vessel typically present clinically much later, often at school age.

Clinical Appearance. Most congenital intrinsic stenoses can hold large volumes of urine due to the high distensibility of the smooth muscle of the renal pelvis and tend to be characterized by a chronic and usually compensated urodynamic state. The risk of progressive renal parenchymal injury is low. Renal pelvic dilatation is usually asymptomatic and is detected in two thirds of cases during the neonatal screening ultrasound examination. In a maximum of 20% of children who

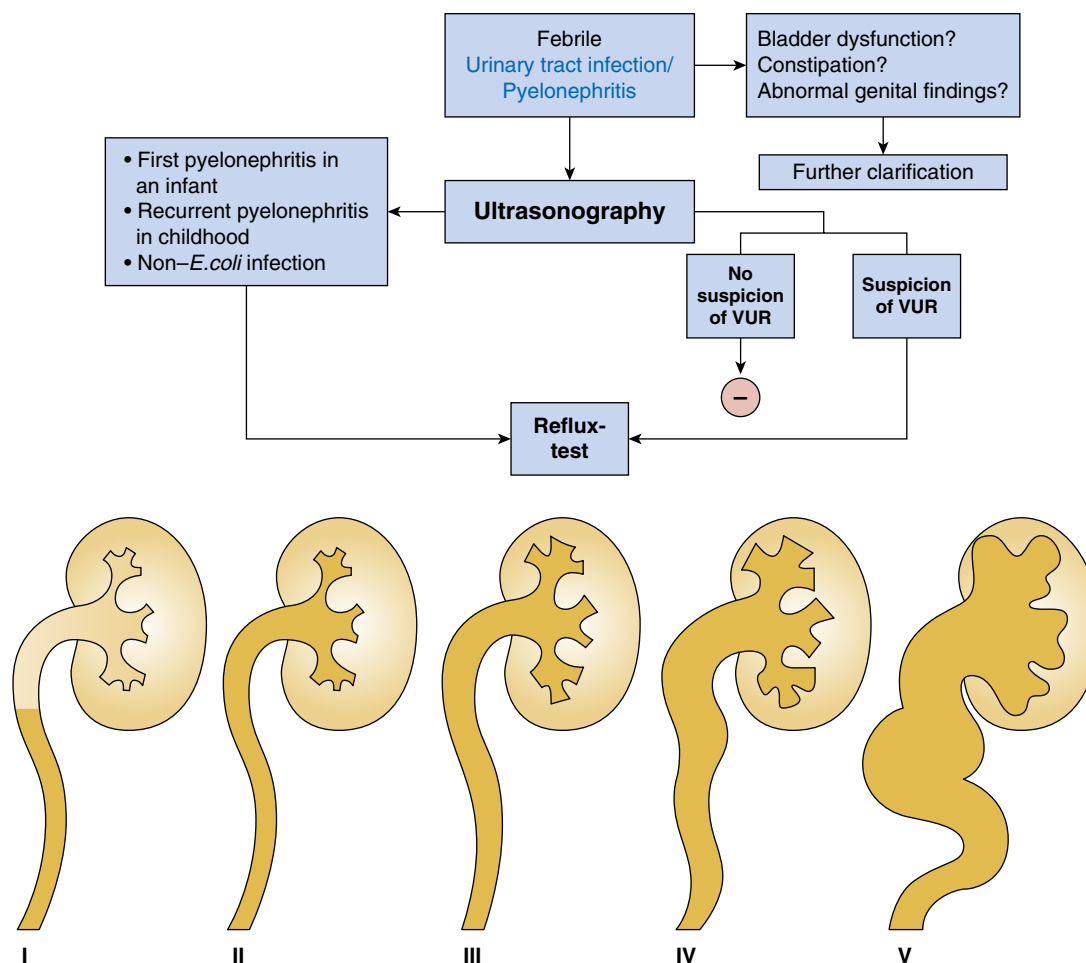


Fig. 71.8 Top: For further testing in infant or toddler with urinary tract infection; bottom: Grading of vesicoureteral reflux. (Top, From Pediatric Urinary Tract Infection -Diagnostics, therapy and prophylaxis. AWMF Register Nr. 2021;166-204. Bottom, From Gargollo P, Diamond D. Therapy insight: what nephrologists need to know about primary vesicoureteral reflux. *Nat Rev Nephrol.* 2007;3:551-563.)

were not screened, clinical symptoms such as urinary tract infection (pyelonephritis, urosepsis), nausea and vomiting or haematuria lead to further investigation. Older children with anomalous lower pole vessels are an exception and often present with recurrent flank pain as a result of acute decompensation. This is caused by large amounts of fluid or diuretic substances. Exceptional presentations of ureteropelvic junction obstruction are the presence of urolithiasis, hypertension or renal dysfunction, especially in the case of a functioning single kidney.

Diagnosis. The aim of all interventions is to identify any relevant obstruction as early as possible to potentially limit renal parenchymal dysfunction. Ultrasonography is the initial diagnostic method of choice. Classic B-scan ultrasonography should be supplemented by colour-coded Doppler sonography, which can be used to assess the vascular supply to the kidneys. In addition, accessory vessels should be looked for and crossing of the ureteropelvic junction as an extrinsic cause of dilatation should be ruled out. If pathological sonographic findings indicate ureteropelvic junction stenosis (especially in children with urinary transport disorder grade $\geq$ III or anterior-posterior diameter of the renal pelvis $\geq$ 15 mm), functional diagnostics using ^{99m}Tc -MAG3 diuresis scintigraphy is also indicated. Worsening of split

renal function and development of urinary obstruction are decisive criteria for further diagnostic and therapeutic procedures.

Therapy. Treatment depends largely on kidney function and drainage pattern. Regular monitoring with sonography and scintigraphy is essential. In compensated conditions, i.e., stable split renal function and functional obstructive outflow, conservative observation is indicated. Antibiotic prophylaxis is not usually required.^{169a} Especially in the first months of life, the outflow conditions improve significantly by stretching and opening of the so-called fetal ureteral loops. Severe renal pelvic calyceal dilatation and a functional loss of $<40\%$ of the affected kidney are indications for surgical intervention. The ureteropelvic junction stenosis should also be surgically corrected if the split renal function changes or the obstructive outflow pattern remains the same over the course of the disease, to prevent further damage to the affected kidney. An initial function of $<15\%$ on the side of the affected kidney may be an indication for temporary percutaneous nephrostomy to assess the possibility of recovery of function of the affected kidney, provided that the kidney is not already severely dysplastic and likely to be non-functional. The creation of a temporary percutaneous nephrostomy should be carefully considered as the complication rate of subsequent

ureteropyeloplasty is higher. The standard procedure is the Anderson-Hynes ureteropyeloplasty. The stenosis at the pyeloureteral junction is resected (dismembered) and the ureter is reanastomosed to the pyelon in a funnel shape. After pyeloplasty, a scintigraphic improvement in urine flow is achieved in 97% to 99% of cases. A functional delay in outflow remains in about 60% of cases, but this has not been shown to affect renal function in the long term. After surgical correction, it is particularly important to monitor renal growth and blood pressure, as hypertension occasionally occurs. In the case of spontaneous improvement or non-surgical findings, it is important to ensure that sonographic monitoring of the pyelon width is carried out every 6 to 12 months during physical growth in order to detect secondary decompensation in good time.

Primary and Secondary Megaureter

The term megaureter initially refers only to an “enlarged” ureter. The aetiology, whether primary or secondary, obstructive and/or refluxing, is crucial. Usually there is a primary obstruction at the ureterovesical junction (primary obstructive megaureter), which may be associated with reflux (primary refluxive-obstructive megaureter) (see Fig. 71.2). The prognosis for primary obstructive megaureter (POM) is very good with today’s essentially conservative management. Only 10% to 15% of children require surgical correction and renal function can almost always be fully preserved.

A megaureter is a ureter which:

- is dilated either prevesically or along its entire course to the renal pelvis
- exceeds 7 mm in diameter^{169b}

Primary forms of megaureter are caused by a pathological ureterovesical junction, either resulting in obstruction, vesicoureteral reflux or a combination of both.

Secondary forms occur when the pathology is distal to the ureterovesical junction, i.e., vesical or sub-vesical (e.g., neurogenic bladder emptying disorder, posterior urethral valves) and a resultant change in the bladder wall has led to vesicoureteral reflux or obstruction at the ureterovesical junction.

Primary and secondary megaureters can be:

- obstructive
- refluxive
- obstructive/refluxive
- non-obstructive/non-refluxive^{169c}

POM is the second most common cause of congenital obstructive uropathies after ureteropelvic junction stenosis, accounting for about 20%. The incidence is approximately 1 in 4,000 births. In primary obstructive megaureter, boys are 4 times more likely to be affected than girls. The megaureter is most common on the left side, but in 20% it is bilateral. About one third (24% to 40%) of patients also have other contralateral urinary tract anomalies, which may present as renal agenesis, renal dysplasia (15%) or reflux into the contralateral kidney. A prenatally diagnosed megaureter does not allow conclusions to be drawn about its cause or postnatal course.^{169d}

Etiology. The primary obstructive megaureter is usually the result of a distal ureteral obstruction at the ureterovesical

junction, possibly in combination with vesicoureteral reflux. The cause of the obstruction is a distal segment of the ureter, usually 0.5 cm to several centimetres long. This interrupts the peristaltic waves, resulting in an obstruction of the outflow. The aetiology of this adynamic segment is unclear. Histologically there is muscle hypotrophy. An overexpression of transforming growth factor- β with a delay in the normal differentiation of the ureteral muscle is suspected.^{169e} Insufficient development of the so-called Cajal cells, and thus reduced motor pacemaker function in this region, may also play an etiological role.^{169f} In the absence of increased intraluminal pressure, primary megaureters usually do not cause renal parenchymal damage and are associated with good renal function (non-obstructive, non-refluxing megaureters).

Clinical Appearance. Nowadays, more than 80% of megaureters are discovered during fetal ultrasound or during screening examinations in infants and young children. The megaureter may remain asymptomatic for a long period, especially if it is purely obstructive. Urinary tract infection is the most common symptom in the presence of associated vesicoureteral reflux. Megaureters are found in about 20% of symptomatic urinary tract infections in children. As part of the CAKUT complex, primary renal dysplasia may be associated on the affected side. Haematuria and stone formation are rare.

Diagnostics. As with all obstructive uropathies, diagnosis is essentially based on sonography and renal function scintigraphy. In the presence of a megaureter, reflux testing is mandatory. 99m Tc-MAG3 diuresis scintigraphy is indicated in asymptomatic infants with ureteral dilatation and renal pelvic calyceal dilatation (urinary transport disorder grade $\geq$ III) or in the presence of ureteral dilatation >10 mm. It allows:

- the determination of split renal function
- an assessment of the urodynamic effectiveness of the urinary outflow obstruction

Therapy. The management of primary obstructive megaureter has changed dramatically over the last few decades. Whereas 30 years ago surgery was performed in more than 80% of cases in the form of ureteral reimplantation, today this condition predominantly managed conservatively, taking advantage of the spontaneous regression of the adnexal segment with corresponding improvement in outflow.^{169g,169h} The mechanism of this regression is not yet understood. Surgical treatment of primary obstructive megaureter should therefore be the exception, but is still necessary in about 10% to 15% of cases.

Prognostically favorable criteria for spontaneous regression include:

- time of diagnosis
- partial renal function > 40%
- non-obstructive washout of urine on diuresis scintigraphy
- active peristalsis of the dilated ureter behind the bladder

The indication for surgical correction depends on the function and outflow pattern and the incidence of urinary

tract infections. Surgical correction is indicated in the presence of initially impaired renal function and a constant obstructive outflow pattern, when outflow conditions deteriorate and/or when there is a loss of function over time, or clinical symptoms arise. Dynamic renal scintigraphy (^{99m}Tc -MAG3) is of value to monitor these changes. If surgical correction is unavoidable, the goal is resection of the distal adnexal segment followed by ureteral reimplantation. Ureteral tapering is only necessary in cases of marked ureteral dilatation. In these cases, a two-stage approach may be required, starting with a ureterocutaneostomy to tone the ureter prior to reimplantation. Other forms of treatment such as endoscopic slitting or balloon dilatation of the ostium or with JJ stent insertion have been reported.

The indication for antibiotic prophylaxis in children with primary obstructive megaureter is controversial. It should possibly be considered in the first six months of life in children with asymptomatic megaureter with simultaneous dilatation of the pelvicoliceal system.

The prognosis is very good if the child is treated conservatively, with strict monitoring of renal function and outflow facility. Urinary outflow improves in about 85% to 90% of children with POM during the first 2 years of life, to the extent that renal injury is no longer a concern. However, if there is ipsilateral high-grade vesicoureteral reflux (primary refluxive-obstructive megaureter), spontaneous maturation of the ureterovesical junction is less likely.

Renal Dysplasia

Renal dysplasia is typically diagnosed via ultrasound antenatally, appearing as increased parenchymal echogenicity in a small kidney. A DMSA radionuclide scan may be performed 4 to 6 weeks after birth to assess the differential function of each kidney, which is useful in decisions regarding surgical interventions.¹⁷⁰ Renal ultrasound can be used to monitor the contralateral kidney over time for compensatory hypertrophy.⁶⁷

The natural history of MCDK is gradual reduction in size; typically an involution occurs in up to 60% by age 2. Renal ultrasound is recommended at an interval of 3 months for the first year of life and then every 6 months up to involution of the kidney, or at least up to age 5 years, with regular monitoring for hypertension.¹⁷¹

Renal Ectopy and Fusion

Ectopic kidney may be located in the pelvis and less commonly on the contralateral side of the body. Abdominal and pelvic ultrasound is used to determine the exact location and presence of a collecting system abnormality. VUCG may be performed in patients with hydronephrosis or horseshoe kidney. No further evaluation is required in the patient with a normal-appearing contralateral kidney and no evidence of hydronephrosis in the ectopic kidney. Elevated serum creatinine or abnormal contralateral kidney are both indications for a DMSA radionuclide scan to assess differential renal function. A diuretic renogram with (^{99m}Tc)-mercaptotriglycylglycine (MAG-3) or technetium-99m Tc -diethylenetriamine pentaacetic acid (DTPA) should be performed to detect obstruction if there is severe hydronephrosis and the VUCG is normal.¹⁷² If the hydronephrosis is mild or moderate and the VUCG is normal, then follow-up ultrasonography should be

performed 3 to 6 months later. If there is progressive hydronephrosis, then an MAG-3 or DTPA diuretic renogram should be performed to detect obstruction. MRI provides an alternative for the evaluation of hydronephrosis in children and provides excellent anatomic detail and clear corticomedullary differentiation; therefore it may be preferred if the facilities exist. GFR and comparative renal function, right and left, can also be determined by MRI and may be helpful to detect severity and progression of disease, as well as to follow-up kidney function post-intervention or over time if indicated.

LONG-TERM OUTCOMES OF RENAL MALFORMATION

Clinical outcomes in CAKUT vary widely from no symptoms whatsoever to CKD resulting in kidney failure during a period ranging from the newborn period to the fourth and fifth decades of life. Risk factors for mortality during infancy and early childhood include coexistence of renal and nonrenal disease, prematurity, low birth weight, oligohydramnios, and severe forms of renal-urinary tract malformation (agenesis and hypodysplasia).¹⁷³ In a case series of 822 children with prenatally detected CAKUT who were followed for a median time of 43 months, Quirino and colleagues¹⁷⁴ reported a mortality of 1.5% and morbidities including UTI, hypertension, and CKD in 29%, 2.7%, and 6% of surviving children, respectively. Sanna-Cherchi and colleagues¹⁷⁵ found that 25% of 312 children born with bilateral CAKUT developed ESKF during the first two decades of life. Elevated serum creatinine, presence of proteinuria, and VUR were found to be associated with a higher risk of renal disease progression. A faster rate of decline of renal function in patients with CAKUT and CKD has been associated with a urine albumin-to-creatinine ratio >200 mg/mmol compared with <50 mg/mmol (eGFR -6.5 mL/min/1.73 m^2 /year vs. -1.5 mL/min/1.73 m^2 /year), and with more than two (vs. <2) febrile UTIs (eGFR -3.5 mL/min/1.73 m^2 /year vs. -2 mL/min/1.73 m^2 /year). A greater decline in eGFR occurs during puberty (eGFR -4 mL/min/1.73 m^2 /year vs. -1.9 mL/min/1.73 m^2 /year).¹⁷⁶ A study examining the risk for dialysis in patients with CAKUT demonstrated a significantly higher risk for patients with a solitary kidney compared with nondisease controls.¹⁹ These results raise the possibility that the prognosis for a solitary apparently normal kidney may not be as “normal” as previously thought. Data from the European Dialysis and Transplant Association Registry showed that the mean age at which patients with CAKUT require dialysis and/or transplantation is 31 years indicates that children with CAKUT are at risk of developing a requirement for dialysis and/or transplantation as adults.¹⁷⁷

GLOMERULAR DISEASES

NEPHROTIC SYNDROME

Nephrotic syndrome (NS) is the most common glomerular disease encountered during childhood, characterized by edema, nephrotic-range proteinuria, hypoalbuminemia, and dyslipidemia. The underlying mechanism of disease is a defective glomerular filtration barrier and loss of podocyte integrity. In most children, the specific cause of disease

Table 71.6 Definitions of the Clinical Forms of Nephrotic Syndrome

Term	Definition
Nephrotic-range proteinuria ^a	Urinary protein creatinine ratio (UPCR) ≥ 200 mg/mmol (2 mg/mg) in a spot urine, or proteinuria ≥ 1000 mg/m ² per day in a 24-h urine sample corresponding to 3+ (300-1000 mg/dL) or 4+ (≥ 1000 mg/dL) by urine dipstick
Nephrotic syndrome	Nephrotic-range proteinuria and either hypoalbuminemia (serum albumin < 30 g/L) or edema when serum albumin is not available
Complete remission	UPCR (based on first morning void or 24-h urine sample) ≤ 20 mg/mmol (0.2 mg/mg) or < 100 mg/m ² per day, respectively, or negative or trace dipstick on three or more consecutive days
Partial remission	UPCR (based on first morning void or 24-h urine sample) > 20 but < 200 mg/mmol (> 0.2 mg/mg but < 2 mg/mg) and serum albumin ≥ 30 g/L
Steroid-sensitive nephrotic syndrome (SSNS)	Complete remission within 4 weeks of PDN at standard dose (60 mg/m ² /day or 2 mg/kg/day, maximum 60 mg/day)
Steroid-resistant nephrotic syndrome (SRNS)	Lack of complete remission within 4 weeks of treatment with PDN at standard dose
Confirmation period	Time period between 4 and 6 weeks from PDN initiation during which responses to further oral PDN and/or pulses of IV MPDN and RAASi are ascertained in patients achieving only partial remission at 4 weeks. A patient not achieving complete remission by 6 weeks, although partial remission was achieved at 4 weeks, is defined as SRNS
SSNS late responder	A patient achieving complete remission during the confirmation period (i.e., between 4 and 6 weeks of PDN therapy) for new-onset NS
Relapse	Urine dipstick $\geq 3+$ (≥ 300 mg/dL) or UPCR ≥ 200 mg/mmol (≥ 2 mg/mg) on a spot urine sample on 3 consecutive days, with or without reappearance of edema in a child who had previously achieved complete remission
Infrequently relapsing nephrotic syndrome	< 2 relapses in the 6 months following remission of the initial episode or fewer than 3 relapses in any subsequent 12-month period
Frequently relapsing nephrotic syndrome (FRNS)	≥ 2 relapses in the first 6 months following remission of the initial episode or ≥ 3 relapses in any 12 months
Steroid-dependent nephrotic syndrome (SDNS)	A patient with SSNS who experiences 2 consecutive relapses during recommended PDN therapy for first presentation or relapse or within 14 days of its discontinuation
Steroid toxicity	New or worsening obesity/overweight, sustained hypertension, hyperglycemia Behavioral/psychiatric disorders, sleep disruption Impaired statural growth (height velocity < 25 th percentile and/or height < 3 rd percentile) in a child with normal growth before start of steroid treatment Cushingoid features, striae rubrae/distensae, glaucoma, ocular cataract, bone pain, avascular necrosis
Sustained remission	No relapses over 12 months with or without therapy
SSNS controlled on therapy	Infrequently relapsing NS or sustained remission while on immunosuppression in the absence of significant drug-related toxicity
SSNS not controlled on therapy	Either frequently relapsing NS despite immunosuppression or significant drug-related toxicity while on immunosuppression
Secondary steroid resistance	SSNS patient who at a subsequent relapse does not achieve complete remission within 4 weeks of PDN at standard dose
Complicated relapse	A relapse requiring hospitalization due to 1 or more of the following: severe edema, symptomatic hypovolemia or AKI requiring IV albumin infusions, thrombosis, or severe infections (e.g., sepsis, peritonitis, pneumonia, and cellulitis)

^aIn adults, nephrotic-range proteinuria is defined by proteinuria > 3.5 g/24 h (or > 3000 mg/g or > 3 g/10 mmol creatinine). These cutoffs should also apply to adolescents (> 16 years). From Trautmann A, Boyer O, Hodson E, et al. IPNA clinical practice recommendations for the diagnosis and management of children with steroid-sensitive nephrotic syndrome. *Pediatr Nephrol.* 2023;38(3):877–919.

remains unknown, although additional causes include glomerular disorders, vasculitides, infections, toxins, and malignancy. Expanding genetic studies are steadily revealing greater numbers of genetic mutations associated with nephrotic syndrome. Nephrotic syndromes are discussed further in [Chapter 30](#).

CLINICAL CLASSIFICATION AND DEFINITIONS

Diagnosis of NS in children requires significant proteinuria (> 40 mg/h/m² or ≥ 1000 mg/m²/day) or urinary protein creatinine ratio (UPCR) ≥ 200 mg/mmol (2 mg/mg) or

3+ on urine dipstick plus hypoalbuminemia (< 30 g/L) (see [Table 71.6](#)). Edema is a characteristic symptom of NS. Idiopathic NS is the most frequent glomerular disease in children and typically manifests between 1 and 10 years of age. Idiopathic NS is clinically classified on the basis of the response to corticosteroid therapy. Clinical remission is defined by a marked reduction in proteinuria (remission: < 4 mg/m²/h, ≤ 20 mg/mmol (0.2 mg/mg) UPCR or urine albumin dipstick of 0 to trace for 3 consecutive days). Reduction of proteinuria is associated with resolution of edema. Patients who enter clinical remission in response

Table 71.7 Potential Causes for Secondary Forms of NS in Childhood

Disease group	Examples
Systemic immunologic diseases	Systemic lupus erythematosus (SLE), IgA-vasculitis (Henoch-Schönlein purpura), IgA-nephropathy, granulomatosis with polyangiitis, panarteritis nodosa, Goodpasture-syndrome, rheumatic fever, sarcoidosis
Infections	chronic bacteremia (e.g., in endocarditis lenta, in foreign body infections, hepatitis B and C, CMV and EBV, HI), malaria, and schistosomiasis
Tumors	Leukemia, Non-Hodgkin lymphoma
Hemodynamic causes	Renal vein thrombosis, heart failure, sickle cell disease
Drugs and toxins	Nonsteroidal antiinflammatory drugs, D-penicillamine, mercury

to daily glucocorticoid treatment alone are referred to as having steroid-responsive or steroid-sensitive NS (SSNS). The majority (80%–90%) of children older than 1 year of age have SSNS, while the remaining 10% to 20% are nonresponsive and classified as having steroid-resistant nephrotic syndrome (SRNS), defined as lack of remission within 4 weeks of treatment with prednisone at standard dose (see later).¹⁷⁸⁻¹⁸⁰ A majority of children who develop NS will experience at least one relapse of their disease. Clinical relapse is characterized by a recurrence of severe proteinuria (exceeding 40 mg/m²/h, ≥ 200 mg/mmol [2 mg/mg] UPCr or urine albumin dipstick registering 3+ for 3 successive days), typically accompanied with edema recurrence. Frequently relapsing NS (FRNS) is characterized by either two or more relapses within 6 months of the initial response or four or more relapses within any 12-month period.¹⁸¹ Steroid-dependent NS (SDNS) describes cases in which a patient with SSNS experiences two consecutive relapses during recommended prednisone therapy (see later) for first presentation or relapse or within 14 days after steroid therapy discontinuation.¹⁷⁸ These patients may require continued low-dose treatment with glucocorticoids or alternative steroid-sparing immunosuppressive/immunomodulatory treatment to prevent development of relapse and are at higher risk of progression to CKD or ESKF.^{178,182,183} The definitions of the various clinical forms of nephrotic syndrome are outlined in [Table 71.7](#).

EPIDEMIOLOGY

Childhood NS occurs at a reported rate of 4.7 (with a range of 1.15–16.9) cases per 100,000 children globally, exhibiting significant variation based on ethnic origin and geographic location.¹⁸⁴ Several studies have investigated the disparities across ethnicities. South Asian children exhibit a greater occurrence of NS compared with the European

population.¹⁸⁵⁻¹⁸⁷ In the United States, NS is more prevalent among African-American children compared with those of European descent.^{188,189} Furthermore, African-American children exhibit a higher propensity for FSGS on kidney biopsy (ranging from 42% to 72%) and, in general, have a higher risk of advancing to ESKF, compared with European children.¹⁹⁰ NS is more common in boys, with a male/female ratio of 1.6:1 (1.7:1 for SSNS and 1.2:1 for SRNS), but this difference declines with age. The peak age of presentation is 1 to 4 years.^{191,192} Age is a potential predictive factor of underlying histology.¹⁹³ The median ages at presentation for minimal change nephrotic syndrome (MCNS), also named minimal change disease (MCD), focal and segmental glomerulosclerosis (FSGS), and membranoproliferative glomerulonephritis (MPGN), are 3, 6, and 10 years, respectively. Additionally, SSNS is more common in first-degree relatives of affected individuals with idiopathic NS reported in identical twins. Congenital nephrotic syndromes are associated with genetic mutations in the NPHS1 (nephrin), NPHS2 (podocin), and WT-1 genes. Familial and sporadic SRNS may be associated with NPHS2 mutations.^{194,195} In patients with monogenic forms of SRNS, immunosuppressive treatment should be withdrawn since there is evidence supporting the ineffectiveness of this treatment.¹⁷⁸

PATHOGENESIS

Idiopathic nephrotic syndrome and genetically caused nephrotic syndrome form the group of primary nephrotic syndromes.

The podocyte is the primary target cell of injury during NS. All forms of this disease involve structural changes in podocytes, including cell swelling, as well as retraction and effacement (spreading) of the podocyte's distal foot processes, leading to the formation of a diffuse cytoplasmic sheet overlying the glomerular basal membrane (GBM). Additional podocyte structural changes include the formation of vacuoles, development of occluding junctions with displacement of the normal slit diaphragms that extend between podocyte foot processes, and in some areas detachment of podocytes from the underlying GBM. These podocyte structural changes, along with detachment from the GBM, result in the proteinuria that characterizes NS.^{188,196,197} Detailed discussion on the podocyte pathophysiology is outlined in [Chapter 4](#).

Extensive investigations over several decades have failed to identify a single unifying cause for idiopathic NS, though there is strong evidence of immune dysregulation, chiefly involving cell-mediated immunity. This theory is supported by the tendency of NS to manifest and relapse after viral infections or atopic episodes, the association with HLA antigens and Hodgkin lymphoma, and the therapeutic response to immunosuppressive therapy, namely glucocorticoids in about 80% to 90% of cases. Evidence for a pathophysiologic role of circulating antinephrin antibodies in idiopathic nephrotic syndrome has emerged.¹⁹⁸

In SRNS patients, up to 30% are diagnosed with a monogenetic disease.

More than 70 genes have now been identified, for which mutations of various types have been found to lead to childhood SRNS.¹⁷⁸ These genes should be included in a next-generation sequencing panel unless the clinical phenotype is suggestive of a specific condition.¹⁷⁸

The secondary glomerulopathies associated with childhood NS include systemic diseases, highlighted in [Table 71.8](#). Examples include conditions such as SLE, IgA-vasculitis (nephritis), diabetes mellitus, sickle cell disease, and sarcoidosis. In addition, other secondary causes of disease include infections such as HIV, hepatitis B, and hepatitis C, malignancies such as leukemia and lymphoma, and drugs such as nonsteroidal antiinflammatory drugs (NSAIDs) and ACE inhibitors (captopril). Finally, morbid obesity, allergies to food, bee stings, and even immunizations have all also been associated with the development of NS.

CLINICAL PRESENTATION

Edema, specifically gravity-dependent edema, is the most common presenting clinical symptom of NS. Edema may be generalized, especially in young children. Periorbital edema is typical and may be mistaken for an allergic reaction. The laboratory evaluation of children presenting with NS always begins with confirmation of the presence of proteinuria, hypoalbuminemia (<30 g/L), hyperlipidemia, and determination of the renal function. Proteinuria may be accompanied by hematuria, which can be either macroscopic or microscopic. Microscopic hematuria was found in 23% of children with MCD versus 48% of children with FSGS versus 59% of children with membranoproliferative glomerulonephritis (MPGN) in the International Study of Kidney Disease study.¹⁹⁷ Hypertension is more common in FSGS or MPGN, with an incidence of approximately 50%, compared with MCD (~21%). The clinical evaluation of children with nephrotic syndrome should emphasize the risk of prerenal acute kidney (AKI) injury due to intravascular hypovolemia in severe cases with severe edema.

DIAGNOSTIC WORKUP, CONSIDERATION OF KIDNEY BIOPSY, AND GENETIC TESTING

Initial diagnostic workup is outlined in [Table 71.8](#) and [Fig. 71.9](#). These summarize the clinical practice recommendations of the International Pediatric Nephrology Association (IPNA).¹⁹⁹

The prognosis for children with nephrotic syndrome is best predicted by the patient's response to initial treatment and frequency of relapse during the first year after treatment. Therefore a kidney biopsy is not usually needed at initial presentation and instead is reserved for children with resistance to therapy or an atypical clinical course.

Kidney biopsy is recommended in all children diagnosed with SRNS unless there is a known genetic or secondary cause. In addition, a kidney biopsy may be indicated in case of a high index of suspicion for a different underlying pathology (e.g., macroscopic hematuria, systemic symptoms of vasculitis, and hypocomplementemia). Beyond the typical age of onset (≥ 12 years of age), a kidney biopsy may be helpful, as the probability of MCD decreases. Kidney biopsy is helpful to determine the diagnoses, as well as to determine the extent of tubular atrophy, interstitial fibrosis, and glomerulosclerosis as prognostic markers.^{200,201} Genetic testing should be performed if possible in all children diagnosed with primary SRNS, particularly in familial cases, cases with extrarenal features, and those undergoing preparation for renal transplantation. This should include a comprehensive gene panel analysis (see earlier

discussion), as well as genetic counseling for patients and their families.¹⁷⁸ The genetic causes of podocyte disorders/nephrotic syndrome are listed in [Table 71.9](#).

HISTOLOGIC CLASSIFICATION

Childhood NS is most often due to primary glomerulopathies, most commonly MCD (~80%), followed with a large distance by FSGS (~10%), as well as mesangial proliferative glomerulonephritis (MPGN), and membranous nephropathy (MN).

FSGS is a complex disease that describes different kinds of kidney defects, not exclusively linked with podocyte defect.²⁰² We regard FSGS as a pattern of histologic injury rather than a disease. Historically it has been divided into primary or secondary etiologies, but integration of more robust clinical, histologic, and genetic knowledge of this disease has led to the proposal that FSGS be categorized into six forms: 1. primary FSGS, 2. (mal)adaptive FSGS, 3. APOL1 FSGS, 4. genetic FSGS, 5. virus-associated FSGS, and 6. medication/toxin-associated FSGS,²⁰³ the most common being primary and adaptive FSGS. All six forms share the common theme of podocyte injury and depletion. In primary FSGS, it has been suggested that the inciting factor of podocyte damage involves a circulating factor, possibly a cytokine. Recurrent FSGS, which occurs in some patients immediately (on a scale of hours to several weeks) post-transplant afterward, supports this theory.^{204,205} In contrast, (mal)adaptive FSGS is thought to be due to persistent glomerular hyperfiltration, which may occur in conditions such as cyanotic congenital heart disease, sleep apnea, or conditions with reduced functional nephron mass such as renal dysplasia, reflux nephropathy, repeated episodes of AKI, or premature or small-for-gestational-age birth.²⁰⁶⁻²⁰⁸

TREATMENT OF NEPHROTIC SYNDROME

Oral corticosteroids such as prednisone and prednisolone are the primary treatment for NS and lead to remission in 80% to 90% of cases, referred to as steroid-sensitive nephrotic syndrome (SSNS).¹⁹⁰ Remission is defined as urinary protein excretion ≤ 4 mg/m²/h or 0-trace of protein on urine dipstick or protein/creatinine ratio <0.02 g/mmol (0.2 mg/mg) for 3 consecutive days.¹⁹⁵ Children who fail to respond or respond with only a partial resolution of their proteinuria are referred to having steroid-resistant nephrotic syndrome (SRNS). Children with SRNS are at a higher risk than those with SSNS for the development of progressive renal disease and often require a renal biopsy to confirm a histologic diagnosis. Regardless of the histologic diagnosis, however, treatment decisions in children are guided primarily by either the success (i.e., SSNS) or failure (i.e., SRNS) of oral glucocorticoids to induce a complete clinical remission.

Supportive therapy includes the provision of general medical care, management of edema, and management of dyslipidemia ([Table 71.10](#)). Sixty percent of NS-associated deaths are attributable to infection; thus prompt identification and treatment of infections is required.²⁰⁹ Peritonitis, bacteremia, and cellulitis are the most common infections, with *Streptococcus pneumoniae* and *Escherichia coli* often implicated.¹⁹⁵ A 3% incidence of thromboembolic events has been reported in children with

Table 71.8 Initial Workup for Child With Nephrotic Syndrome

Investigations	Comments
Clinical Evaluation	
Relevant patient history	(Grade A, strong recommendation)
Presence of gravity-dependent edema	Consider especially in patients from endemic areas before starting immunosuppressant medications (grade C, weak recommendation)
Fever episodes, pain, abdominal discomfort, fatigue	
Search for risk factors for secondary causes (e.g., sickle cell disease, HIV, systemic lupus erythematosus, hepatitis B, malaria, parvovirus B19, and medications)	
Screen for tuberculosis	
Physical examination	(Grade A, strong recommendation)
Blood pressure, assess volume status and extent of edema (ascites, pericardial and pleural effusions), lymphadenopathy	Further workup is recommended (grade A, strong recommendation)
Signs of infection (respiratory tract, skin, peritonitis, urinary tract)	
Extrarenal features (e.g., dysmorphic features, ambiguous genitalia, or eye abnormalities) (microcoria, aniridia), rash, arthritis	
Anthropometry	We recommend comparing data with appropriate national standards or WHO-MGRS charts (grade A, strong recommendation)
Growth chart: height/length, weight, and head circumference (<2 years)	
Vaccination status	This is recommended before starting immunosuppressant medications other than PDN (grade B, moderate recommendation)
Check/complete according to national standards, especially for encapsulated bacteria: pneumococcal, meningococcal, <i>Haemophilus influenzae</i> , Hep B, SARS-CoV-2, influenza vaccine, and varicella	
Family history	(Grade A, strong recommendation)
Kidney disease in family members	
Extrarenal manifestations	
HIV or tuberculosis in endemic regions	
Consanguinity	
Biochemistry	Recommended at least once before starting treatment of the first episode (grade B, moderate recommendation)
Spot urine	
Protein/creatinine ratio (in first morning void)	
Urinalysis including hematuria	
Blood	eGFR (mL/min/1.73 m ²) = k height (cm)/serum creatinine (mg/dL), where k is a constant = 0.413
Complete blood count, creatinine, eGFR, urea, electrolytes, albumin	eGFR (mL/min/1.73 m ²) = k height (cm)/serum creatinine (pmol/L), where k is a constant = 36.5N ¹⁷ , N ¹⁸
Complement C3, C4, antinuclear and anti-streptococcal antibodies, and ANCA	Recommended in patients with macroscopic hematuria (grade A, strong recommendation)
Varicella and MMR specific IgG, in nonimmunized children	Consider before start of PDN treatment (grade D, weak recommendation)
Imaging	Consider a kidney ultrasound in all children with INS to exclude kidney malformations and venous thrombosis and in patients with reduced eGFR, hematuria, or abdominal pain and always before kidney biopsy (grade D, weak recommendation)
Kidney ultrasound	
Chest radiograph	Recommended in case of suspected lymphoma (grade D, weak recommendation)
Histopathology	Recommended in patients with atypical features including macroscopic hematuria, low C3 levels, AKI not related to hypovolemia, sustained hypertension, arthritis, and/or rash (grade A, strong recommendation)
Kidney biopsy	Consider in patients with infantile-onset NS if genetic screening is not available (age 3-12 months) (grade B, weak recommendation)
	Consider in patients >12 years of age on a case-by-case basis (grade C, weak recommendation)
	Consider in patients with persistent microscopic hematuria in specific populations with a high incidence of glomerular diseases, such as IgA-nephropathy in East Asia (grade C, weak recommendation)

Table 71.8 Initial Workup for Child With Nephrotic Syndrome

Investigations	Comments
Genetic testing	Recommended in patients diagnosed with SRNS (grade A, strong recommendation) Recommended in patients with congenital NS, extrarenal features, and/or family history suggesting syndromic/hereditary SRNS (grade A, strong recommendation) Consider in patients with infantile-onset NS (age 3-12 months) (grade C, weak recommendation) (fig. 2) Recommended in patients diagnosed with SRNS (grade A, strong recommendation)

AKI, Acute kidney injury; eGFR, estimated glomerular filtration rate; ANCA, antineutrophil cytoplasmic antibodies.
From Trautmann 2022.¹⁹⁹

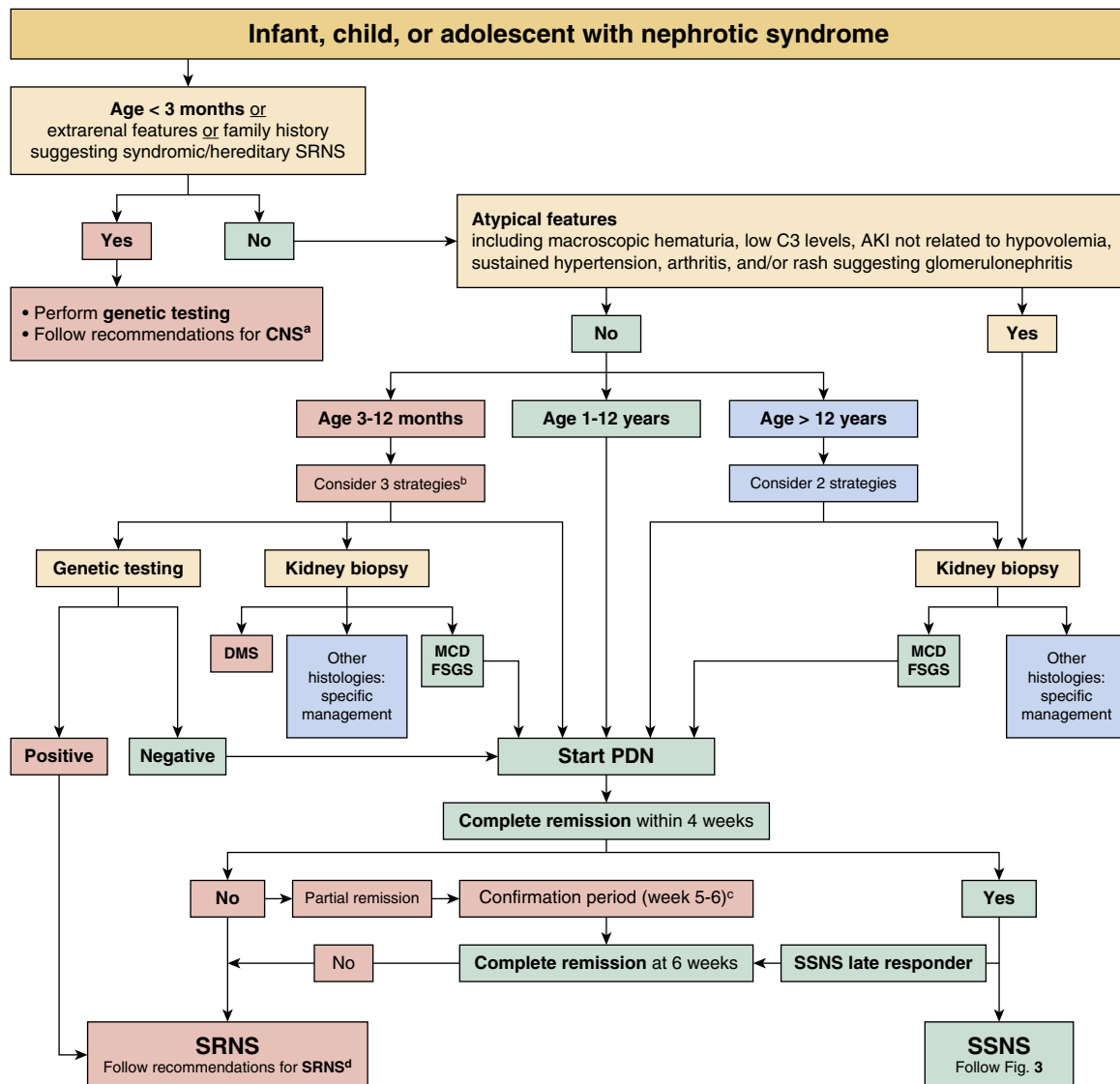
**Fig. 71.9 Algorithm for the initial management of a child with nephrotic syndrome.** (From Trautmann 2022.¹⁹⁹)

Table 71.9 Genetic Disorders of Podocytes

Gene (Inheritance Pattern)	Product	Clinical Manifestations
COL4A3/4/5 (AD, AR, XL)	Type IV collagen	Alport syndrome/spectrum
NPHS1 (AR)	Nephrin	Early onset SRNS, Finnish-type CNS
NPHS2 (AR)	Podocin	Early- or late-onset SRNS
WT1 (AD)	Wilms tumor 1	Denys-Drash syndrome, Wilms tumor, Frasier syndrome
PLCE1 (AR)	Phospholipase C ϵ 1	Early onset SRNS
CD2AP (AD)	CD2-associated protein	Early onset SRNS, FSGS
ACTN4 (AD)	α -Actinin 4	Early- or late-onset SRNS, FSGS
TRPC6 (AD)	Transient receptor potential cation channel 6	Late-onset SRNS
INF2 (AD)	Inverted formin 2	Charcot-Marie-Tooth disease, late-onset SRNS
LMX1B (AD)	LIM homeobox transcription factor 1- β	Nail-patella syndrome, Collagenofibrotic glomerulopathy
SCARB2 (AR)	Lysosomal integral membrane protein 2	Action myoclonus-renal failure syndrome: ataxia, myoclonus
CUBN (AR)	Cubilin: intrinsic factor-cobalamin receptor	Megaloblastic anemia secondary to vitamin B12 deficiency, SRNS
COQ6 (AR)	Coenzyme Q6	Sensorineural hearing loss
MYH9 (AD)	Nonmuscle myosin 11a	Bleeding diathesis, macrothrombocytopenia, progressive sensorineural deafness, $\uparrow$ liver enzyme, cataract
SMARCA1 (AR)	SMARCA-like protein	Schimke immune-osseous dysplasia

AD, Autosomal dominant; AR, autosomal recessive; SRNS, steroid-resistant nephrotic syndrome; tRNA, transfer RNA; XL, X-linked.

Adapted from Woojin A, Bombardier AS. Approach to diagnosis and management of primary glomerular diseases due to podocytopathies in adults: core curriculum. *Am J Kid Dis.* 2020;75(6):955–964.

Table 71.10 Major Components of Supportive Therapy for Children With Steroid-Resistant Nephrotic Syndrome

General Medical Care	Management of Edema	Management of Dyslipidemia
Identification and treatment of suspected infection • Pneumonia • Cellulitis • Peritonitis • Sepsis	Avoidance of excessive fluid intake Elevation of extremities Moderate dietary salt restriction Judicious use of diuretics for severe edema Head-out water immersion	Avoidance of high fat intake Regular exercise (>30 min/day of moderately intense activity) Consideration of hydroxymethylglutaryl-coenzyme A reductase inhibitors (statins) Consideration of low-density lipoprotein apheresis
Identification and treatment of suspected thrombosis • Gross hematuria $\pm$ acute oliguria (renal venous thrombosis) • Respiratory distress (pulmonary thrombosis) • Asymmetric extremity swelling (deep venous thrombosis) • Neurologic symptoms (cerebral venous thrombosis)		
Maintenance or restoration of intravascular volume		
Maintenance of adequate protein intake (130%–140% of recommended dietary allowances)		
Alteration of immunizations while immunosuppressed		

NS, due to disease-related hypercoagulability, underlying thrombophilic predisposition, infections, and vascular access lines. Anticoagulation is recommended for those with previous history of venous thromboembolic events and should be considered for those with additional risk factors such as (indwelling central venous lines, hereditary thrombophilic predisposition, infection, or risk of dehydration).¹⁷⁸

STEROID-SENSITIVE NEPHROTIC SYNDROME

Over the past few decades, multiple protocols have been reported for the initial treatment of children presenting with NS. The initial widely accepted approach was reported by the International Study for Kidney Diseases in Children, which suggested treatment with prednisone in divided doses for 4 weeks at 60 mg/m²/day (or 2 mg/kg/day up to a maximal dose of 80 mg/day), followed by a taper to 40 mg/m²/day (or 1.5 mg/kg/day up to a maximal dose of 60 mg/day) for an additional 4 weeks.¹⁸⁰ KDIGO guidelines recommend that oral glucocorticoids be given for 8 weeks (4 weeks of daily glucocorticoids followed by 4 weeks of alternate-day glucocorticoids) or 12 weeks (6 weeks of daily glucocorticoids followed by 6 weeks of alternate-day glucocorticoids).²¹⁰ The 2022 IPNA clinical practice recommendations suggest a single daily morning dose rather than divided to increased adherence to therapy and reduce risk of hypothalamic–pituitary–adrenal (HPA) axis suppression and sleep disturbances: prednisone 4 weeks at 60 mg/m² or 2 mg/kg (maximum dose 60 mg/day), followed by alternate-day prednisone at 40 mg/m² or 1.5 mg/kg (maximum dose of 40 mg on alternate days) for 4 weeks, or prednisone 6 weeks at 60 mg/m² or 2 mg/kg [maximum dose 60 mg/day] followed by alternate-day prednisone at 40 mg/m² or 1.5 mg/kg (maximum dose of 40 mg on alternate days) for 6 weeks.¹⁹⁹ Hodson and colleagues²¹¹ conducted a meta-analysis of randomized control trials investigating corticosteroid therapy in NS and concluded a therapy duration of 3 to 7 months resulted in fewer relapses. This supported the previously held notion that a longer duration of treatment was more important than total dose in reducing the risk of relapse. This has been challenged by trials that postulate extending initial prednisolone treatment with an increasing dose and or longer treatment time does not improve clinical outcomes.^{212–214} The occurrence of relapses in follow-up may rather reflect the “natural” course of the disease, which may not be influenced by duration and cumulative dose of corticosteroid treatment.

Daily dipstick testing until remission is recommended. In follow-up urine dipstick testing, twice weekly is suggested in the first year. In episodes of fever, infections, or suspected relapse (e.g., edema and positive dipstick), daily testing is recommended (Table 71.11).¹⁹⁹

STEROID-DEPENDENT AND FREQUENTLY RELAPSING NEPHROTIC SYNDROME

A majority of children with NS relapse within the first 6 months of initial therapy and approximately 50% to 60% develop frequently relapsing nephrotic syndrome (FRNS) or steroid-dependent nephrotic syndrome (SDNS).¹⁹⁹ Certain characteristics increase the likelihood of relapses, such as age younger than 3 years at onset, male gender, delayed time

to remission (after 7 to 9 days), and occurrence of an early relapse (in the first 6 months after initial treatment).^{215,216} Additionally, upper respiratory tract infections (URTIs) precede 71% of relapses, with respiratory syncytial virus (RSV) being the most commonly implicated viral trigger.²¹⁷ The IPNA clinical practice recommendations do not recommend the routine use of a short course of low-dose daily PDN at the onset of an URTI for prevention of relapses but suggest considering a short course of low-dose daily PDN at the onset of an URTI in children who are already taking low-dose alternate-day PDN and have a history of repeated infection-associated relapses.¹⁹⁹

Relapses (urine dipstick $\geq 3+$ (≥ 300 mg/dL) or UPCR ≥ 200 mg/mmol (≥ 2 mg/mg) on a spot urine sample on 3 consecutive days, with or without reappearance of edema) are managed by reinitiating a single daily dose of prednisolone at 60 mg/m² until the urine is protein free for 3 days. The dose is then reduced to 40 mg/m² on alternate days for 4 weeks, without a tapering schedule.^{181,199} Children in whom the aforementioned measures fail to adequately control the disease can be classified as FRNS or SDNS. Those patients are at risk of developing significant signs and symptoms of steroid toxicity. In this case, several alternative treatments may be used (Fig. 71.10). While these agents as a group can often reduce the frequency of relapses and are steroid sparing, many of them have significant toxicities of their own (Table 71.12).

Cyclophosphamide is an alkylating agent used in both FRNS and SDNS at a dose of 2 mg/kg/day for 12 weeks (maximum cumulative dose of 168 mg/kg).²¹⁰ It has been shown to induce remission at 2 years in 70% and 25%, for FRNS and SDNS, respectively. The benefit of cyclophosphamide is related to the length of treatment, with the optimal treatment being 8 to 12 weeks. Adverse effects include bone marrow suppression, alopecia, hemorrhagic cystitis, infection, gonadal dysfunction/infertility, hyponatremia, SIADH, and hypocalcemia. Chlorambucil is an alternative alkylating agent used at a dose of 0.1 to 0.2 mg/kg/day for 8 weeks with adverse effects including bone marrow suppression, seizures, infertility, and teratogenicity.^{181,218}

Levamisole is an antihelminthic drug with immunostimulatory properties, which has been shown to significantly reduce relapse rates in both FRNS and SDNS, although some patients develop relapses shortly after discontinuation. Typical dosing includes administration of 2–2.5 mg/kg PO on alternate days (maximum dose 150 mg/day), and a typical treatment duration ranges from 4 months to a year. Adverse effects include leukopenia, agranulocytosis, hepatotoxicity, vasculitis, and encephalopathy.^{210,215,219}

Mycophenolate mofetil (MMF) is a purine synthetase inhibitor. Treatment at doses of 25 to 30 mg/kg daily (1200 mg/m²/day in two divided doses), significantly reduced relapse rates and spare the use of corticosteroids, though Bagga and colleagues²²⁰ noted cessation of therapy was associated with relapses. A significantly high interindividual variability of mycophenolic acid (MPA) exposure at a given dose of MMF exists. Gellermann and colleagues²²¹ showed that at a dose of 800 to 1200 mg/m², the achieved MPA-AUC varied from 24 to 120 $\mu\text{g}\cdot\text{h}/\text{mL}$, making therapeutic drug monitoring beneficial. An MPA area-under-the-concentration time curve of >50 mg $\times$ h/L should be aimed for.²¹⁰ Adverse effects include serious infection, malignancy, embryo-fetal toxicity, leukopenia, abdominal pain, and

Table 71.11 Monitoring During Acute Phase and Follow-up of Child With Nephrotic Syndrome

Investigations	Comments
Home Monitoring	
Dipstick assessment (preferably in first morning void)	We recommend daily home urine dipstick testing until remission (grade X, moderate recommendation) We suggest home urine dipstick testing, at least twice weekly in the first year, individualize thereafter (grade D, weak recommendation) We recommend daily testing if 1 + or more or during episodes of fever, infections, and/or suspected relapse (edema) (grade X, moderate recommendation)
Clinical evaluation	We suggest outpatient visits every 3 months within the first year, individualized thereafter with more frequent visits in cases of relapse (grade D, weak recommendation)
Frequency of outpatient visits	Recommended at every visit. Points to infection or drug toxicity (grade A, strong recommendation)
Patient history	
Fever episodes, pain, abdominal discomfort, swelling, fatigue, increased appetite, weight gain, sleep disturbances, behavioral changes	
Physical examination	Recommended at every visit (grade A, strong recommendation)
Blood pressure	Recommended at every visit in patients in relapse (grade A, strong recommendation)
Assessment of volume status, including edema (ascites, pericardial and pleural effusions)	Recommended at every visit in patients on medication (grade A, strong recommendation)
Drug toxicity (e.g., striae, Cushingoid features, avascular necrosis, acne, tremor, hirsutism, and gum hyperplasia)	Recommended at every visit (grade A, strong recommendation)
Signs of infection (respiratory tract, skin, peritonitis, urinary tract)	Recommended yearly in patients on PDN (grade A, strong recommendation)
Ophthalmologic examination (glaucoma, cataract)	Recommended at every visit; data should be compared with appropriate national standards or WHO-MGRS charts (grade A, strong recommendation)
Anthropometry	
Growth chart: height/length, weight, and head circumference (<2 years)	Recommended in patients who received PDN treatment within the last 12 months (grade A, strong recommendation)
Calculation of BMI and annual height velocity	Suggested as appropriate (grade D, weak recommendation)
Vaccination status	
Check/complete according to national standards, especially for encapsulated bacteria: pneumococcal, meningococcal, <i>Haemophilus influenzae</i> , Hep B, SARS-CoV2, influenza, and varicella-zoster	
Biochemistry	Suggested as appropriate (positive dipstick) (grade C, weak recommendation)
Spot urine	
Protein/creatinine ratio (preferably in first morning void)	
Blood	Recommended as appropriate in patients on medication or with complicated relapses (grade A, strong recommendation)
Complete blood count, creatinine, eGFR, urea, electrolytes, albumin	
MPA, CsA, TAC	We recommend (pharmacokinetic) blood monitoring in patients on medication as given in Table 5 (grade B, moderate recommendation)
25-OH-vitamin D	Annually in patients with SDNS or FRNS (after 3 months of remission); aiming for levels >20 ng/mL (>50 nmol/l) (grade C, weak recommendation)
Imaging	Recommended before kidney biopsy (grade A, strong recommendation)
Kidney ultrasound	
Histopathology	We recommend considering a kidney biopsy in patients with SSNS during follow-up if the findings may potentially influence therapy or help assess prognosis (grade X, moderate recommendation)
Kidney biopsy	
CsA, Cyclosporin A; MPA, mycophenolate acid; TAC, tacrolimus. From Trautmann 2022. ¹⁹⁹	

diarrhea. MMF is not associated with nephrotoxicity, which is an advantage over the calcineurin inhibitors (CNI).²¹⁸ Because of its efficacy and favorable toxicity profile compared with CNIs, a commentary to the German best

practice guideline on INS in children suggests MMF as a primary alternative steroid-sparing agent.²²²

CNIs include tacrolimus and cyclosporin, both of which are effective in the induction and maintenance of remission

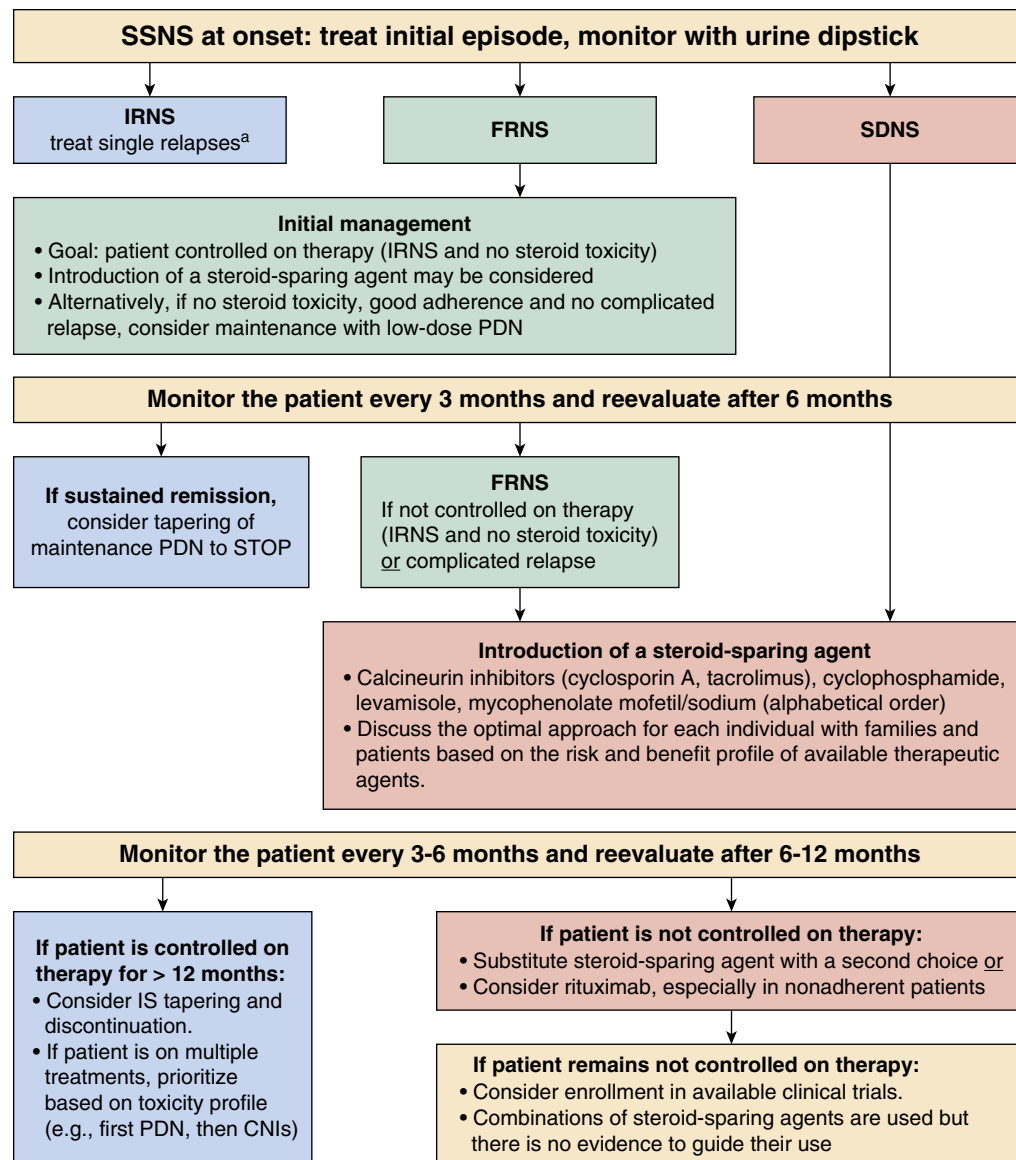


Fig. 71.10 Algorithm for management of children with steroid-sensitive nephrotic syndrome. (From Trautmann 2022.¹⁹⁹)

in FRNS and SDNS. CNIs exert their effects by decreasing T cell activity and stabilizing the podocyte actin cytoskeleton. Adverse effects of the CNIs include hirsutism, diabetes, hypertension, hyperkalemia, dyslipidemia, and gingival hyperplasia. Additionally, there is an increased risk of nephrotoxicity manifesting as irreversible interstitial fibrosis. Its incidence and severity appear to correlate generally with the duration and dose of drug exposure. Patients often relapse after discontinuation, so they are usually prescribed over a prolonged period. Patients on long-time therapy with cyclosporin or tacrolimus should be considered for kidney biopsy during follow-up.^{181,190,219}

Rituximab is an anti-CD20 monoclonal antibody that may be used for SRNS, as well as SSNS and FRNS, and is given as 375 mg/m² per dose for up to 4 weekly doses. The efficacy of this treatment has been reported in mostly small case series and uncontrolled trials, where it has been found generally to induce complete remission in approximately 25% of children with SRNS and partial remission in approximately

25% of children. Adverse effects include infusion-related reactions, which vary from mild (fever or rash) to severe (hypotension and bronchospasm). Rituximab-associated lung injury (RALI) is pulmonary toxicity that is typically temporary and reversible but may lead to potentially fatal complications including pulmonary fibrosis and interstitial pneumonitis and may present between 1 and 3 months after initiation of rituximab. Thus a chest radiograph should be obtained before initiation and periodically during treatment.^{210,223} Some recipients of rituximab and other anti-CD20 monoclonal antibodies experience prolonged depletion of CD20 lymphocytes and significant impairment of immunoglobulin production with attendant attenuation of vaccine responses and susceptibility to certain pathogens, sometimes requiring regular supplementation with external immunoglobulins until immune system recovery.²²⁴ For FRNS and SDNS, rituximab is advised to be used as a second-line steroid-sparing agent in children due to its uncertain long-term safety profile.

Table 71.12 Glucocorticoid-Sparing Therapies in Children With Nephrotic Syndrome

Treatment	Dose and Duration	Clinical Tips
First Line:		
• Oral cyclophosphamide	2 mg/kg/d for 12 weeks (maximum cumulative dose 168 mg/kg)	Cyclophosphamide should not be started until the child has achieved remission with glucocorticoids. Moreover, second courses of alkylating agents should not be given. Weekly CBCs are recommended during the treatment course to assess for severe leukopenia or overall bone marrow suppression prompting dose reduction or treatment cessation
• Oral levamisole	2.5 mg/kg on alternate days, with a maximum dose of 150 mg	Monitor CBC every 2-3 months and alanine and aspartate aminotransferases every 3-6 months during therapy with levamisole. Check ANCA titers every 6 months, if possible, and interrupt treatment in case of ANCA positivity, skin rash, or agranulocytosis. Maintaining low-dose alternate-day glucocorticoid dosing on the days not taking levamisole may be effective in some children. Levamisole should be continued for at least 12 months.
Alternative agents:		
• Mycophenolate mofetil	Starting dose of 1200 mg/m ² /d (given in 2 divided doses)	Target area under the curve >50 µg·h/mL.* Mycophenolate mofetil should be continued for at least 12 months, as most children will relapse when it is stopped. In children experiencing significant abdominal pain on mycophenolate mofetil, other mycophenolic acid analogs (MPAAs), such as sodium mycophenolate, may be employed at equivalent doses (360 mg of sodium mycophenolate corresponds to 500 mg of mycophenolate mofetil)
• Rituximab	375 mg/m ² IV x 1-4 doses	Rituximab may be used as a treatment for steroid-sensitive nephrotic syndrome in children who have continuing frequent relapses despite optimal combinations of prednisone and glucocorticoid-sparing oral agents, and/or who have serious adverse effects of therapy. Current trials report 1 to 4 doses of rituximab. There are insufficient data to make a recommendation for specific number of needed doses. Where available, CD20 levels should be monitored. Hepatitis B surface antigen, hepatitis B core antibody, and a QuantiFERON test for tuberculosis must be checked before rituximab administration. Monitoring IgG levels both before and after rituximab therapy may allow for earlier identification of risk for developing significant infection and identify patients who may benefit from immunoglobulin replacement.
• Calcineurin inhibitor†		CNI should be continued for at least 12 months as most children will relapse upon discontinuation. Monitor CNI levels during therapy to limit toxicity
- Cyclosporine	4-5 mg/kg/d (starting dose) in 2 divided doses	Cyclosporine may be preferable in patients at risk for diabetic complications. Target 12-hour trough level of 60-150 ng/mL (50-125 nmol/L) aiming for lowest levels to maintain remission and avoid toxicity
- Tacrolimus	0.1 mg/kg/d (starting dose) given in 2 divided doses	Tacrolimus may be preferred over cyclosporine in patients for whom the cosmetic side effects of cyclosporine are unacceptable. Target 12-hour trough level of 5-10 ng/mL (6-12 nmol/L) aiming for lowest levels to maintain remission and avoid toxicity

From KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases Kidney Disease: Improving Global Outcomes (KDIGO) Glomerular Diseases Work Group²¹⁰.

STEROID-RESISTANT NEPHROTIC SYNDROME

Despite optimal management, approximately 10% of children with NS will initially present with or subsequently develop SRNS.²²⁵ Notably, before labeling a child as having SRNS, it is essential to exclude several potentially treatable

causes for treatment failure including poor medication adherence, poor medication absorption, possible underlying infection, and possible malignancy. Once diagnosed with SRNS, a child's risk for both the development of extrarenal complications of NS and the development of ESKF and/or

CKD increases, with SRNS patients comprising 15% of all children with CKD requiring RRT.²²⁶ Furthermore, in cases where partial or complete remission is not attained, there is a 50% likelihood of progressing to ESKF within 5 years of diagnosis.²²⁷ For this reason, both the alternative treatments described earlier, as well as additional treatments, are often employed in an attempt to induce a complete remission of proteinuria.

The prognosis of SRNS is based on histopathologic findings at the time of diagnosis. ESKF-free survival rates in children with minimal change glomerulopathy (MCGN) were 92% at 5 years and 79% at 10 and 15 years compared with 69% 5-year, 52% 10-year, and 37% 15-year renal survival rates in children diagnosed with FSGS. The poorest outcomes were seen in patients with diffuse mesangial sclerosis (DMS) who have an 80% ESKF risk at 5 years after diagnosis.

The overall management of children with SRNS can be divided into three major categories: 1. supportive therapy, 2. immunosuppressive therapy, and 3. nonimmunosuppressive therapy.

SUPPORTIVE THERAPY

Supportive therapy in SRNS and severe cases of SSNS includes control of edema, maintaining nutrition, and preventing complications including hyperlipidemia, arterial hypertension, thrombosis, and infections. Management of chronic edema includes moderate restriction of fluid intake, restriction of salt intake, and the judicious use of diuretics. Albumin infusions may be used in addition to diuretics for patients with severe refractory edema at dosages up to 1 g/kg given as 20% to 25% albumin over a period of at least 4 hours, followed by IV furosemide at a dose of 1 to 2 mg/kg to induce diuresis.²²⁸ SRNS is associated with dyslipidemia, which is associated with cardiovascular mortality. It is recommended to consider lipid-lowering treatment in children with SRNS and persistent LDL cholesterol levels >130 mg/dL (3.4 mmol/L), commencing initially with lifestyle changes including dietary modifications, enhanced physical activity and weight control, followed by pharmacologic therapy when the aforementioned approaches fail, and more recently the use of low-density lipoprotein (LDL) apheresis. Prophylactic anticoagulation and antiinfective therapies may be considered in patients at risk.¹⁹⁹

IMMUNOSUPPRESSIVE THERAPY

Immunosuppressive therapy is the most widely employed for SRNS (Table 71.13). The KDIGO guidelines recommend usage of a CNI such as tacrolimus or ciclosporin, in combination with low-dose corticosteroid therapy as initial therapy for SRNS for a minimum of 6 months.²¹⁰ These agents induce complete remission in a significant percentage of children with SRNS (33%–59%), as well as partial remission in many of the remaining patients.^{216,228-232} Long-term toxicity, however, remains a concern.

NONIMMUNOSUPPRESSIVE THERAPY

First-line nonimmunosuppressive treatment in children with SRNS includes inhibition of the renin-angiotensin-aldosterone-system (RAASi) with angiotensin-converting

enzyme inhibitors (ACEi) or angiotensin receptor blockers (ARBs). This therapy decreases intraglomerular pressure, reduces proteinuria, and decelerates progression of CKD.¹⁷⁸

Potential future therapies include Sparsentan, which is an orally active dual-acting ARB combined with a highly selective endothelin type A receptor antagonist. It was found to decrease proteinuria by 45% compared with 19% in the group treated only with irbesartan.²³³ Another alternative treatment for SRNS is LDL apheresis, which is not a drug but a device providing extracorporeal therapy. It has been found to increase steroid or immunosuppressive therapy responsiveness in patients with FSGS.²³⁴ Furthermore, it has been reported to not only reduce the hyperlipidemia of NS effectively but also induce complete remission in approximately 25% of adults and children diagnosed with SRNS and to induce partial remission in another 25% of patients.²³⁵

OUTCOME

The long-term outcome for childhood NS is excellent overall. The vast majority of children who develop NS will have SSNS, among whom approximately one-half of children will have no relapses or will have infrequent relapses (i.e., <4 relapses per year), and the other half will have frequent relapses (≥ 4 relapses per year, or FRNS). The risk for the development of CKD or ESKF among children with SSNS is <5%. This risk rises dramatically for children who present with or subsequently develop SRNS, where the risk of CKD and/or ESKF approaches 50% within 5 years.¹⁸³ Newer follow-up studies have reported higher rates of relapse occurring in adulthood. In studies from Finland, the United Kingdom, Switzerland, and France, the rate of relapse during adulthood was reported to be 14%, 19%, 33%, and 42%, respectively.¹⁹⁵

MEMBRANOUS NEPHROPATHY

Membranous nephropathy (MN) is a relatively rare cause of glomerular disease among children. It may present with proteinuria, hematuria, or in more severe cases with AKI and/or NS. MN can occur either as a primary disease (i.e., idiopathic MN) or as a disease secondary to a variety of causes such as infections, medications, or autoimmune diseases.

EPIDEMIOLOGY

MN has an estimated incidence of 0 to 12 per million in North America and 2 to 17 per million in Europe, with a 2:1 male predominance and equal incidence among regions and ethnicities.²³⁶ Among adults, it occurs at a frequency of about 1.2/100,000 population per year with a mean age of diagnosis at 50 to 60 years. In contrast, among children, MN occurs at only about 0.05 to 0.1/100,000 population per year, or approximately 10 to 20 times less commonly than in adults, and is more common in adolescents than younger children.^{197,236-241}

PATHOGENESIS

In MN the glomerulus, particularly podocyte antigens, are targeted by circulating autoantibodies, which result in the formation of immune complexes (ICs), activation of

Table 71.13 Treatment Options of SRNS in Children

Treatment	Dose and Duration	Clinical Tips
Calcineurin inhibitors	• Oral cyclosporine 5 mg/kg/d (starting dose) in 2 divided doses. Target 12-h trough level of 60–150 ng/mL (50–125 nmol/L) aiming for lowest levels to maintain remission and avoid toxicity or • Oral tacrolimus 0.1 mg/kg/d (starting dose) given in 2 divided doses for a minimum of 6 months. Target 12-h trough level of 5–10 ng/mL (6–12 nmol/L) aiming for lowest levels to maintain remission and avoid toxicity	CNIs should be continued for at least 12 months as 70% of those who achieve a complete response or partial response will relapse upon discontinuation. They should be discontinued in those without at least a partial response by 6 months. Tacrolimus may be preferred over cyclosporine in patients for whom the cosmetic side effects of cyclosporine are unacceptable. Cyclosporine may be preferable in patients at risk for diabetic complications. There are no studies that investigate differences in long-term outcomes in SRNS on the basis of treatment duration. Median time to complete response or partial response is variable. Response can be seen as long as 6 months following treatment initiation. Trough levels could be measured to minimize nephrotoxicity
Glucocorticoids	• IV methylprednisolone bolus of 500 mg/m²/d for 3 days before starting CNI. Followed by taper: alternate-day oral prednisolone to be tapered gradually over 6 months • Low-dose prednisone (<0.25 mg/kg/d alternate-day dosing)	Most clinical trials and observational studies have included low-dose glucocorticoids in combination with CNIs to induce remission. No studies compare the outcomes between children treated with CNIs alone or in combination with low-dose glucocorticoids
Cyclophosphamide	• Not recommended	Two randomized control trials provide moderate-level data demonstrating no benefit using cyclophosphamide to treat children with SRNS. However, in countries with limited resources where CNIs are not available, this approach may be considered
Mycophenolate mofetil	• Starting dose of 1200 mg/m²/d (given in 2 divided doses) for 1 year	This approach may be employed in children who have achieved stable remission on a CNI, to maintain remission without accumulating nephrotoxicity
Rituximab	• 375 mg/m² IV	Giving 2 infusions (day 1 and day 8) at this dose may be preferable in the presence of nephrotic-range proteinuria to achieve complete B cell depletion. Hepatitis B titers must be checked before rituximab administration. Monitoring IgG levels both before and after rituximab therapy may allow for earlier identification of risk for developing significant infection and identify patients who may benefit from immunoglobulin replacement

From KDIGO 2021 Clinical Practice Guideline for the Management of Glomerular Diseases Kidney Disease: Improving Global Outcomes (KDIGO) Glomerular Diseases Work Group.²¹⁰

complement, and massive proteinuria.²³⁶ MN can be either primary or idiopathic or secondary to a wide variety of causes. The many causes of MN in children are shown in Table 71.14.^{239,242}

Idiopathic MN is characterized by IC formation associated with the subepithelial layer of the GBM. Disease progression is characterized by subepithelial deposition, extending, and incorporation into the GBM, ultimately causing GBM remodeling.²⁴³ Idiopathic MN is predominantly driven by antibodies targeting the M-type phospholipase A2 receptor (anti-PLA2R).²⁴⁴ This receptor is a transmembrane glycoprotein located on the surface of podocytes and is implicated in approximately 85% of cases.²⁴⁵ Anti-PLA2R antibody titers have been demonstrated to be an excellent biomarker of disease activity in patients with MN, with levels being increased during relapses, absent during remissions, and rituximab-induced depletion of anti-PLA2R titers correlating with clinical responses.^{246,247} Additionally, 3% to 5% of idiopathic MN cases are associated

with thrombospondin type 1 domain containing 7A (THSD7A) (3%–5%) and 10% remain due to unidentified mechanisms.²⁴⁵ Secondary MN may arise secondary to lupus, cancer, drug reactions, or infectious disease and may be differentiated from idiopathic MN by atypical features such as mesangial proliferation and specific immunoglobulin and complement deposits.²⁴⁸

CLINICAL MANIFESTATIONS

The most common manifestation of MN is proteinuria, which may be accompanied by microscopic hematuria, though one study reported an incidence of macroscopic hematuria in almost 40% of patients.²⁴⁹ Additionally, 40% to 75% of patients present with nephrotic syndrome.²⁵⁰ Although hematuria is more frequently observed alongside proteinuria in children when compared with adults, hypertension and thromboembolic events are significantly less common in pediatric cases.^{210,236,251} A kidney biopsy is the standard diagnostic method for MN, although this

Table 71.14 Causes of Membranous Nephropathy in Children

Primary	Secondary				
	Infections	Medications	Autoimmune Diseases	Neoplasms	Miscellaneous
Idiopathic membranous nephropathy	Hepatitis B	Captopril	Systemic lupus erythematosus	Neuroblastoma	Sickle cell disease
	Hepatitis C	Nonsteroidal antiinflammatory drugs	Sjögren syndrome	Angiomatoid fibrous histiocytoma	De novo, postrenal transplantation
	Streptococcal infections	Penicillamine	Sarcoidosis	Ovarian tumor	Mercury exposure
	Syphilis	Infliximab	Autoimmune hepatitis		
	Tuberculosis	Tiopronin	Primary biliary cirrhosis		
	Epstein-Barr virus Cytomegalovirus Malaria	Etanercept Gold			

is not required to confirm the diagnosis in patients with nephrotic syndrome and a positive anti-PLA2R antibody test.

TREATMENT

Due to the rarity of MN in the pediatric population, there is no standardized approach to therapy in children; thus treatment decisions for children with MN are usually derived from data published in adult populations. Notably, however, children typically have different comorbid factors compared with adults and inherently different life expectancies. These factors must be considered in making treatment decisions. Supportive care should be initiated in all patients, consisting of blood pressure control, ACE inhibitors/ARBs for proteinuria, statins for hyperlipidemia, and salt restriction with diuretics for edema, and a higher-protein diet may be advised to compensate for urinary protein losses.²⁴⁵

Children with asymptomatic, nonnephrotic proteinuria (urine protein-to-creatinine ratio <2.0 mg/mg), normal renal function, and minimal sclerosis on kidney biopsy are typically managed conservatively with ACEi or ARB as this group is at low risk of progressive renal disease.²⁵⁰ This approach can generally be expected to achieve only approximately 30% reduction in proteinuria, however, and thus would be insufficient for patients with more severe proteinuria.²⁵²⁻²⁵⁴

The therapeutic approach for children with MN differs from adult therapy. Adults presenting with nephrotic-range proteinuria (urine protein-to-creatinine ratio >2 mg/mg) are recommended to be treated with a combination of oral corticosteroids and cyclophosphamide.²⁵⁵ In children, many patients may well have already had steroid treatment started as presumed initial management of childhood NS, which complicates the decision about the use of immunosuppression early in the disease course. Although no controlled trials have examined steroid treatment of MN in children, a report from 1966 found that steroid treatment of children with MN (80% with NS) resulted in a 50% remission rate for those with NS.²⁵⁶

Early outcomes of rituximab in adults have shown when given alone or in combination with steroids can induce remission in 60%, with reduction or disappearance of anti-PLA2R antibodies.²⁴⁸ The Membranous Nephropathy Trial of Rituximab (MENTOR) study comparing rituximab to cyclosporine in adults with MN found similar results in both therapies in achieving proteinuria remission at 12 months, though rituximab showed superiority at 24 months in sustaining long-term remission with a reduced incidence of relapses. Furthermore, in patients with anti-PLA2R-associated MN, the rituximab group exhibited a more rapid and substantial decline in anti-PLA2R autoantibody titer compared with the cyclosporine group.^{251,257} In children, there is no concrete evidence to guide management in children with MN. KDIGO guidelines recommend treatment with prednisone for at least 8 to 12 weeks with the use of rituximab and CNIs.²¹⁰ It is important to exclude secondary causes of MN such as SLE, chronic hepatitis B virus infection, or neoplasm. A newer treatment for MN is adrenocorticotrophic hormone, which seems in adults to have relatively similar efficacy to that of steroids combined with alkylating agents, although no data are available for children.²⁵⁸⁻²⁶⁰

In summary, while conservative treatment with ACE inhibitor or ARB therapy may suffice for children with modest proteinuria, immunosuppressive treatment appears necessary to induce remission in the majority of patients. There are few data to guide the treatment of MN specifically among children, therapy combining steroids with alkylating agents or CNIs appears to be reasonably effective, and the more recent introduction of rituximab appears to offer the potential for somewhat improved outcomes, potentially with fewer side effects.

OUTCOMES

Data documenting long-term outcomes in children with MN are scarce. However, reports from the North American Paediatric Renal Trials and Collaborative Studies (NAPRTCS) group and the USRDS suggest that the incidence of MN among children with CKD is 0.5% and that

MN represents approximately 0.5% of all pediatric patients with ESKF.^{2,5,261} MN is discussed in further detail in [Chapter 30](#).

NEPHRITIC SYNDROME AND RELATED DISEASES

DIFFERENTIAL DIAGNOSIS OF HEMATURIA

As the differential diagnosis of hematuria is wide ([Table 71.15](#)), including both glomerular and nonglomerular causes, investigations must encompass microscopic urinalysis, detailed review of the personal and family medical history, a thorough clinical examination with focus on extrarenal disease manifestations and urologic causes (e.g., kidney stones, urethritis), and a complete laboratory workup (including blood and urine) to evaluate glomerular causes and renal ultrasound to exclude structural causes.

Glomerular hematuria is a manifestation of renal pathology and is characterized by urine sediment that consists of >5 red blood cells/high-power field together with acanthocytes, red cell casts, or mixed red and white cell casts. Nephritic syndrome presents with hematuria, impaired renal function, hypertension, decreased urine output, nonnephrotic proteinuria, and edema. It is typically a result of glomerular inflammation, which may be caused by postinfectious glomerulonephritis, infective endocarditis, IgA-nephropathy, lupus nephritis, Goodpasture disease, and vasculitides. The syndrome may range from an acute self-limiting disease to rapidly progressive glomerulonephritis (RPGN) or may progress to CKD.^{262,263} Pathophysiology and treatment of the individual diseases are discussed later and in the relevant chapters elsewhere in this book.

ALPORT SYNDROME AND THIN BASEMENT MEMBRANE NEPHROPATHY

Pathogenesis

Alport syndrome and thin basement membrane nephropathy (TBMN) define a spectrum of hereditary diseases affecting type IV collagen, the major component of the glomerular basement membrane (GBM). Type IV collagen is crucial for basement membrane stability and is composed of 6 genetically distinct $\alpha 1$ to $\alpha 6$ type IV collagen chains that assemble to form 3 unique heterotrimers, $\alpha 1\alpha 1\alpha 2$, $\alpha 3\alpha 4\alpha 5$, and $\alpha 5\alpha 5\alpha 6$. Alport syndrome is characterized by mutations in the *COL4A3*, *COL4A4*, or *COL4A5* genes, causing the production of faulty type IV collagen $\alpha 3$, $\alpha 4$, or $\alpha 5$ chains, respectively.²⁶⁴ There are varied forms of inheritance for Alport syndrome including an X-linked form (85% of the cases; mutations in *COL4A5*), an autosomal recessive form (most of the other patients; compound-heterozygous or homozygous mutations in *COL4A3* or *COL4A4*), and an autosomal dominant form (rare; dominant mutations in *COL4A3* or *COL4A4*).²⁶⁵ TBMN is an autosomal dominant disease, caused by mutations in *COL4A3* and *COL4A4*.²⁶⁶ Defective type IV collagen manifests as GBM thinning and variable segmental GBM thickening and splitting, as well as a “basket-weave” appearance on electron microscopy (EM) rather than the normal ribbon-like structure.²⁶⁷ Other genes also known

Table 71.15 Differential Diagnosis of Hematuria

Glomerular Hematuria

- a) Glomerular diseases due to immune-mediated damage to the glomerular structure
 - IgA-nephropathy (IgAN)
 - IgAVN (former: Henoch-Schönlein purpura-associated glomerulonephritis)
 - Postinfectious nephritis
 - Primary glomerulonephritis (e.g., membranoproliferative GN, membranous GN, and C3GN)
 - Secondary nephritis due to systemic diseases (like systemic lupus erythematosus, ANCA vasculitis, etc.)
- b) Glomerular diseases due to an inherited abnormality of basement membrane collagens
 - Alport syndrome: X-linked, autosomal, and digenic
- c) Glomerular diseases due to thrombotic microangiopathy
 - Hemolytic uremic syndrome (HUS)
 - EHEC induced, *Streptococcus pneumoniae*-related HUS, H1N1 and influenza-related HUS
 - Atypical HUS (complement gene mutations, complement factor H antibody, DGKE mutations, cobalamin C defects)
 - HUS with coexisting disease condition (malignancy, solid organ and stem cell transplantation, drug-induced)
 - Thrombotic thrombocytopenic purpura (TTP)
- Postglomerular hematuria
 - a) Hematuria associated with crystal formation
 - Hypercalciuria and other crystallurias
 - Nephrolithiasis (NL)
 - b) Hematuria associated with mechanical damage
 - Trauma
 - Severe hydronephrosis
 - c) Hematuria associated with cyst formation
 - Autosomal dominant polycystic kidney disease (ADPKD)
 - Solitary kidney cyst
 - d) Hematuria associated with vascular damage
 - Nutcracker syndrome (NCS)
 - Hemangioma
 - Arteriovenous malformation
 - Renal vein thrombosis
 - e) Tubulointerstitial nephritis (TIN) (infectious, immune mediated)
 - f) Medications (cyclophosphamide, aspirin, anticoagulants)
 - g) Tumor
- Extrarenal systemic causes of hematuria
 - a) Coagulopathies
 - b) Hemoglobinopathies

Modified from Horváth O, Szabó AJ, Reusz GS. How to define and assess the clinically significant causes of hematuria in childhood. *Pediatr Nephrol.* 2023;38(8):2549–2562. Erratum in *Pediatr Nephrol.* 2023;38(8):2901.

to cause familial microscopic hematuria (FMH) include CFHR5, MYH9, and FN1.²⁶⁸ Further discussion on genetic kidney diseases can be found in [Chapter 44](#).

In addition to the defective type IV collagen, Alport syndrome is marked by other features. Macrophage elastase MMP12 plays a role in the proteolysis of collagen IV protomers, ($\alpha 1\alpha 1\alpha 2$), and it has been shown that the expression of macrophage elastase MMP12 is >40 times higher in Alport syndrome. Altered type IV collagen causes a compensatory overproduction of atypical laminin

isoforms (laminin $\alpha 1$ and $\alpha 5$), which further contributes to the increased permeability of the GBM. Additionally, transforming growth factor $\beta 1$ (TGF- $\beta 1$) produced by podocytes contributes to progressive fibrosis. It has been suggested that podocytes are able to detect altered collagen protomers via discoidin domain receptor 1 (DDR1) tyrosine kinase transmembrane receptors. Detection results in an inflammatory response and upregulation of cytokines and growth factors ultimately results in chronic inflammation, disease progression, and further injury.^{267,269-270}

Clinical Manifestations

Alport syndrome is characterized by the triad of familial nephritis, deafness, and ocular changes. Hematuria is the earliest and most common manifestation, often occurring in the first 2 decades of life. Women with X-linked Alport syndrome are often undiagnosed; however, on average twice as many women are affected as men. Furthermore, up to one-third of these women will develop renal failure. Edema, proteinuria, and hypertension typically do not develop in childhood, and the incidence and severity of proteinuria increases with age and level of renal impairment. Bilateral, high-frequency sensorineural deafness is common and onset generally develops in adolescence before the onset of renal failure. Ocular changes in Alport syndrome include dot-and-fleck retinopathy and anterior lenticonus, which is defined as the conical protrusion of the lens into the anterior chamber due to a thin basement membrane of the lens capsule, occurring in 85% and 25% of X-linked AS patients, respectively.²⁷¹ Disease severity of Alport syndrome is determined by the type of mutation with a strong genotype–phenotype correlation (point or missense mutations versus deletions, frameshift, or truncating mutations) and the type of inheritance. In a large cohort of X-linked Alport syndrome patients, missense mutations were the most common, occurring in 51% of cases, and conferred the most favorable prognosis with a median time to ESKF of 37 years, as compared with patients with deletions, who reach ESKF by age 22 (95% confidence interval, 16–23 years).²⁷²

THIN BASEMENT MEMBRANE NEPHROPATHY

TBMN is the most common cause of persistent glomerular hematuria in adults and children and has a prevalence of approximately 1%.²⁷³ In contrast to Alport syndrome, TBMN typically presents with isolated hematuria, with renal biopsy showing mild mesangial cell proliferation and slight matrix expansion.²⁷³ Additionally, TBMN is due to heterozygous COL4A3 or COL4A4 mutations and often represents the carrier state of autosomal recessive Alport syndrome. As a result, a spectrum of collagen IV–related disorders encompasses Alport syndrome at the severe end and TBMN at the milder end.²⁷⁴ There have also been proposals to classify Alport syndrome and benign familial hematuria/thin basement membrane nephropathy as forms of collagen IV–related renal disease or by Alport kidney disease or Alport spectrum.²⁷⁵

Establishing the Diagnosis of Alport Syndrome and Thin Basement Membrane Nephropathy

AS and TBMN may exhibit similar clinical and ultrastructural characteristics, yet it is important to differentiate them due to the distinct risks of renal failure and other complications they pose to affected individuals and their family members. The diagnosis of AS is suspected when a patient presents with hematuria, as well as a positive family history of hematuria, chronic renal failure, or both; high-frequency SNHL; or pathognomonic ocular anomalies (dot-and-fleck retinopathy and anterior lenticonus). Confirmation of the diagnosis requires genetic testing or a renal biopsy revealing a lamellated GBM on EM in the right clinical setting. GBM thinning is associated with early stages of disease, while late stages of the disease are associated with thickening and multilamination of the GBM (Fig. 71.11).^{274,276} In contrast, EM in TBMN demonstrates diffuse thinning of the GBM with a width <250 nm in the majority of capillary loops and at least 50% of individual capillaries. Notably, TBMN specimens will not demonstrate thickening or lamellation.^{273,274}

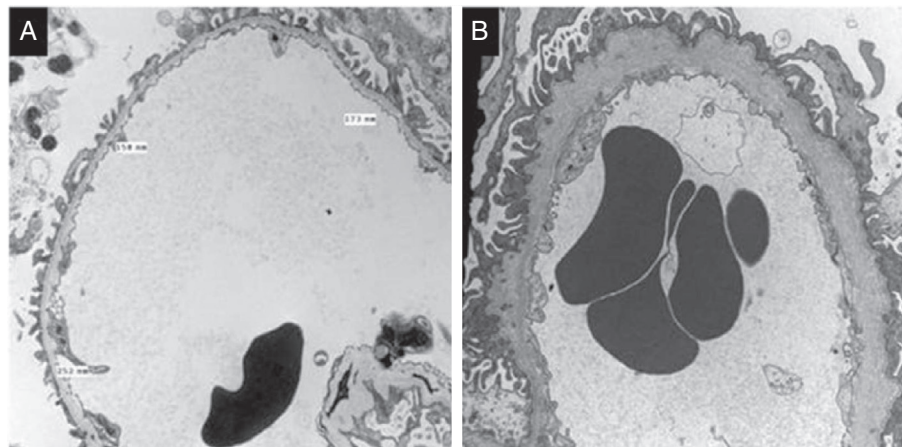


Fig. 71.11 Electron microscopy findings in thin basement membrane disease (TBMN) and Alport syndrome (AS). (A) TBMN shows diffuse thinning of the glomerular basement membrane (GBM) lamina densa compared with that of age-matched controls (best established in the same laboratory). (B) AS shows an irregular GBM with thinning and thickening, the latter becoming more prominent over time and with splitting/fragmentation of the lamina densa resulting in a multilaminated/basket-weave appearance often containing stubs of cellular processes with electron-lucent areas and small electron-dense particles. (A and B: electron microscopy $\times 10,000$.)

A new classification of Alport syndrome has been proposed that does not require a family history or evidence of disease progression.²⁷⁷ Women with heterozygous *COL4A5* variants are classified as having X-linked Alport syndrome, not as carriers. It also eliminates the diagnosis of “thin basement membrane nephropathy” and considers individuals with heterozygous *COL4A3* or *COL4A4* variants to have autosomal dominant Alport syndrome. However, the classification is controversial due to the variability in Alport syndrome phenotypes, with some patients developing only mild, kidney-limited disease. Notably, patients with FSGS on kidney biopsy and a pathogenic variant in any of the *COL4A3-5* genes would be considered to have Alport syndrome rather than a genetic form of FSGS under the proposal.

Treatment

Current guidelines for the treatment of Alport syndrome suggest the initiation of ACE inhibitors and/or ARBs at the onset of overt proteinuria to prolong renal survival and delay the need for RRT.²⁷⁷ ACE inhibitors should be first line, with ARB or aldosterone inhibition used as second-line agents.²⁷⁸ A prospective RCT showed ramipril decreased the risk of progression by approximately 50% and highlighted the safety of preemptive ramipril therapy (defined as presence of microscopic hematuria but negative urine dipstick for albuminuria) before kidney damage in children older than age 2 without hypertension.²⁷⁹ Similarly, the angiotensin receptor blocker (ARB) losartan has been shown to reduce proteinuria to a greater extent than a placebo or amlodipine.^{267,278,280} Small studies have been performed investigating combination therapies and have found that combination of an ACE inhibitor and ARB together reduces proteinuria to a significantly greater extent than either agent alone. Additionally, combination of an ACE inhibitor with spironolactone suppresses proteinuria to a greater degree than the combination of an ACE inhibitor with an ARB.²⁸¹⁻²⁸³

CNIs such as cyclosporine should not be used in Alport syndrome.²⁸¹ CNIs have shown a significant and sustained improvement in proteinuria, although this was associated with a significant reduction in GFR, calcineurin toxicity on renal biopsy, and hypertension.²⁸⁴⁻²⁸⁶ Several novel therapies are under investigation for potential use in Alport syndrome. Sodium glucose cotransporter 2 in 6 Alport syndrome patients showed a 40% decline in the mean urine albumin-creatinine ratio (UACR), with no detrimental effect on eGFR.^{281,287} Other agents under investigation include aminoglycoside analogs, endothelin A receptor antagonists, lipid-modifying drugs, hydroxychloroquine, and gene replacement therapy.^{278,279,288-291}

SYSTEMIC DISEASES WITH RENAL INVOLVEMENT

IgA-VASCULITIS NEPHRITIS AND IgA-NEPHROPATHY

INTRODUCTION

IgA-nephropathy (IgAN) is the most common form of chronic glomerulonephritis in children.²⁹² IgA-vasculitis

(previously known as Henoch-Schönlein purpura) is the most common vasculitis in children, with glomerulonephritis being its most common chronic manifestation.²⁹³ First described by Berger and Hinglais in 1968 among patients with macroscopic hematuria during viral upper respiratory infections, IgAN is now recognized as the most common primary glomerulopathy in the world.^{294,295} While early on it was considered a benign disease, it is increasingly recognized that between 20% and 50% of adults and children ultimately developed ESKF.²⁹⁵⁻²⁹⁹ In contrast to IgAN, which is limited to the kidneys, IgA-vasculitis is a multisystem disease primarily affecting the skin, joints, gastrointestinal tract, and kidneys.^{300,301} Because these diseases share many pathogenic and clinical features, however, their relevance in pediatric kidney disease is discussed together. For in-depth discussion of IgA-nephropathy and IgA-vasculitis, please see Chapter 33.

EPIDEMIOLOGY

IgAN is the most common glomerulonephritis worldwide with an incidence of approximately 0.06 to 4.2 per 100,000 population per year.³⁰² Globally, there are distinct geographical and sex-based variations. There is increased prevalence in Far East Asia compared with Europe and North America and a low prevalence in Africa. Additionally, the male-to-female ratio is 3:1 in Europeans but 1:1 in East Asians.³⁰³ While IgAN occurs in both children and adults, it presents most commonly during the second and third decades of life.²⁹⁵ Among children, IgAN appears to generally occur with a male-to-female ratio of approximately 2:1.^{298,300,304}

The worldwide incidence of IgA-vasculitis has been described between 3.5 and 26.7 per 100,000, thus occurring much more commonly than IgAN.³⁰⁵ However, only a minority (~30%) of children with IgA-vasculitis develop glomerulonephritis, which is most commonly mild, and the incidence of clinically significant IgA-vasculitis-associated nephritis has been estimated to be potentially lower than that of IgAN.^{298,306-308} It typically impacts children aged 5 to 15 years and is rarely observed in adults and infants. The age at which it manifests has been deemed an important factor in disease severity and prognosis.³⁰⁹

PATHOPHYSIOLOGY

IgAN is influenced by a combination of genetic and environmental factors, and the pathogenesis is described as a multi-“hit” process involving genetically susceptible variants encoding galactosylation, dysfunctional mucosal immune responses, and environmental triggers such as infection, alteration of microbiota, and food antigens.³¹⁰ A key observation in patients with IgAN is the presence of circulating and glomerular ICs composed of galactose-deficient IgA1, an IgG autoantibody directed against the hinge region O-glycans, and C3. These pathways ultimately contribute to glomerulosclerosis and tubulointerstitial fibrosis, leading to loss of renal function.³¹¹ Serum levels of IgA are increased in 50% to 70% of patients with IgAN during active disease but return to normal during recovery.³¹²⁻³¹⁴ Evidence suggests that the precipitating event in IgAN, abnormal glycosylation of IgA1, appears to be an inherited trait rather than an acquired abnormality, but that different families may have differing molecular

mechanisms of disease development.³¹⁴ The inheritance pattern suggests an autosomal dominant trait with variable penetrance. Genome-wide association studies (GWASs) have identified 15 common risk variants, collectively accounting for approximately 6% to 8% of the overall disease risk.^{303,315} The exact etiology of IgA-vasculitis nephritis remains unknown, yet all IgA-vasculitis nephritis patients have been found to have IgA1-circulating ICs of small molecular mass. Only those with nephritis have additional large-molecular-mass IgA1-IgG-containing circulating ICs.³¹⁶

CLINICAL PRESENTATION

The presentation of IgAN ranges from isolated hematuria to significant proteinuria to AKI and even CKD with a 10-year risk of progression to ESKF of 26%, though the 2 most common features are asymptomatic hematuria and progressive kidney disease.^{311,317} Macroscopic hematuria is the initial presentation of approximately 75% of childhood cases in the United States, although only in 26% of cases in Japan.^{297,318-320} The hematuria is often associated with an upper respiratory infection and resolves within 2 to 4 days. Less commonly it occurs in association with other infections associated with the mucosal system, such as sinusitis or diarrhea. Other less common presentations include microscopic hematuria with proteinuria, isolated microscopic hematuria, and isolated proteinuria. Even less commonly (~10% of cases), IgAN can present with macroscopic hematuria with reversible AKI, CKD, or rarely with RPGN.^{297,318,321,322} Hypertension is present in approximately 31% to 58% of children at presentation, although it is usually not severe unless AKI is also present.³²³ Prognostic factors that confer a favorable outcome include minimal proteinuria with no proteinuria, whereas hypertension, persistent hematuria, and decreased eGFR at diagnosis are associated with poorer outcomes.^{295,311,324,325}

Children with IgA-vasculitis nephritis typically present far earlier (~4–6 years) than those with IgAN (~10–30 years), although like IgAN, IgA-vasculitis nephritis can present at any age. Most cases of IgA-vasculitis occur in the autumn and winter with proposed triggers including URTIs, medications, vaccinations, and malignancies.³²⁶ The typical presentation includes a lower extremity purpuric rash, often associated with colicky abdominal pain, and sometimes arthritis. In a review of ²⁶⁰ adults, joint involvement, glomerulonephritis, and gastrointestinal involvement were found in 62%, 70%, and 53% of patients, respectively.³²⁷ The renal manifestations of IgA-vasculitis nephritis at presentation may range from no involvement to severe nephritic and/or NS.^{292,328} Abnormal urinalysis indicating hematuria and/or proteinuria is seen in approximately 35% of children with IgAV. Approximately 25% of children with an abnormal urinalysis will also have macroscopic hematuria.³²⁹ Notably, children with IgA-vasculitis nephritis are approximately twice as likely to develop NS than those with IgAN, usually occurring within 1 to 2 months of onset of IgA-vasculitis.^{326,328,330-332}

LABORATORY EVALUATION

Because no specific serologic biomarker exists to diagnose IgAN, confirmation of the diagnosis depends on kidney biopsy and identification of IgA deposits in

the glomerular mesangium. A large review of published biopsy studies found 100% of IgAN biopsies had positivity for IgA, 43% had positivity for IgG, and 54% for IgM.³³³ Laboratory evaluation generally includes the workup for glomerulonephritis including serum creatinine, serum albumin, serum C3/C4 levels, serum antinuclear antibody (ANA), antinuclear cytoplasmic antibody (ANCA), antistreptolysin O Titre, as well as a complete urinalysis and urine protein-to-creatinine ratio determination. Diagnosis of IgA-vasculitis nephritis includes the clinical features of IgA-vasculitis in combination with evidence of renal involvement on urinalysis and urine protein-to-creatinine ratio. Serum IgA levels are not reliable biomarkers because they are elevated in both IgAN- and IgA-vasculitis. Serum C3 and C4 levels typically remain within the normal range for both diseases. This distinction helps differentiate IgAN from conditions like PIGN, MPGN, or SLE nephritis, which often present with hypocomplementemia.³³⁴ The pediatric IgAN risk score is modified from the adult International IgAN Prediction Tool and can accurately predict the risk of a 30% decline in eGFR or ESKD in children with IgAN.³³⁵ IgAN KDIGO guidelines recommend performing kidney biopsies with symptoms (hematuria, proteinuria, normal C3) to confirm the diagnosis (and rule out other diagnoses) and assess the degree of inflammation/presence of necrosis, but they do not suggest a timeframe.³³⁵ The mesangial hypercellularity (M), endocapillary proliferation (E), segmental sclerosis/adhesion (S), tubular atrophy/interstitial fibrosis (T), and cellular or fibrocellular crescents (C) (MEST-C) score in conjunction with the Oxford Classification of IgAN in children can be used to predict risk of progression and support treatment decision making.³³⁵

TREATMENT

The treatment of IgAN differs between children and adults. The pediatric population with IgAN is more limited and more complex to study, and whether children should be treated the same as adults remains uncertain.³³⁶ Suggested treatments for adults with IgAN have placed a greater focus on conservative therapies, focusing more on nonimmunosuppressive treatment with ACE inhibitors and ARBs, whereas suggested treatments for children with IgAN tend to be more active, often including immunosuppression. This field is changing rapidly and there may be more focused therapies in the near future. For detailed discussion of the therapeutic approach to IgA, see [Chapter 33](#).

For those children who remain asymptomatic but who have more active disease, as reflected by a first morning urine protein-to-creatinine ratio of >0.5 mg/mg (>56.5 mg/mmol), ACE inhibitors and/or ARBs are generally indicated. The use of ACEI and/or ARB in pediatric patients with IgAN has been shown to be safe and to reduce proteinuria.³³⁷ The sodium glucose cotransporter 2 (SGLT2) inhibitors are indicated in adults and are likely a useful addition, especially in children with persistent proteinuria.³³⁸ Blood pressure control is important. Children with IgAV tend to receive nonsteroidal antiinflammatory drugs for joint pains and inflammation; these drugs should be discontinued if renal disease develops.

In children who present with nephrotic-range proteinuria or NS, additional treatment is required. Immunosuppressive

treatment including corticosteroid therapy and possibly addition of cyclophosphamide, azathioprine, mycophenolate mofetil, or tacrolimus may be indicated.³³⁹ The most common protocol is oral prednisone 1 to 2 mg/kg/day (maximum dose 60 mg/day) for 4 weeks tapered over 4 to 6 months, though intravenous methylprednisolone may also be used.³⁴⁰ Though typically reserved for patients with more active disease, it has been suggested corticosteroids reduce the risk of progression to ESKF, regardless of initial eGFR and in direct proportion to the extent of proteinuria.³⁴⁰ A Japanese RCT investigating immunosuppressive treatments for IgAN found the addition of prednisone, azathioprine to heparin-warfarin, and dipyridamole was associated with a significant reduction in proteinuria, serum IgA concentration, mesangial IgA deposition, and glomerulosclerosis.³⁴¹ Short courses of low-dose steroids are given in IgAV periodically for abdominal or joint symptoms. These have no impact on the development of kidney disease as doses are not high enough for treatment effects.

Other immunosuppressive therapies including MMF, cyclophosphamide, CNIs, and rituximab are less often used due to a relative lack of clear evidence to support more widespread use. KDIGO guidelines do not recommend mycophenolate mofetil in IgAN.³⁴² Cyclophosphamide has shown promise in adult IgAN but not pediatric cases,³⁴³ and a limited effect has been seen with CNIs³⁴⁴ and rituximab.^{210,345-347} TRF-budesonide has shown efficacy as a potential second-line treatment in pediatric IgAN, though further large-scale studies are required.³⁴⁸ Other future therapies currently under investigation include hydroxychloroquine combined with RAS inhibitors, inhibitors of endothelin-1 receptors, inhibitors of the BAFF-TNF receptor family, and several complement inhibitors including eculizumab.³⁴⁰

Some nonimmunosuppressive treatments for IgAN have reported varied success. Fish oil, vitamin E, and tonsillectomy had uncertain effects on improving proteinuria or preventing eGFR decline.³⁴⁹ The KDIGO guidelines suggested tonsillectomy not be performed in patients with IgAN without a clinical indication beyond IgAN.³⁵⁰⁻³⁵² The Chinese herbal medicine Sairei-to (TJ-114), although not widely used, was found to be more effective than no treatment in a randomized controlled trial by the Japanese Pediatric IgA Nephropathy Treatment Study Group, conducted in children with mild IgAN disease treated for 2 years.³⁵³

OUTCOME

The long-term outcome for children with IgAN is highly variable. Approximately one-third of children will experience a complete remission of disease. Long-term studies have reported the development of progressive CKD in 9% at 15 years from diagnosis and ultimately ESKF in 20% to 27% of children.³⁵⁴ However, mild cases of IgAN in children might be missed with manifestation of irreversible damage only decades after the true onset, as 50% of subjects with IgAN enter renal replacement treatment before the age of 50 years.

In general, poor clinical prognostic indicators include persistent high-grade proteinuria and hypertension, while poor histologic prognostic indicators include higher percentages of glomeruli with crescents, adhesions

or sclerosis, significant tubulointerstitial changes, subepithelial electron-dense deposits, or lysis of the GBM.³⁵⁵⁻³⁵⁷

For children with IgA-vasculitis nephritis, long-term outcomes are less clear due to marked variability in disease severity among the reported cases. The risk of progression to end-stage renal disease requiring dialysis in children varies from 2.5% to 25%, but it is on average around 8%.

RENAL DISEASES WITH SYSTEMIC VASCULITIS

Vasculitis, or inflammation of the blood vessel walls, is an uncommon cause of glomerular injury in children but can result in severe glomerulonephritis and even death. Vascular inflammation can lead to stenosis and/or aneurysm of the vessel walls and result in various types of glomerulonephritis, depending on the size of the vessel walls and the organs most affected by disease. [Table 71.16](#) summarizes the current classification of vasculitides affecting children. The most common types of vasculitis seen in children are Kawasaki disease (KD) and IgA-vasculitis (IgAV; i.e., HSP). IgAV and IgAN have already been discussed in detail earlier in this chapter, while large vessel vasculitides, Takayasu arteritis, and giant cell arteritis are extremely rare in children. The medium vessel vasculitides include two primary diseases that include KD, which is common but only rarely has renal manifestations, and polyarteritis nodosa (PAN), which is rare in both children and adults but commonly has renal manifestations. The small vessel vasculitides, which include lupus nephritis and ANCA vasculitis (discussed briefly later) and anti-glomerular basement membrane disease and immune complex vasculitides (discussed elsewhere in this book; see [Chapters 31 and 32](#)).

KAWASAKI DISEASE

KD is a vasculitis that typically involves the mucosa and skin and is known to also preferentially affect coronary vessels. This common vasculitis is defined by a child having at least 5 days of fever, accompanied by four of the following five criteria: 1. edema, erythema, or desquamation of the hands, feet, or perineal area; 2. bilateral nonpurulent conjunctival injection; 3. polymorphous exanthema (primarily truncal); 4. oral and pharyngeal mucosal injection; and 5. cervical lymphadenopathy (typically unilateral). KD only rarely has renal manifestations, which can include sterile pyuria, proteinuria, AKI with tubulointerstitial changes, and rarely renal artery aneurysms. Nephrotic syndrome has also been reported, although it appears to remit spontaneously without glucocorticoid therapy.³⁵⁸⁻³⁶¹

POLYARTERITIS NODOSA

PAN is now defined as a necrotizing arteritis involving small or medium arteries, without evidence of antineutrophil cytoplasmic antibodies or vasculitis in venules, capillaries, or arterioles.³⁶² While uncommon in both children and adults, specific criteria for diagnosing pediatric PAN have been published.^{363,364} This disease is specifically distinguished from microscopic polyangiitis (MPA) pathologically by its lack of ANCAs or necrotizing glomerulonephritis and clinically by its higher incidence of gastrointestinal symptoms

Table 71.16 Classification of Pediatric Vasculitides by Primary Vessel Involvement

Small Vessel Vasculitis	Medium Vessel Vasculitis	Large Vessel Vasculitis ^a
• Antineutrophil cytoplasmic antibody (ANCA)-associated vasculitis (AAV) (i.e., pauci-immune small vessel vasculitis) • Granulomatosis with polyangiitis (GPA; can also involve medium vessels) • Microscopic polyangiitis (MPA) - Eosinophilic granulomatosis with polyangiitis (Churg-Strauss syndrome, EGPA)^a • Immune complex small vessel vasculitis • IgA-vasculitis (Henoch-Schönlein) (IgAV) • Antiglomerular basement membrane (anti-GBM) disease^a • Hypocomplementemic urticarial vasculitis (anti-C1q vasculitis)^a - Cryoglobulinemic vasculitis (CV)^a	• Kawasaki disease (KD) • Polyarteritis nodosa (PAN, can also involve small vessels)^a	• Takayasu arteritis^a • Giant cell arteritis^a

^aRarely seen in children.

Modified from Jennette JC, Falk RJ, Bacon PA, et al. 2012 revised International Chapel Hill Consensus Conference nomenclature of vasculitides. *Arthritis Rheum.* 2013;65:1–11.

and peripheral neuropathy.³⁶⁵ Clinical presentation typically includes fever, skin lesions, and/or myalgia, with involvement of the kidneys, heart, and/or lungs occurring in 15% to 20% of children.^{366–368} Notably, the major renal manifestations result from involvement of medium-sized vessels and include renovascular hypertension and aneurysms with the potential for rupture or hemorrhage. Treatment generally includes oral and/or IV glucocorticoids combined with cyclophosphamide.^{369–371}

SYSTEMIC LUPUS ERYTHEMATOSUS

Systemic lupus erythematosus (SLE) is a common systemic inflammatory autoimmune disease that can affect many organ systems. While renal involvement may not be present at the initial disease presentation, most children with SLE will develop renal involvement at some point in their disease, and lupus nephritis often becomes among the most significant predictors of morbidity and mortality among children with SLE.^{372–376} For example, data from adults have reported that approximately 10% of patients who develop lupus nephritis will progress to ESKF.³⁷⁷ This section does not address the initial diagnosis of SLE but focuses instead on the diagnosis and treatment of lupus nephritis among children previously diagnosed with lupus. Lupus nephritis is discussed in more detail in [Chapter 31](#).

Approximately 20% of lupus cases are diagnosed during childhood with an annual incidence of 0.3 to 2 cases per 100,000 childhood population and a higher prevalence in certain ethnic groups, such as African American, Hispanic, and Asian populations.^{378,379} Lupus nephritis occurs in 50% to 82% of children in comparison with 20% to 40% of adults with SLE.³⁸⁰ SLE is notable for its strong predilection for females, with the female-to-male ratio ranging from 1.3:1 among prepubertal children to 4.5:1 among adolescents to 8 to 15:1 among adults.^{381–384} The age of onset in pediatric SLE varies, but it most commonly presents during adolescence, with a peak incidence between 12 and 16 years of age.³⁸⁵

The diagnosis of lupus nephritis in patients with SLE may be clinically subtle, making screening urinalyses (at least yearly) of all patients with SLE important. The most common clinical manifestations of lupus nephritis include

proteinuria (100%), which may range from mild to NS (50%), followed by microscopic hematuria (80%), tubular abnormalities (70%), and AKI (60%).³⁸⁶ In children with SLE in whom nephritis is suspected, the renal evaluation should include a urinalysis and renal function panel and renal biopsy should be considered.

The approach to the treatment of lupus nephritis is similar in adults and children as outlined in [Fig. 71.12](#). Hydroxychloroquine is recommended for all patients with lupus to reduce the risk of flares, ESKF, and mortality, with a daily dose of <5 mg/kg to avoid retinopathy.³⁸⁷ Children on hydroxychloroquine require regular ophthalmologic follow-up. Angiotensin-converting enzyme inhibitors and angiotensin receptor blockers are the preferred supportive renoprotective therapies.

Immunosuppressive therapy for lupus nephritis is determined by biopsy-determined class and the degree of activity versus chronicity.^{388,389} The Single Hub and Access Point for Paediatric Rheumatology in Europe (SHARE) guidelines for class I and II lupus nephritis suggest low-dose oral glucocorticoids, tapering over 3 to 6 months, with disease-modifying antirheumatic drugs (DMARDs) like MMF or AZA for persistent proteinuria.³⁹⁰ For children with class III and IV ± V lupus nephritis with signs of activity on the kidney biopsy, induction therapy should include high-dose glucocorticoids combined with either MMF or cyclophosphamide. MMF is favored due to its improved safety profile, particularly for pediatric patients. The regimen typically consists of pulse intravenous methylprednisolone for 1 to 3 days, followed by daily oral prednisone at 1 to 2 mg/kg/day tapered over a few months down to a maintenance dose.³⁹¹ MMF dosing is typically 1200 to 1800 mg/m²/per day (maximum dose 1500 mg twice daily) for 6 months followed by maintenance dosing.^{392,393} Therapeutic drug monitoring of MMF is beneficial in optimizing therapeutic efficacy as there exists high interindividual variability of mycophenolic acid (MPA) exposure at a given dose of MMF. Cyclophosphamide may be given intravenously (500–1000 mg/m² monthly for ~6 months), or less commonly orally (2 mg/kg daily for ~6 months), to be most commonly followed by MMF as maintenance therapy. Newer therapies such as belimumab and voclosporine may be used in combination with other therapies

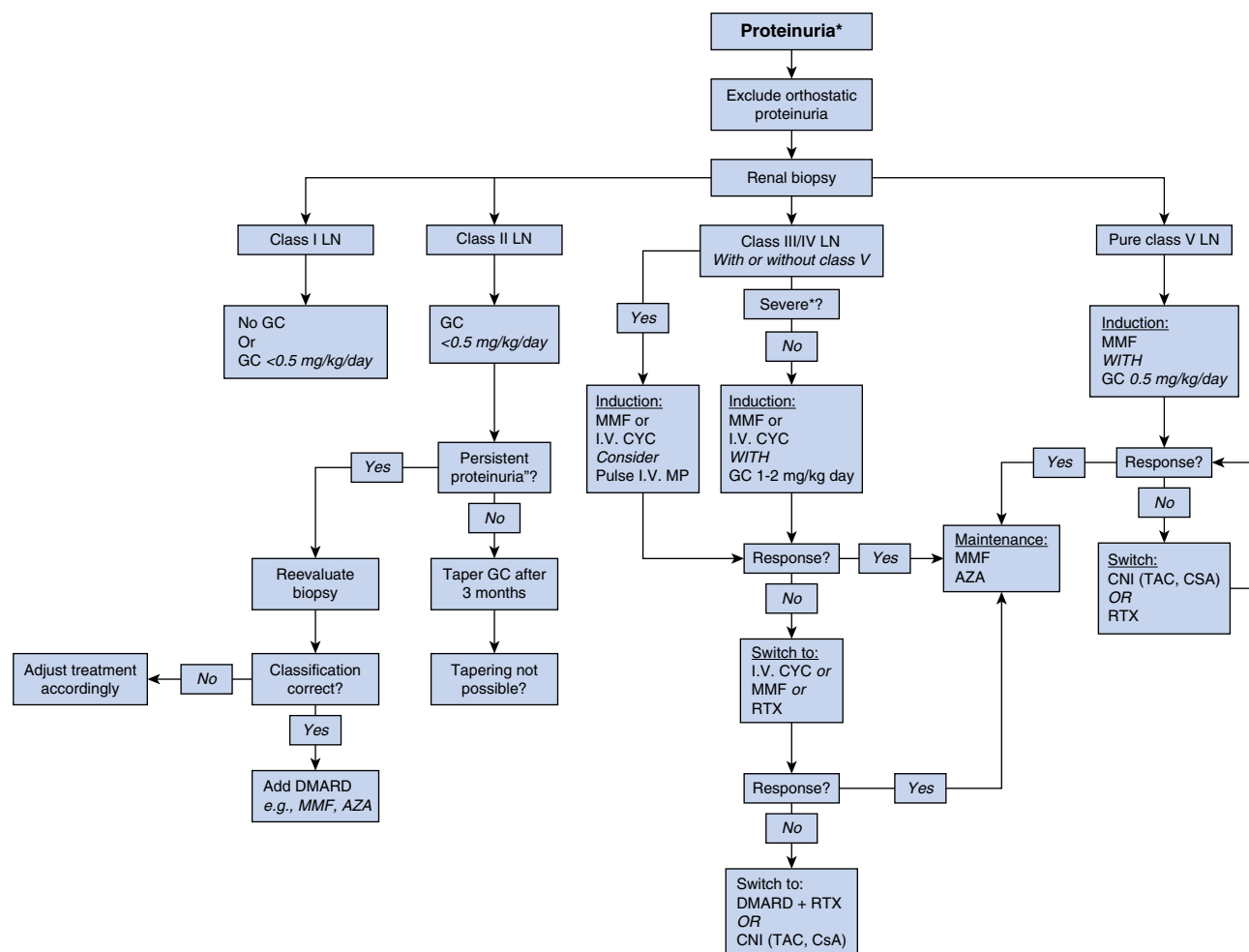


Fig. 71.12 Treatment strategies for the different classes of LN definitions. *Proteinuria: 0.5 g/24 hour or UP:CR >50 mg/mmol in a urine sample; **persistent proteinuria: presence of proteinuria for >3 months; DMARD: MMF, AZA, CNI, intravenous CYC; +severe disease, e.g., impaired eGFR, estimated glomerular filtration rate (1 g/m²/day), biopsy-proven crescentic glomerulonephritis. AZA, Azathioprine; CNI, calcineurin inhibitors; CsA, ciclosporin; CYC, cyclophosphamide; DMARD, disease-modifying antirheumatic drug; GC, corticosteroids; LN, lupus nephritis as classified by the ISN/RPS 2003 classification system; MMF, mycophenolate mofetil; MP, methylprednisolone; RTX, rituximab; TAC, tacrolimus. (From Groot N, de Graeff N, Marks SD, et al. European evidence-based recommendations for the diagnosis and treatment of childhood-onset lupus nephritis: the SHARE initiative. *Annals of the Rheumatic Diseases*. 2017;76[12], 1965–1973.)

from induction. Belimumab is a monoclonal antibody that inhibits the activity of the B cell survival cytokine BAFF (B cell activating factor). In the large BLISS-LN RCT in adults, it has shown favorable outcomes of primary renal efficacy (43% vs. 32%; $P=0.03$) and complete renal response (CRR) (30% vs. 20%; $P=0.02$) compared with the placebo group when added to MMF or reduced-dose cyclophosphamide.³⁹⁴ Voclosporin is a cyclosporin analog that has shown to be superior in adults to standard induction therapy (with MMF and corticosteroids) at both 24 weeks (CRR: 32.6% [low-dose] vs. 27.3% [high-dose] vs. 19.3% [placebo]) and 48 weeks (CRR: 49.4% [low-dose] vs. 39.8% [high-dose] vs. 23.9% [placebo]).³⁹⁵ In 2021, voclosporin was approved for the treatment of lupus nephritis in adults in combination with background immunosuppression. If voclosporine is not available, other CNIs (i.e., tacrolimus or cyclosporine) can be used.

Maintenance-phase treatment typically includes low-dose oral glucocorticoids, along with agents such as MMF, with or without CNIs or azathioprine if MMF is not available or not tolerated.³⁹¹ Low-dose cyclosporine and tacrolimus have

shown promise, with complete renal remission occurring in 45.5% patients at 6 months and 71.4% after 3 years.³⁹⁶ The optimal duration for maintenance-phase treatment of lupus nephritis remains uncertain, although suggested durations have ranged from approximately 3 years to as long as 8 years.^{388,397} SHARE and KDIGO both recommend immunosuppression for at least 3 years.

Because of therapeutic advances in recent decades, 10-year survival rates are now approximately 90%.³⁹⁸ Despite these advances, only 55% of children with class III and IV lupus nephritis achieve renal remission and a 25% to 50% rate of lupus nephritis flares on therapy.³⁷⁸ The rate of kidney failure in high-income countries from pediatric lupus nephritis is about 15%.³⁹⁹ Long-term follow-up and close collaboration among pediatric rheumatologists, nephrologists, and other specialists are essential for optimizing outcomes and managing potential complications.³⁹¹ Key issues must be considered including adherence, which may support the use of intravenous medications; growth, which encourages limiting glucocorticoid exposure; fertility, particularly

as patients near adolescence, which favors minimizing cyclophosphamide use; prophylaxis against infections during the induction phase, minimization of the long-term effects of glucocorticoid use, and psychosocial factors, such as school attendance and peer socialization.

ANCA-ASSOCIATED VASCULITIS

Childhood-onset ANCA-associated vasculitis (AAV) is rare, typically occurring between ages 10 and 15.⁴⁰⁰ AAV subtypes include granulomatosis with polyangiitis (GPA) and microscopic polyangiitis (MPA).⁴⁰¹ Autoantibodies targeting two key neutrophil proteins, proteinase 3 (PR3-ANCA) or myeloperoxidase (MPO-ANCA), in GPA and MPA, can prime and activate neutrophils, which, along with other inflammatory cells like macrophages, monocytes, and the complement system, contribute to endothelial injury. In childhood, females are more affected than males, unlike in adulthood where both genders are equally impacted. Kidney involvement is more common in children, especially in those who are MPO-ANCA+, while cardiovascular and ENT involvement is less frequent. About 65% to 78% of children who are PR3-ANCA+ will have renal involvement, while 88% to 95% of children who are MPO-ANCA+ will have renal involvement.⁴⁰¹ KDIGO guidelines suggest the use of glucocorticoids, MMF, rituximab, or cyclophosphamide depending on the severity of kidney disease for induction therapy.²¹⁰ Currently, the American College of Rheumatology/Vasculitis Foundation and KDIGO guidelines conditionally recommended against routine addition of plasma exchange to remission induction therapy, except for patients with life-threatening organ involvement.^{210,402} Maintenance therapy should be continued with MMF, rituximab, or azathioprine and low-dose corticosteroids. Close follow-up is required to detect and treat relapses early. Disease activity scores such as the Birmingham Vasculitis Score have not been well validated in children but may be of use in detecting early relapse. The optimal duration of maintenance therapy is not known but is suggested for at least 18 months to 4 years after attainment of remission.²¹⁰ Relapsing disease must be treated with induction, followed by maintenance therapy again. ANCA vasculitis is discussed in detail in [Chapter 32](#).

COMPLEMENT-ASSOCIATED KIDNEY DISEASES

Complement dysregulation plays a role in a wide spectrum of kidney diseases. These diseases are characterized by excess complement activation as a result of genetic variants or circulating autoantibodies.⁴⁰³ Two of the most prominent examples of complement-associated kidney diseases, MPGN/C3G and atypical hemolytic uremic syndrome (aHUS), are explored in greater depth in the following sections. For more on these disorders, see [Chapter 36](#).

THE COMPLEMENT SYSTEM

The classical pathway (CP) of complement is initiated by immunoglobulins (Igs) or ICs and thus is especially involved in autoimmune diseases and the Ig/IC-mediated

form of MPGN.⁴⁰⁴ The lectin pathway (LP) is activated by repetitive carbohydrate structures found (e.g., on the surface of bacteria).⁴⁰⁴ By difference, the alternative pathway (AP) of complement is constitutively active via a process called “tick over” and requires tight regulation.⁴⁰⁵ All three activation pathways converge in the activation of the central complement component C3 to C3b. Together with complement factor B (CFB)—activated by complement factor D—and properdin (CFP), C3b forms the AP C3 convertase (C3bBb), which rapidly amplifies C3 activation. The resulting activation products are involved in inflammatory responses (C3a; anaphylatoxin) or deposits on bacterial surfaces (C3b; opsonization), rendering them a target for phagocytosis. C3b also initiates C5 activation via the formation of the C5 convertase (C3bBbC3b), thus contributing to inflammation (C5a; anaphylatoxin) and initiation of the terminal pathway (TP) of complement (C5b) with the formation of the membrane attack complex (MAC) C5b-9.⁴⁰⁶⁻⁴¹⁰

Complement regulation is provided by soluble and membrane-anchored proteins, which limit complement activation in a timely and spatial fashion both in fluid phase and on cell surfaces. The most important complement regulator, fluid-phase CFH (FH), regulates complement via 1. limiting C3b binding to surfaces and formation of the C3 convertase, C3bBb (competition); 2. inactivating/cleaving C3b (cofactor activity); and 3. accelerating the natural decay of the C3 convertase, C3bBb (decay-acceleration activity).

IMMUNE COMPLEX MEMBRANOPROLIFERATIVE GLOMERULONEPHRITIS AND C3 GLOMERULOPATHY

Membranoproliferative glomerulonephritis (MPGN) accounts for approximately 7% to 10% of all biopsy-confirmed glomerulonephritis.⁴¹¹ It is characterized by the deposition of Ig/ICs and/or complement proteins in the mesangium and/or along the capillary wall of the glomerulus, resulting in a histopathologic pattern of endocapillary and mesangial hypercellularity, mesangial matrix expansion, and lobulated capillary morphology.⁴¹² It is classified as either a primary complement-mediated MPGN (C-MPGN), also known as C3 glomerulopathy (C3G), or a secondary immune-complex-mediated (IC-MPGN) with predominant IgG/IgM deposition.⁴¹³ C3G is considered a primary complement disease characterized by C3 deposition that is ≥ 2 orders of magnitude higher than accompanying immunoglobulin, whereas IC-MPGN may be idiopathic or secondary to a spectrum of underlying diseases or conditions including infections, autoimmune diseases, malignancies, or monoclonal gammopathy ([Table 71.17](#)).⁴¹⁴⁻⁴²⁰ C3G may be further divided into C3 glomerulonephritis (C3GN) and dense deposit disease (DDD), which may appear on light microscopy (LM) as MPGN, diffuse proliferative/mesangioproliferative glomerulonephritis, and necrotizing or crescentic glomerulonephritis. EM may be used to differentiate between the two with C3GN characterized by discrete C3 deposits in the mesangium and along the capillary walls and DDD by C3 deposits in the mesangium and within the GBM.^{421,422}

Table 71.17 Causes of Secondary Membranoproliferative Glomerulonephritis

Infectious diseases: bacterial/viral/protozoal	- Hepatitis B, C, Epstein-Barr virus, human immunodeficiency virus - Endocarditis/visceral abscesses - Infected ventriculoatrial shunts/empyema - Malaria, schistosomiasis, mycoplasma - Tuberculosis, leprosy - Epstein-Barr virus infection - Brucellosis
Systemic immune diseases	- Cryoglobulinemia - Systemic lupus erythematosus - Sjögren syndrome - Rheumatoid arthritis - Hereditary deficiencies of complement components - X-linked agammaglobulinemia
Neoplasms/ dysproteinemias	- Plasma cell dyscrasia - Fibrillary and immunotactoid glomerulonephritis - Light-chain deposition disease - Heavy-chain deposition disease - Light- and heavy-chain deposition disease - Leukemias and lymphomas (with cryoglobulinemia) - Waldenstrom macroglobulinemia - Carcinomas, Wilms tumor, malignant melanoma
Chronic liver disease	- Chronic active hepatitis (B and C) - Cirrhosis α-1-antitrypsin deficiency
Miscellaneous	- Thrombotic microangiopathy - Sickle cell disease - Partial lipodystrophy - Transplant glomerulopathy - Niemann-Pick disease (type C)

PATHOGENESIS

Complement plays a central role in the pathogenesis of IC-MPGN and C3G. The enhanced rate of C3 activation is evidenced by low serum C3 and increased levels of MAC/C5b-9 seen in patients with C3G and MPGN.⁴²³ Control of C3 convertase is normally regulated by the inhibitors: complement factor H (CFH), factor I (FI), and membrane-cofactor protein (MCP). It is now believed that central to the pathogenesis of these diseases is dysregulation of the alternative pathway (AP)

C3 convertase in the circulation via one or more of the following mechanisms:

- Autoantibodies to components of the AP C3 convertase such as C3 nephritic factor stabilizing this complex, thus prolonging its natural decay.
- Mutations in or autoantibodies to CFH resulting in the absence or loss of function of CFH.
- Mutations in C3 or CFB resulting in their gain of function with an exceedingly stable AP C3 convertase.

Overactivation of the C3 convertase leads to the deposition of C3 degradation products, such as C3b and iC3b, in the kidney, along with downstream products like C3d, which likely play a critical role in the development of C3G. However, the possibility that C3G may partially result from impaired clearance of these C3 degradation products from the glomeruli, especially in patients without clear signs of systemic or local complement alternative pathway (CAP) activation, warrants further investigation.⁴²⁴ In <10% of cases, dysregulation of AP C3 convertase is due to inactivating mutations in the genes coding for CFH, FI, or MCP or activating mutations in the genes coding for the two main components of the C3 convertase, C3 and factor B.⁴²⁵

The complexity of the complement system and increasing understanding of the pathogenesis and availability of targeted therapies highlight the crucial role of complete genetic and biochemical analyses in the advanced management of IC-MPGN/C3G patients.

CLINICAL MANIFESTATION

MPGN and C3G (C3GN and DDD) are rare diseases with an annual incidence of 1 to 2 per million (both pediatric and adult).^{237,426} The median age of affected patients spans from 12 to 15 years (DDD) up to 28 years, with approximately 38% to 40% of cases manifesting the disease during childhood, with a slight male predominance.⁴²⁵ The clinical manifestations of MPGN and C3G typically include hematuria, proteinuria (up to nephrotic range), hypertension, and impaired kidney function.^{427,428} There is notable symptom overlap between different subtypes of the diseases. Classic MPGN often manifests with features of both acute nephritic syndrome and nephrotic syndrome. Nephritic phenotypes are common in early-stage presentations with proliferative lesions on biopsy, while nephrotic phenotypes are more prevalent in advanced cases with repair and sclerosis. Crescentic MPGN is associated with rapidly progressive glomerulonephritis (RPGN).⁴¹⁴ Hypocomplementemia is a key feature in MPGN, with low levels of both C3 and C4 typically seen in IC-MPGN and low C3 and normal C4 levels seen in C3 glomerulopathy and alternative-pathway dysfunction. It should be noted that a normal C3 level does not necessarily exclude the possibility of alternative-pathway dysfunction.^{414,428} The risk of ESKF is comparable in C3G and IC-MPGN (4%–41% vs. 9%–41%).⁴²⁹

Extrarenal features may be associated with C3G (DDD and C3GN) and MPGN. Ocular abnormalities including drusen-like deposits, which appear similar in both the Bruch membrane and the glomerular basement membrane, have been described with an associated risk of long-term

visual impairment of approximately 10%.^{430,431} Additionally, acquired partial lipodystrophy (aPL), which is characterized by the loss of subcutaneous fat in the upper half of the body, may precede the renal manifestations of C3G and may be associated with decreased C3 levels and the presence of C3NeF.^{432,433}

DIAGNOSIS

MPGN and C3G most commonly present as acute nephritis or NS, though there is considerable clinical overlap with presentations of other forms of renal disease. Complement analysis (e.g., C3 and C4 levels, CFB levels, C3b breakdown products, and soluble C5b-9) may be useful; however, normal results do not exclude the diagnosis.⁴²³ The measurement of nephritic factors or other complement autoantibodies (e.g., anti-CFH, CFB, and C3b) may also be supportive; however, these assays are neither standardized nor widely available and lack diagnostic specificity.⁴²⁴ Genetic complement testing may be valuable and is—particularly in the background of the increasing number of identified disease-causing mutations in the CFHR locus—highly recommended and discussed further in [Chapter 36](#). Ultimately, the diagnosis is finalized by renal biopsy. The renal pathologic characteristics on biopsy vary significantly between subtypes of MPGN. Additionally, within the same subgroup of MPGNs, and even in a given patient, these features can differ from one kidney biopsy to a subsequent one. Kidney biopsy findings and eGFR are the best predictors of disease progression.⁴²⁵

The differential diagnosis of a setting of nephritic/NS with hypocomplementemia (in particular, low C3 and C4 levels) is narrow and mainly includes IC-MPGN/C3G and SLE. Features that support a diagnosis of SLE include arthralgias; rashes; and neurologic, hepatic, lung, or cardiac involvement combined with laboratory evidence of immune dysregulation including antibody-mediated hemolysis, leukopenia, coagulopathy, myositis, or hepatitis. Postinfectious glomerulonephritis (PIGN) represents an acute form of C3G, possibly accompanied by elevated antistreptolysin (ASO) titer or anti-DNase titers, which may resolve spontaneously. Some patients initially present with PIGN and later develop C3G, necessitating careful differentiation during the diagnostic process.⁴²⁴ Persistence of low C3 for >3 months in a child thought to have PIGN should prompt consideration of C3G and kidney biopsy. As MPGN and C3G may present with normal complement levels, they should also be considered in any differential diagnosis for glomerular disease even with normal C3 and C4 levels. CFHR5 nephropathy, caused by a heterozygous mutation in the *CFHR5* gene, leads to autosomal dominant inheritance of glomerulonephritis and kidney failure and has so far been reported only in individuals from Cyprus.⁴³⁴ However, it should also be considered in patients outside Cyprus who present with persistent microscopic hematuria and synpharyngitic gross hematuria, with or without a family history of ESKF.

TREATMENT

Currently, there is no universally effective treatment for primary C3G and IC-MPGN, and current management

strategies aim to control renal inflammation and target symptoms. General management is similar to other proteinuric glomerulopathies including a low-salt diet, treatment of hypertension, and treatment of dyslipidemia. Additionally, ACEi or ARBs are used in many patients with IC-MPGN/C3G due to their antiproteinuric and antihypertensive effects, as well as their association with better renal survival.^{423,435} Treatment of arterial hypertension and CKD/ESKF follows established standards.

Current KDIGO guidelines suggest a trial of mycophenolate mofetil (MMF) with low-dose oral glucocorticoids for patients with moderate severity primary C3G described as proteinuria >0.5 g/day despite supportive therapy or moderate inflammation on renal biopsy or abnormal kidney function.²¹⁰ Among pediatric patients with MPGN I treated with MMF and prednisone for a mean of 40 months, 56% achieved complete or partial remission.⁴¹³ However, treatment failed in all patients with decreased C3 levels at diagnosis.^{414,436,437}

Various alternative treatment protocols, including CNIs, cyclophosphamide, intravenous methylprednisolone boluses, monoclonal antibodies, and plasmapheresis, have been suggested for patients who do not respond initially. However, there are limited data available regarding the outcomes of these approaches.

CNIs have been used with variable success. One study demonstrated efficacy of the combination of low-dose steroids and CNI in patients with refractory MPGN with respect to reduction of proteinuria and decline in renal function in 94% after 2 years.⁴³⁸ In five adult steroid and cyclophosphamide-resistant MPGN patients, the combination of steroids and tacrolimus resulted in complete or at least partial remission in all patients after 24 weeks or longer.^{439,440} In two children with MPGN and suboptimal response to long-term prednisone, a rapid and complete remission was achieved with tacrolimus.⁴⁴¹ In another two patients with DDD, low-dose prednisone and cyclosporine A were able to induce remission.^{442,443} In another study no treatment benefit of CNI was found in DDD patients.⁴³¹

Rituximab has been suggested for treatment of C3G and Ig-MPGN, although this too has not shown consistent benefit. Several case reports of MPGN (in particular, IC-MPGN) patients describe partial or complete remission after administration of rituximab (in 50% combined with steroids).⁴⁴⁴⁻⁴⁴⁸ By contrast, two DDD patients did not show any improvement in proteinuria or renal function but were rescued by eculizumab therapy.^{449,450}

Ecuzumab is an anti-C5 monoclonal antibody that prevents C5 activation and TP induction with the formation of the terminal complement complex (C5b-9).⁴⁵¹ It has also been investigated in treatment of C3G with mixed outcomes. A retrospective trial in France showed 23% of patients resulted in a global clinical response, 23% had a partial clinical response, and 54% had no response.⁴⁵² It has been suggested that ecuzumab may be beneficial in treating inflammatory aspects of the disease with no or limited impact on C3 and immunoglobulin deposits.⁴²⁵ Some patients with specific complement mutations or

dysregulation may benefit. Cost is usually a barrier. If successful, patients require lifelong vaccination and prophylaxis against meningitis.

Long-term outcome data for MPGN patients—in particular, applying the new classification—are limited. Disease progression occurs regardless of the underlying MPGN subtype in 40% to 50% of patients within 10 years of diagnosis.^{427,428} Renal function (eGFR), nephrotic-range proteinuria, hypertension, and age at presentation are negatively correlated with long-term renal outcome.^{423,426-428} In addition, the presence of glomerular crescents or DDD on biopsy are independent predictors of disease progression.⁴²⁶

Recurrence of MPGN and DDD post-transplantation has been well described with recurrence rates ranging from 10% to 100%.⁴⁵³ However, the impact of disease recurrence on allograft survival remains controversial. Applying the new classification, MPGN recurrence rates for renal grafts are reported as 43%, 55%, and 60% for MPGN, DDD, and C3GN, respectively.⁴⁵⁴ The impact on allograft survival was not reported.⁴²³ In another study with 13 transplants, all of the 6 DDD and 4 of the 7 C3GN patients had disease recurrence, with graft failure occurring in 3 patients with DDD and 3 with C3GN.⁴²⁶ Overall, 5-year allograft survival years was 69%. Recent studies have shown eculizumab therapy is associated with lower rates of recurrence of C3GN and allograft loss.⁴⁵⁵

POSTINFECTIOUS (POSTSTREPTOCOCCAL) GLOMERULONEPHRITIS

PATHOGENESIS

The MPGN disease spectrum also includes acute postinfectious (PIGN) or poststreptococcal glomerulonephritis (PSGN).⁴²¹ The underlying etiology of PIGN is an extrarenal infection triggering immune-mediated glomerular injury. PIGN is characterized by strong mesangial C3 with codominant IgG staining on immunofluorescence and formation of mesangial, subendothelial, and subepithelial humps on EM.⁴²¹ Of note, some patients present with C3 deposition only without IgG deposition and/or humps. Causative agents include, but are not limited to, *Streptococcus*, *Staphylococcus*, *Pneumococcus*, *Chlamydia*, *Mycoplasma*, herpes simplex virus, cytomegalovirus, Epstein-Barr virus, mumps, RSV, influenza, hepatitis B virus, toxoplasma, and *Plasmodium malariae*. Further discussion of infection-associated glomerulonephritis is in Chapter 34.

PSGN is the classic example of this relatively common condition with an estimated global incidence of 472,000 cases per year.⁴⁵⁶ PSGN typically follows 7 to 14 days after a β -hemolytic group A *Streptococcus pneumoniae* infection and is associated with an increased antistreptolysin (ASO) titer. Several strains of streptococci have been identified and classified as nephritogenic in association with nephritis-associated plasmin receptor (NAPr) and streptococcal pyrogenic exotoxin B (SPeB) proteins. Both NAPr and SPeB lead to activation of the AP resulting in low complement levels and promote circulating ICs formation, which deposit in the kidney subendothelium and cause an inflammatory response with leukocyte recruitment and further complement activation via both the CP and AP.⁴⁵⁶⁻⁴⁵⁸

Clinical Manifestations

PIGN is the most common cause of acute glomerulonephritis in childhood. Age at presentation is

typically between 4 and 14 years and rarely in children younger than 2. PIGN is more common in males. The latent period between onset of primary infection and PIGN is approximately 3 to 5 weeks for skin infections and 1 to 2 weeks for URTIs. PIGN presents clinically with acute nephritic syndrome with hematuria (macroscopic in a third of patients), hypertension (60%–80%), edema (90%), oliguria (<50%), low C3 levels, or nephrotic syndrome (2%–4%).^{459,460} PIGN usually resolves spontaneously and has a good prognosis in children.^{457,458} In rare cases, PIGN can also be associated with AKI, ESKF, and death.^{457,461-464}

Treatment

PIGN typically follows a course with spontaneous recovery and thus does not require specific treatment. Resolution of proteinuria occurs in weeks, whereas microscopic hematuria may resolve over months. In some cases, PIGN may progress without remission to crescentic glomerulonephritis.⁴⁶⁵ The KDIGO guidelines recommend that individuals affected by classic PSGN, which typically occurs 1 to 3 weeks after initial signs of pharyngitis or impetigo, should receive treatment with penicillin (or erythromycin if penicillin allergic).⁴⁶⁶ This treatment is advised even in the absence of persistent infections to reduce the antigenic load. Patients with nephritic syndrome should be treated with fluids and supportive management including the use of diuretics, antihypertensive medications, and, if necessary, renal replacement therapy with dialysis. The value of high-dosed glucocorticoids remains unproven. In most cases, PIGN caused by agents other than *Streptococcus* is self-limiting, and the evidence base for the treatments used is not robust.^{456,467}

THROMBOTIC MICROANGIOPATHY

Thrombotic microangiopathy (TMA) defines a spectrum of diseases, which manifest clinically with microangiopathic hemolytic anemia, nonimmune thrombocytopenia, and organ dysfunction. Central to the development of TMA is the activation and injury of the microvascular endothelium, which result in platelet and neutrophil recruitment, thrombus formation, inflammation, thromboembolic occlusion of the dependent microvasculature, and subsequent organ failure.⁴⁶⁸⁻⁴⁷⁰ Thrombotic thrombocytopenic purpura (TTP) and hemolytic uremic syndrome (HUS) are primary forms of TMA.⁴⁷¹ Further discussion of thrombotic microangiopathies can be found in Chapter 36.

TTP is defined by severe deficiency of ADAMTS13, a protease that cleaves von Willebrand factor (vWF) multimers upon release from endothelial cells.⁴⁷² Deficiency may be due to acquired autoantibodies or genetic mutations and results in unusually large multimers of vWF- and platelet-rich thrombi in capillaries and arterioles causing the clinical pentad of fever, purpura/hemorrhage with thrombocytopenia, hemolytic anemia, neurologic manifestations, and variable degrees of renal dysfunction.⁴⁷³⁻⁴⁷⁶

HUS classically presents as a triad of microangiopathic hemolytic anemia, thrombocytopenia, and renal impairment and may involve organ systems such as the brain, heart, lungs, gastrointestinal tract, and skin. HUS is associated with

multiple conditions as outlined in Table 71.18. There are four types of HUS:

1. HUS associated with infection, commonly implicated pathogens, including Shiga toxin-producing *E. coli* (STEC), *Streptococcus pneumoniae*, influenza A, H1N1, and HIV
2. HUS associated with secondary conditions such as hematopoietic stem cell or solid organ transplantation; malignancy; autoimmune diseases; drugs (most commonly quinine, cyclosporine, and tacrolimus); malignant hypertension; and preexisting nephropathy
3. HUS associated with cobalamin C defect
4. Atypical HUS (aHUS) due to dysregulation of alternative complement pathway and mutations of the gene, diacylglycerol kinase ϵ (DGKE)⁴⁷⁷⁻⁴⁷⁹

Previously, the term aHUS was applied to any TMA that was not TTP or STEC-HUS. It is now used to describe variants of TMA caused by abnormalities of the alternative complement pathway resulting in endothelial cell dysfunction and formation of microvascular thrombi.⁴⁷⁹ aHUS is associated with poorer clinical outcomes compared with infection-associated typical HUS, with a 25% mortality rate, approximately 50% of patients progressing to ESKF, significantly higher rates of renal and extrarenal morbidity, and high risk of postransplant disease recurrence.⁴⁸⁰⁻⁴⁸³

DIAGNOSIS

The similar presentation of aHUS with other conditions can pose a diagnostic challenge. Prompt diagnosis and early initiation of therapy are essential for both TTP and aHUS to avert end-organ damage. The diagnosis of TMA is defined by clinical criteria including hemolytic anemia, thrombocytopenia, and renal and extrarenal disease manifestations. The three most likely diagnoses are STEC-HUS, aHUS, TTP, and less commonly, secondary-HUS from genetic abnormalities or vitamin B₁₂ metabolism defects.^{436,484-486} Fig. 71.13 outlines the diagnostic algorithm. Chapter 36 provides an overview of complement activation markers relevant to the identification of diagnoses/conditions possibly underlying secondary TMA.

STEC-HUS has a peak incidence in children <5 years old and typically follows a specific clinical sequence: Diarrhea developing 2 to 12 days after ingestion of the disease-causing bacteria (e.g., *E. coli* O157:H7) is initially nonbloody but becomes bloody in 60% to 90% of cases after 1 to 3 days. Microbiologic identification of *E. coli* in the stool via culture or polymerase chain reaction (PCR), detection of antilipopolysaccharide (LPS) antibodies specific for disease-causing bacteria, and Shiga toxin (ST) detection in blood via PCR supports the diagnosis.^{468,487-489} TTP affects 0.1 per million children per year and is typically characterized by

Table 71.18 Spectrum of Thrombotic Microangiopathies

Thrombotic Microangiopathy (TMA)	Role of Complement	Diagnostic Test
Thrombotic thrombocytopenic purpura	Elevated C5b-9	Biochemistry
Hemolytic uremic syndrome	Transient activation possible	Biochemistry; if persistent C activation, consider genetics
STEC-HUS/eHUS	Continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
Atypical HUS		
Other hereditary forms of TMA	N/A	Biochemistry; if persistent C activation, consider genetics
DGKE		
INF2		
TMA postsolid organ transplantation	Transient or continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
De novo TMA		
TMA recurrence		
TMA posthematopoietic stem cell transplantation/bone marrow transplantation	Transient or continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
TMA associated with pregnancy	Transient or continuous (mutation and abs.)	Biochemistry and genetics and antibody screen
TMA associated with glomerulonephritis	Continuous C activation possible (mutation and abs.)	Biochemistry and genetics and antibody screen
TMA associated with drugs	Transient C activation possible	Biochemistry; if persistent C activation, consider genetics
TMA associated with metabolic disease	C activation possible	Biochemistry; if persistent C activation, consider genetics
TMA associated with infections (e.g., <i>Streptococcus pneumoniae</i> , human immunodeficiency virus, and influenza)	Transient C activation possible	Biochemistry; if persistent C activation, consider genetics
TMA associated with malignant hypertension	Transient C activation possible	Biochemistry; if persistent C activation, consider genetics

Persistent: after initial cause has resolved; Biochemistry: Complement activation tests (see Table 71.3).
eHUS, *Escherichia coli*-induced hemolytic uremic syndrome; STEC, Shiga toxin *Escherichia coli*.

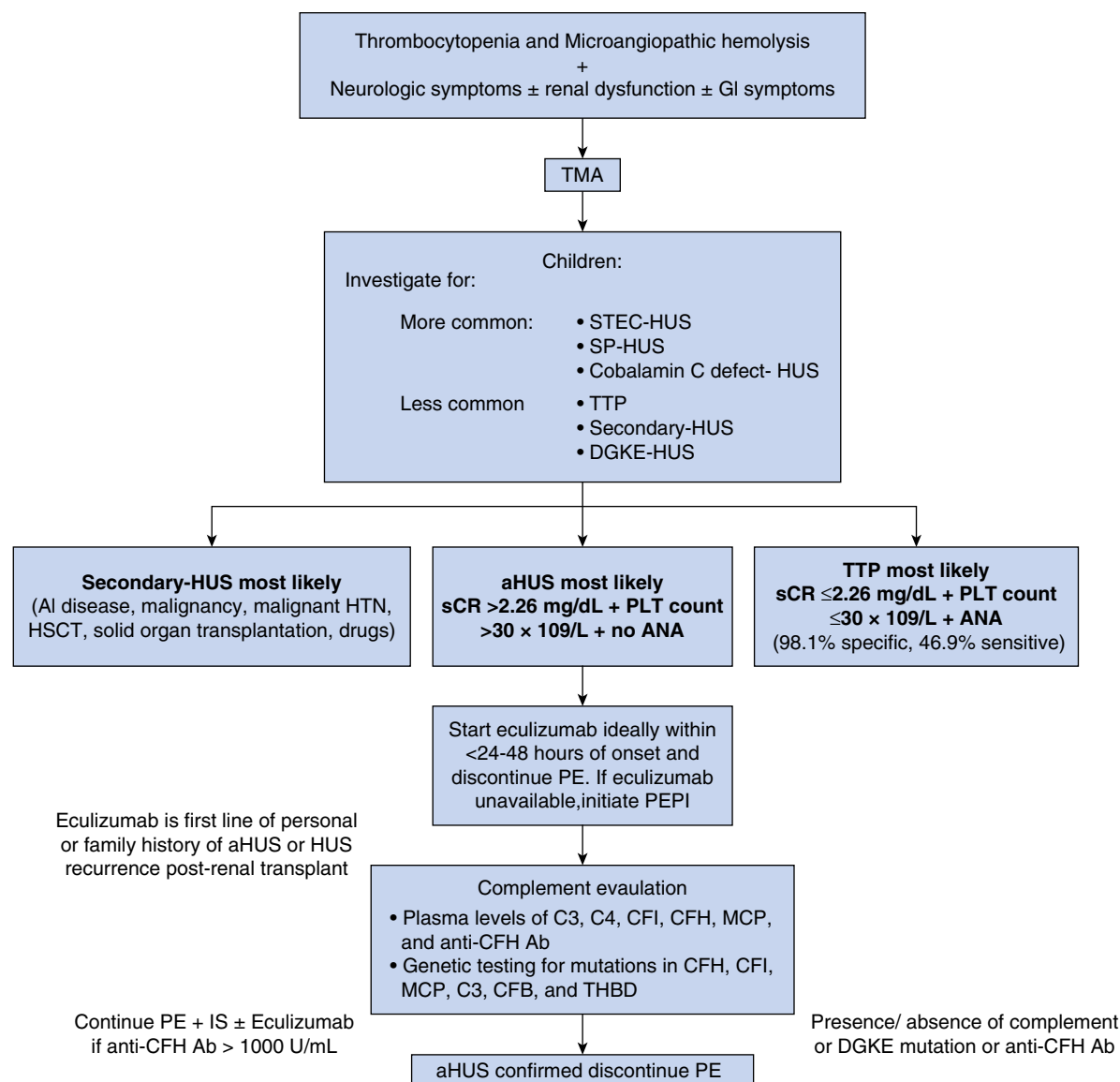


Fig. 71.13 Algorithm for workup of atypical hemolytic uremic syndrome. (From Raina R, Krishnappa V, Blaha T, et al. Atypical hemolytic-uremic syndrome: an update on pathophysiology, diagnosis, and treatment. *Ther Apher Dial.* 2019;23[1]:4–21.)

an ADAMTS13 activity level of <10%.⁴⁷⁰ Detection of anti-ADAMTS13 autoantibodies (inhibitors) or identification of an ADAMTS13 mutation confirms the diagnosis of TTP.^{473,474} Secondary-HUS must be considered in patients with associated conditions/drugs outlined in Table 71.18 in which the treatment is withdrawal of an offending drug or treating a triggering condition.

In contrast, aHUS is a diagnosis of exclusion, which can be made once secondary TMA, STEC-HUS, and TTP are ruled out, and it requires relatively comprehensive evaluation of the complement pathway. Enzyme-linked immunosorbent assays (ELISA) allow for the individual assessment of the complement CP, LP, and AP.^{490–492} This involves assessing blood levels of C3, factor H, I, and B, as well as screening for anti-factor H antibodies. Flow cytometry is employed to examine membrane-cofactor protein (MCP) (CD46). Additionally, next-generation sequencing is conducted

for genes, including CFH, CFI, CFB, C3, MCP, DGKE, and THBD.⁴⁷⁹

INFECTIOUS FORMS OF THROMBOTIC MICROANGIOPATHY

STEC-HUS

Pathogenesis. HUS is the most common cause of childhood AKI in the western world and is a common cause of acute kidney failure in children.⁴⁹³ Enterohemorrhagic Shiga toxin-producing *E. coli* (STEC) is the most common infectious agent causing HUS and has an incidence of approximately 2/100,000, with a peak of 6.1/100,000 in children younger than 5 years of age.^{493–495} Previously, it was thought that the O157:H7 STEC serotype was the most prevalent. However, other serotypes, such as O26, O55, O80, O103, O104, O111, and O145, are also known to cause STEC-HUS.⁴⁹⁶ In Europe

and North America, non-O157:H7 serotypes are now found to be as frequent as O157:H7. However, O157:H7 remains the predominant (>70%) serotype in Latin America, where STEC-HUS occurs up to 10 times more frequently elsewhere.⁴⁹⁴

Enterohemorrhagic *E. coli* (EHEC) is typically acquired through ingestion, leading to initial episodes of secretory diarrhea, which may progress to bloody diarrhea. The primary reservoir for STEC is the intestinal tract of (healthy) cattle, as well as the consumption of raw or undercooked beef, unpasteurized milk, juices, cheese, well water, and wading pools with person-to-person transmission possible.^{497,498} Shiga toxin production is the primary virulence factor in STEC, linked to HUS. Endothelial injury is the initial step in Stx-HUS pathogenesis. Diarrhea, including bloody diarrhea, results from STEC adhering to enterocytes via the intimin protein.

Clinical manifestations. STEC-HUS is characterized by a typical sequence of 1. ingestion of the STEC; 2. diarrhea, abdominal pain, fever, and vomiting; 3. bloody diarrhea; and 4. manifestation of HUS (approximately 15% vs. spontaneous resolution in approximately 85%). The incubation period ranges between 2 and 12 days.⁴⁸⁷ Ninety-four percent of patients show a diarrheal prodrome, with two-thirds having bloody diarrhea. Symptoms usually appear on days 6 to 7 after EHEC ingestion. Many children present with bloody diarrhea followed by fatigue, pallor, dizziness, petechiae and bruising, shortness of breath, edema, and decreased urine volume. Extrarenal manifestations include CNS involvement with neurologic symptoms, pancreatitis, elevated transaminases, acute respiratory distress syndrome, and cardiac involvement. The kidney is the most commonly involved organ, and dialysis is required in two-thirds of patients, with a median treatment duration of 10 days.^{493,499} At admission, most children have the complete triad of thrombocytopenia, hemolytic anemia (hemoglobin <10 g/dL, schizocytosis, undetectable haptoglobin, and increased lactate dehydrogenase levels), and AKI.⁴⁸⁹

Mortality is <5% in high-income settings, most often due to neurologic complications.⁵⁰⁰ Prognostic markers for poor clinical outcome and persistence of long-term sequelae include high white blood cell (WBC) count (>25,400/mL), hemoconcentration (hematocrit $\geq$ 20% at the acute stage), longer time on dialysis (median 15 days), anuria lasting more than 5 days, and arterial hypertension at disease onset.^{489,493,499}

It is recommended to screen for STEC in all patients, especially children, who present with HUS. EHEC can be isolated from stool specimens or rectal swabs, and the serotype can be determined by PCR or enzyme immunoassay. Stool samples for STEC testing should be collected promptly, as toxin excretion in infected patients generally lasts only a few days after the onset of diarrhea. Antilipopolysaccharide (LPS) antibodies can be detected in the serum, useful when stool screening for STEC is negative, often due to late stool collection or prior antibiotic therapy.^{489,501} Infection with EHEC belonging to one of the five traditional high-risk serogroups (O157, O26, O145, O111, and O103) or Stx2-producing EHEC is associated with a higher risk of developing HUS. Nonsorbitol-fermenting O157:H7/H- is the most common serotype (50%), but non-O157 serotypes

are emerging and occur more often in patients 1 year of age.^{390,399} In patients with positive testing for EHEC, Stx, or LPS antibodies, the diagnosis of STEC-HUS is confirmed and no further investigations are required.⁵⁰² In children with bloody diarrhea but negative EHEC, Stx, or LPS testing, STEC-HUS still remains the most likely diagnosis. In patients without bloody diarrhea and negative testing, other differential diagnoses for TMA/aHUS should be considered.

Treatment. There is no specific treatment for STEC-HUS, thus therapy is focused on supportive care including fluid management, treatment of arterial hypertension, dialysis, and, if indicated, ventilatory support. In children, supportive fluids and strict fluid status monitoring are critical as dehydration may be exacerbated by fluid losses in diarrhea and vomiting. Routine use of antibiotics has historically not been recommended due to suspicion of their association with a higher risk of HUS and currently the role of antibiotics in STEC-HUS remains unclear. Karch and colleagues⁵⁰³ showed trimethoprim-sulfamethoxazole increased the release of Shiga toxin by cultured *E. coli* O157:H7 and Wong and colleagues⁵⁰⁴ showed that children who received trimethoprim-sulfamethoxazole or β -lactams had a significantly higher risk of HUS than those who did not receive antibiotics. In contrast, research by Nitschke and colleagues⁵⁰⁵ showed azithromycin treatment in the acute phase shortened STEC carriage and Ikeda and colleagues⁵⁰⁶ have suggested fosfomycin treatment within 2 days of the appearance of the first symptom of *E. coli* O157 infection may prevent the development of HUS.

Eculizumab is a monoclonal antibody that targets complement C5b and has shown potential in treating STEC-HUS. In a systematic review in 381 patients who received eculizumab for STEC-HUS, there was no significant positive medium to long-term outcome.⁵⁰⁷ However, there was a risk of bias because only severely ill patients received eculizumab. In phase 3 clinical trial, 100 children received eculizumab for STEC-HUS, with no significant difference in KRT rates but a statistically significant lower percentage of children experiencing renal sequelae at 1 year.⁵⁰⁸

Neurologic sequelae were reported in 4% and proteinuria in 18% of patients after 1-year follow-up, and arterial hypertension was present in 9% of patients after 5-year follow-up.⁵⁰⁹ Long-term follow-up data indicate that some asymptomatic patients after 1 year will develop hypertension or proteinuria 2 to 5 years after first onset of HUS.⁴⁹⁹ More than 25% to 30% of STEC-HUS patients who do not fully recover from the acute disease experience long-term renal sequelae including CKD.⁵¹⁰ This important finding warrants yearly follow-up in all patients at least up to 5, potentially up to 10 years, to monitor for proteinuria, decline in renal function, and hypertension.

Pneumococcal HUS

Pathogenesis. Following EHEC-associated HUS, pneumococcal HUS (P-HUS) accounts for around 5% to 15% of all infection-associated HUS.^{468,511} P-HUS is caused by infections with *S. pneumoniae* (P-HUS), and the most frequent serotype associated with TMA is *S. pneumoniae* 19A. *S. pneumoniae* contains many virulence factors such as autolysin, which

aids in the spread of cell wall components and enzymes including pneumolysin. Pneumolysin causes hemolysis and contributes to anoxia.⁵¹² Pneumococcal neuraminidase is an enzyme responsible for cleavage of N-acetyl neuraminic acid from cell surfaces and thereby exposure of the so-called Thomsen–Friedenreich antigen (T-antigen) on the surface of various cell types including vascular endothelial cells and erythrocytes. Historically, it was hypothesized to be the main contributor of pneumococcal HUS via antibody binding to T-antigen resulting in Coombs positive hemolysis (i.e., erythrocytes), cellular injury (e.g., vascular endothelial cells), and subsequent thrombosis with the clinical picture of HUS.⁴⁶⁸ This has led to caution in administration of fresh frozen plasma to children with P-HUS as donor plasma can contain anti-T antibodies.⁵¹³ However, further studies have suggested pneumococcal neuraminidase is only partially responsible for development of HUS.^{514,515} Meinel and colleagues⁵¹⁶ reported increased expression of specific variants of the bacterial pneumococcal surface protein C (PspC) in HUS patients, which may contribute to disease by binding plasminogen.

Clinical manifestations. P-HUS may develop in patients with severe pneumococcal infections including septicemia and pulmonary or pleural infection and in 20%–30% of cases, with meningitis, typically presenting 3 to 13 days after a pneumococcal infection.⁵¹⁷ In contrast to STEC-HUS, individuals with SP-HUS are typically younger, experience more severe initial hospital stays, have prolonged episodes of oliguria, necessitate more transfusions, and have longer periods of thrombocytopenia.⁵¹⁸ The overall mortality rate of SP-HUS is 2% to 12%, although in patients who present with meningitis, mortality rates are as high as 37%.⁵¹⁸ ESKF was observed in 10% to 16% of patients, while others experienced persistent proteinuria and hypertension.⁵¹⁸ To date, no instances of disease recurrence have been reported. Vaccination against *Pneumococcus* may reduce the risk of pneumococcal HUS.⁵¹⁹

Treatment. Treatment of patients with P-HUS includes infection-specific treatment (e.g., antibiotics and antiviral drugs) and supportive care (e.g., KRT and ventilatory support). Dialysis is required in 43% to 84% of patients with P-HUS.⁵²⁰ The American Academy of Pediatrics suggests initiating treatment for invasive pneumococcal infection with vancomycin and an extended-spectrum cephalosporin where resistant *Pneumococcus* species are prevalent. Antibiotic choices may be made based on culture results or local pneumococcal susceptibilities, if necessary.⁵²¹ Patients with P-HUS exhibit dysregulated complement activity, suggesting a potential benefit from eculizumab. Case series and individual reports have documented mixed outcomes in severe P-HUS cases after eculizumab administration, and this is an area of ongoing research.⁵¹⁹ Eculizumab may not improve the course of P-HUS, with kidney outcomes strongly correlated to the duration of dialysis and mechanical ventilation instead.⁵²²

NONINFECTIOUS, NONCOMPLEMENT-MEDIATED FORMS OF THROMBOTIC MICROANGIOPATHY

Over the past 20 years, extensive genetic screening has identified the underlying genetic cause in about 50% of patients with

TMA.⁵²³ Newer techniques such as whole-exome sequencing and whole-genome sequencing have allowed for a broader approach to genetic screening, and several mutations in genes not associated with the complement system were discovered in patients and families with aHUS/TMA. Although their prevalence is low, genetic testing applying next-generation sequencing should be considered in patients without complement alterations, who are unresponsive to complement-targeted therapy, such as eculizumab, and in patients presenting with specific clinical characteristics. Identification of the underlying genetic defect will guide treatment decisions and inform discussions about inheritability, long-term outcomes, transplantation, and posttransplant recurrence risk. Table 71.19 highlights the approach to various forms of TMA in children. Further discussion on workup and therapies for complement-related disorders can be found in Chapter 36.

Diacylglycerol Kinase Epsilon Thrombotic Microangiopathy

Pathogenesis. A small proportion of patients with aHUS have homozygous or compound-heterozygous variants in diacylglycerol kinase epsilon (DGKE).⁵²⁴ DGKE aHUS is rare, with an estimated incidence of 0.009/million/year.⁵²⁵

DGKE plays an important role in cellular signaling by phosphorylating diacylglycerol, which activates protein kinase C (PKC). This activation triggers a cascade of downstream effects, including alterations in vascular tone, the release of both prothrombotic and antithrombotic factors, platelet activation, and modifications in the actin cytoskeleton. DGKE is present in podocytes, and it is hypothesized its deficiency leads to activation of podocyte calcium channels with rearrangement of the actin-cytoskeletal structure, reduced VEGF receptor expression, and downregulation of the decay accelerating factor.^{524,526} The precise involvement of complement activation in DGKE nephropathy remains uncertain. While certain patients have exhibited markers indicative of complement activation, two individuals experienced disease relapses even after receiving complement-inhibiting therapy.⁵²⁵

Clinical Manifestations. DGKE aHUS patients typically present within the first year of life with clinical and histologic features of TMA, hypertension, and progressive proteinuric nephropathy with 20% requiring dialysis or transplant within 10 years of diagnosis.⁵²⁵ In a case series of 12 patients, 10/12 patients had nephrotic syndrome and 1/12 patients had a membranoproliferative pattern of glomerular injury (MPGN).^{524,527} Disease recurrence was observed in 70% of patients with approximately half of these patients experiencing multiple relapses.⁵²⁶ Persistent proteinuria is common in DGKE TMA, with proteinuria persisting beyond resolution of acute TMA in up to 80% of patients (up to nephrotic range in 24%). Of note, 88% of patients also showed persistent hematuria during follow-up. Progression to ESKF occurred in 7 of 35 patients, with a median age of 11 years. Ten-year renal survival was reported as 89%.⁵²⁴

Treatment. Supportive therapy is the current mainstay of DGKE aHUS, which has been associated with improvement of renal function. Although the optimal treatment of DGKE aHUS is not yet clear, complement-inhibiting therapy is

Table 71.19 Thrombotic Microangiopathies and Treatments in Children

Thrombotic Microangiopathy (TMA)	Diagnostics	Treatment
Thrombotic thrombocytopenic purpura	Diagnosis is established by a severe ADAMTS13 deficiency (plasma activity <10 %) in the absence of inhibitory antibodies in the blood. Molecular studies for a pathogenic mutation in the <i>ADAMTS13</i> gene confirm the diagnosis	Plasma exchange, caplacizumab; steroids/rituximab; bortezomib if refractory
Hemolytic uremic syndrome STEC–HUS/eHUS	Suggested to confirm the diagnosis by detection of fecal Shiga toxin, as well as by culture. Atypical HUS is a diagnosis of exclusion.	Supportive therapy with dialysis and initiation of appropriate antibiotic therapy in STEC–HUS. Plasma exchange, pulse steroid therapy, and eculizumab can be considered.
Atypical HUS	Genetic testing for complement mutations and anti Factor H antibodies is indicated if available	Complement inhibitor therapy or plasma exchange may be indicated
Other hereditary forms of TMA DGKE INF2	Genetic testing	If persistent complement activation, consider genetics. Only nephroprotective strategies are available.
TMA postsolid organ transplantation De novo TMA TMA recurrence	Diagnostic approach and investigations are similar to the ones discussed for aHUS.	Supportive measures that may include withdrawal or minimization of potential triggering agents (i.e., calcineurin inhibitors), hypertension, eculizumab, and defibrotide.
TMA posthematopoietic stem cell transplantation/bone marrow transplantation	Features are similar to other forms of TMA including hemolytic anemia, thrombocytopenia, hypertension, and renal dysfunction	Supportive measures that may include withdrawal or minimization of potential triggering agents (i.e., calcineurin inhibitors), treatment of coexisting conditions such as infections and GVHD, managing hypertension, eculizumab, and defibrotide.
TMA associated with pregnancy	Abnormal uterine Doppler analysis, low circulating PlGF, and an elevated sFlt-1:PlGF ratio may aid in confirming the diagnosis	Plasma infusion, plasma exchange, immunosuppression
TMA associated with glomerulonephritis	Measurement of complement levels	Eculizumab or ravulizumab; plasma exchange; rituximab or cyclophosphamide
TMA associated with drugs	High suspicion in patients taking high-risk drugs	Cessation of causal drug
TMA associated with metabolic disease	Genetic studies	Cobalamin
TMA associated with infections (e.g., <i>Streptococcus pneumoniae</i> , human immunodeficiency virus, influenza)	Identifying underlying infection	Treating underlying infection
TMA associated with malignant hypertension	In patients presenting with malignant hypertension and TMA, diagnostic and treatment algorithms should be followed as detailed for aHUS	Blood pressure control

Modified from Genest DS, Patriquin CJ, Licht C, John R, Reich HN. Renal thrombotic microangiopathy: a review. *Am J Kid Dis*. 2023;81(5):591–605 and Mangani M, Kini P. Thrombotic microangiopathy in children: redefining hemolytic uremic syndrome, thrombotic thrombocytopenic purpura and related disorders. *Pediatr Hematol Oncol J*. 2024;9(1):45–53.

likely not beneficial. Patients presenting with only nephrotic-range proteinuria may benefit from corticosteroids. Patients with DGKE aHUS who progress to ESKF should undergo kidney transplantation, with no reported cases of remission post-transplant.

Cobalamin C–Associated Thrombotic Microangiopathy

Pathogenesis. Cobalamin C (CblC) deficiency is the most commonly inherited error of vitamin B₁₂ metabolism. A

deficiency of Cbl and its metabolites can arise from genetic mutations or other factors impacting its intake, absorption, or storage. Molecular defects in MMACHC (methylmalonic aciduria and homocystinuria type C protein) cause autosomal recessive CblC deficiency. Deficiency of Cbl or dysregulation of its metabolism results in hyperhomocysteinemia, decreased plasma methionine level, and methylmalonic aciduria, which generate reactive oxygen species leading to endothelial dysfunction, platelet activation, and increased tissue factor expression.^{528,529}

Clinical Manifestations. CblC deficiency is characterized by multisystem involvement with severe neurologic, hematologic, renal, and cardiopulmonary manifestations.^{530,531} Onset of symptoms typically commences in infancy with failure to thrive, macrocytic anemia, macular degeneration, developmental delay, muscular hypotonia, microcephaly, seizures, and other neurologic manifestations. The most common renal manifestation is TMA, though tubulointerstitial nephritis and proximal renal tubular acidosis have been reported.⁵²⁹ Infantile onset of CblC disease is associated with poor outcomes and has a mortality rate of 30, which increases to 56% with TMA.⁵³² Involvement of the lungs, the need for dialysis, and very early disease onset are negative prognostic markers.⁵³¹

Treatment. Timely diagnosis is crucial as it guides treatment. Cobalamin-mediated TMA is treated by systemic administration of hydroxocobalamin, folinic acid, betaine, and carnitine.⁵³² Patients who do not respond to oral cyanocobalamin should be treated with intramuscular hydroxocobalamin.⁵³³ Because CblC deficiency is not complement mediated, eculizumab is not beneficial except for patients with additional complement abnormalities.⁴⁶⁸

Interferon 2-Mediated Thrombotic Microangiopathy

Mutations in interferon-2 (INF2) are associated with TMA, often presenting with FSGS. INF2 is a ubiquitously expressed formin protein, which accelerates actin polymerization and depolymerization, thus regulating a range of cytoskeleton-dependent cellular functions including the secretory pathway.^{494,534} It is hypothesized that impaired VEGF secretion from podocytes results in endothelial injury leading to the formation of platelet microthrombi and a thrombotic microangiopathy.⁵³⁴ INF2-associated TMA has been described in two families presenting with proteinuria and progressive renal failure with a varied range of severity. Additionally, mutations in INF2 are the commonest cause of familial autosomal dominant nephrotic syndrome, often resulting in a syndromic form of FSGS associated with the demyelinating peripheral neuropathy, Charcot-Marie-Tooth (CMT).⁵³⁵ Treatment with complement-targeted therapy including plasma and eculizumab is not beneficial in these patients.⁵³⁴

NONINFECTIOUS, COMPLEMENT-MEDIATED THROMBOTIC MICROANGIOPATHY

Atypical HUS

Pathogenesis. aHUS is a rare variant of TMA that is caused by alternative complement pathway dysfunction resulting in endothelial cell impairment and formation of microvascular thrombi. In aHUS, the alternative complement pathway becomes excessively activated, either as a result of the production of FH autoantibodies or due to mutations in key complement proteins such as FH, FI, FB, C3, and thrombomodulin.⁵³⁶ CFH mutations account for 15% to 25% of aHUS cases.⁵³⁷ In individuals with mutation in *CFH*, *CFHR3*, *MCP*, *CFI*, *CFB*, and *C3* genes, a secondary trigger (second

hit) is usually necessary for aHUS to manifest, such as pregnancy, viral infection, cancer, organ transplantation, or specific medication use.⁵³⁸ The primary mechanism underlying familial aHUS is complement dysregulation at the cell surface. C3 deposits in the glomeruli and arterioles lead to further activation of complement and local C3 utilization amplifying the injury loop. Activation of the MAC (C5b-9) results in microvascular thrombosis, especially within the kidneys.⁴⁷⁹

In aHUS patients, complement mutations have also been described as combinations of two or even three defects.⁵³⁹⁻⁵⁴⁴ In a U.S. cohort, 12% of patients had mutations in more than one gene, and in a Dutch (pediatric) cohort 9% of patients carried combined mutations.^{540,543,545} The number of patients with combined mutations in other complement genes may be as high as 25%.⁵⁴⁶ In addition, genetic haplotypes and SNPs in complement genes have been identified in aHUS patients and may act as disease-susceptibility factors.^{547,548} These variations can be additive,⁵⁴⁹ and a single SNP—for example, in MCP (CD46)—is unlikely to be sufficient to cause disease.^{550,551} Evidence has shown that the combination of complement gene risk haplotypes can increase disease penetrance.⁵⁵² Abnormalities in the *C3* gene were associated with an increased (>twofold) risk of disease, whereas CFI and CD46 gene mutations, besides being responsible for less severe disease, are also less likely to recur in family members. These disorders are discussed in more detail in Chapter 36.

Clinical Manifestations. aHUS is rare with an incidence of 0.23 to 1.9 per million and a prevalence of 2 to 10 per million population annually, depending on region and age group.⁵⁵³ Data from the global aHUS registry reveal a two-phase pattern in disease manifestation.⁵⁵⁴ In individuals aged 30 and older, females show a higher prevalence, whereas in younger age groups, males predominate (with a median onset age of 10.0 years in males compared with 25.6 years in females). It is hypothesized that the biphasic pattern found in females might reflect the enhanced risk for aHUS manifestation caused by pregnancy. Patients with CFI mutations typically manifested symptoms in adulthood, whereas those with MCP/CD46 mutations tended to do so in childhood. Among the pediatric cohort, patients with anti-CFH autoantibodies presented at a notably older age than those with other complement abnormalities (median age of 6.4 years). Notably, between the ages of 6 and 17 years, anti-CFH autoantibodies were the most prevalent cause of aHUS.⁵³⁹

Besides renal manifestations, up to 27.2% of pediatric aHUS patients display neurologic complications, 7% display cardiovascular complications, and 10% to 80% present with gastrointestinal complications.⁵⁵⁵ Children are more likely to experience gastrointestinal symptoms compared with adults (47% vs. 33%, respectively).⁵³⁹ Hypertension was present at first presentation in 71% of patients and in 84% of patients, a triggering event precipitated aHUS, such as gastrointestinal symptoms (74%), URTI (45%), or fever (32%).⁵⁴³

Serum creatinine at first presentation is a significant predictor of the 1-year outcome.⁵⁴⁴ Renal outcome is closely tied to the age of initial disease manifestation.⁵³⁹

Treatment. Treatment of aHUS includes supportive care (dialysis, antihypertensive medications, fluid and electrolyte management), plasma therapy (plasma infusion or plasma exchange, PLEX), and more specific complement-targeted therapy (anti-C5 antibody eculizumab).^{436,486,556} Treatment of aHUS caused by anti-CFH autoantibodies (DEAP-HUS: deficiency of CFH-related proteins and anti-CFH autoantibody-positive HUS) includes eculizumab, PLEX, and immunosuppression (i.e., steroids, cyclophosphamide, MMF, rituximab, and combinations thereof).^{436,486,557-561} For other cases of aHUS, the mainstay of treatment is C5 complement inhibitors and supportive treatment, with plasma exchange showing an unclear benefit. Before the advent of eculizumab, plasma therapy was the mainstay of treatment for aHUS. However, treatment efficacy was limited (about 50%) and risk of progression to ESKF was high (about 50%).⁵⁶²

Eculizumab blocks production of C5a and C5b-9, reversing acute aHUS, and maintenance therapy can prevent aHUS progression and recurrence. Eculizumab is also effective in preventing disease recurrence postrenal transplantation and may lead to some recovery of renal function even in chronic aHUS.^{563,564} Initiation of eculizumab within 24 to 48 hours of onset is associated with significantly greater recovery of kidney function.⁴⁸⁶ This therapy may be required long term, which is associated with significant cost and biweekly infusions. With eculizumab, event-free status and complete response were observed, respectively, in >92% and 50% to 85% of the aHUS patients who were respectively resistant to plasma therapy or dependent. More recently, the reengineering of the drug ravulizumab from eculizumab has shown promise in treating aHUS with a fourfold longer duration of action, thus reducing frequency of maintenance doses.⁵⁶⁵ Both therapies required updated vaccination against *Meningococcus*, and some centers continue antibiotic prophylaxis.

Mutations in genes encoding CFH, followed by mutations in CFI genes, have been most commonly reported in patients with aHUS recurrence after transplant. Less than 6% of patients who received eculizumab prophylaxis presented with aHUS recurrence, and recurrence was more likely to occur when eculizumab was discontinued. The KDIGO workgroup recommends the prophylactic use of eculizumab in kidney transplant patients at high risk of recurrence based on their genetic mutations.⁴³⁶

SECONDARY FORMS OF THROMBOTIC MICROANGIOPATHY WITH POSSIBLE COMPLEMENT CONTRIBUTION

Secondary TMA is a complication that can arise in various medical conditions including renal and stem cell transplantation, autoimmune diseases, and drug therapies. In transplant patients, both de novo and recurrent TMA can occur, often related to genetic mutations affecting the complement pathway, and treatment options include complement-targeted therapies like eculizumab. Patients with autoimmune conditions such as SLE and ANCA-associated vasculitis may also develop TMA, with complement dysregulation playing a central role in its pathogenesis. Drug-induced TMA is another subtype, often triggered by immunosuppressive or chemotherapeutic agents and managed through drug withdrawal and immunomodulatory

treatments. Across these conditions, early diagnosis, genetic testing, and targeted therapy are crucial to improving outcomes.

Thrombotic Thrombocytopenic Purpura

The pathophysiology of TTP stems from a severe deficiency of ADAMTS13 (a disintegrin and metalloproteinase with thrombospondin motifs 13) (activity <10%), which is the only biologic marker specific for TTP. This causes accumulation of platelet-hyper adhesive ultralarge VWF multimers, leading to the formation of platelet-rich microthrombi within small arterioles.⁵⁶⁶ ADAMTS13 deficiency can also result from homozygous or compound-heterozygous ADAMTS13 mutations, which constitute the hereditary or congenital form of TTP (cTTP; Upshaw-Schulman syndrome), or from inhibiting anti-ADAMTS13 autoantibodies, which constitute the acquired, immune-mediated form of TTP (iTTP).^{549,567-569} TTP is more common in adults and occurs with a standardized annual incidence rate of 2.88×106 per year in adults versus 0.09×106 per year in children.⁵⁷⁰ Adults are more likely to have iTTP, and children are more likely to have cTTP.^{474,476}

Acquired TTP occurs in around 1 per million children; occurs preferentially in girls, with a median age of 13 (range of 4 months to 17 years old); and has a mortality rate of up to 25%.⁵⁷¹ Without treatment, the mortality rate is up to 90%.⁵⁷² Acquired forms of TTP were found to be idiopathic in 55% of cases (median age of onset: 15 years) and associated with a clinical context in 45% of cases (median age of onset: 8 years) and largely related to anti-ADAMTS13 autoantibodies.^{573,574} In contrast, cTTP comprises approximately one-third of all childhood-onset TTP cases with a prevalence estimated at roughly 0.8 cases per million children. There is equal incidence between sexes, with initial presentation before 5 years old, often in the neonatal period.⁵⁷⁴ TTP is discussed in detail in Chapter 36.

Clinical Features. The classical pentad of TTP historically found in 88% to 98% of patients involves a nonimmune microangiopathic hemolytic anemia, thrombocytopenia, neurologic symptoms (often transient; ranging from headache or mental changes to focal signs, seizures, and coma), variable degrees of renal dysfunction and fever.^{474,476,575} Disease onset is often preceded by prodromal manifestations including fatigue, arthralgia, myalgia, and abdominal or lumbar pain. Cardiac involvement may include myocardial infarction, congestive heart failure, arrhythmias, cardiogenic shock, and sudden cardiac arrest. Digestive tract involvement may include abdominal pain, nausea, vomiting, and diarrhea.^{474,476} TTP carries a lifelong risk of recurrence often occurring within 1 to 2 years of initial presentation but may occur decades later. As with initial presentations, relapses are associated with decreased ADAMTS13 activity.⁴⁷⁶

It is critical to differentiate TTP from its major differential diagnosis of aHUS to initiate proper treatment. ADAMTS13 activity levels of <10% are considered highly suggestive of TTP and are generally used as a diagnostic guide.⁴⁸⁶ Additional diagnostic criteria include the degree of thrombocytopenia with platelet levels in TTP of typically $<30 \times 109/L$ (as opposed to aHUS with $>30 \times 109/L$), and serum

creatinine levels reflecting the degree of renal impairment being less elevated in TTP than in aHUS.^{474,476}

Treatment. Child-onset TTP is an emergency. Severe ADAMTS13 deficiency is required to confirm the diagnosis, but the test should not delay the initiation of treatment.⁵⁷⁶ First-line acute therapy for both idiopathic and congenital TTP is daily plasma replacement therapy, with plasma infusion or PLEX. These therapies replete ADAMTS13 levels; however, PLEX additionally removes anti-ADAMTS13 autoantibodies in acquired forms of TTP.⁵⁷⁴ Therapy is typically continued to achieve a normal platelet count ($\geq 150 \times 10^9/L$) for 2 consecutive days, alongside the normalization of LDH and notable clinical improvement.⁵⁷³

In addition to plasma replacement and/or exchange, therapy for acquired TTP management includes immunosuppression. Corticosteroids are recommended as an adjunct to plasma therapy to target antibody production to allow for recovery of circulating levels of ADAMTS13, with higher doses appearing to be more efficacious.^{577,578} Rituximab, a monoclonal antibody against CD20, is recommended for acute iTTP, as it not only lowers the relapse rate versus control but also reduces mortality and appears to be effective in patients with refractory TTP or poor response to plasma replacement therapy.⁵⁷⁹ ADAMTS13 replacement therapy in the form of recombinant ADAMTS13 has become available as a promising treatment strategy for both cTTP and iTTP. Recombinant ADAMTS13 effectively restores vWF cleaving activity in cTTP.⁵⁸⁰ Caplacizumab, a nanobody targeting the A1 domain of von Willebrand factor (VWF), inhibits platelet adhesion. There is a suggestion that its inclusion in the standard initial treatment regimen, alongside PEX, corticosteroids, and rituximab, may confer additional benefits.^{579,581}

TUBULAR DISORDERS

TUBULOINTERSTITIAL DISEASE

Acute tubulointerstitial nephritis (AIN) is a relatively uncommon cause of AKI in children, accounting for <10% of cases, and is characterized by inflammation of the renal interstitium.⁵⁸² Known causes include hypersensitivity reactions to medications, infections, and autoimmune conditions like TINU (tubulointerstitial nephritis with uveitis). Most patients present with nonspecific symptoms, such as vomiting, lethargy, and joint pains, alongside severe renal impairment.

Tubulointerstitial nephritis and uveitis (TINU) syndrome was first described in 1975, with fewer than 200 identified cases in the world literature.⁵⁸³ It is defined as the occurrence of tubulointerstitial nephritis and uveitis in a patient in the absence of other systemic diseases that can cause either interstitial nephritis or uveitis. It is a diagnosis of exclusion. Diagnosis is suggested with evidence of leukocyturia and tubular dysfunction including glycosuria, phosphaturia, and tubular proteinuria. Ophthalmologic examination should be performed and may reveal uveitis, which requires independent treatment and follow-up. The current hypothesis is that TINU arises from an interaction of an

environmental trigger (such as a drug or rarely an infection) with a susceptible genetic background and that this triggers an autoimmune cascade, although many cases are likely idiopathic. Most series suggest that TINU only accounts for 0.1% to 2% of patients seen in specialized uveitis centers, but the syndrome is likely to be underdiagnosed.⁵⁸³

Treatment of interstitial nephritis typically involves immunosuppressive therapy, such as intravenous or oral corticosteroids, and most patients showed improvement in renal function, though 70% had an eGFR of <90 mL/min/1.73 m² at last follow-up.⁵⁸⁴

CYSTIC KIDNEY DISEASE

Cystic kidney diseases in children range from 0.44 to 4.1 cases per 10,000 births.⁵⁸⁵ They are related mainly to dysfunction of primary cilia, located on the apical surface of the renal tubular epithelial cells, which result in structural and/or functional disruptions to the cilia contributing to kidney cyst formation and abnormal kidney architecture. Cystic kidney diseases are discussed in more detail in [Chapter 45](#).

AUTOSOMAL DOMINANT POLYCYSTIC KIDNEY DISEASE

Autosomal dominant polycystic kidney disease (ADPKD) is the most common genetic cystic kidney disease. Genes involved in ADPKD include *PKD1*, *PKD2*, and less commonly *ALG5*, *ALG8*, *ALG9*, *DNAJB11*, *GANAB*, *IFT140*, and *NEK8*.⁵⁸⁶ Prognosis is based on allelic heterogeneity, genetic complexity, and the presence of cystic kidney disease modifier genes. ADPKD can be diagnosed in children due to symptomatic presentation, family screening, or incidental findings during abdominal imaging performed for unrelated reasons. Symptomatic presentations of ADPKD in children include nocturnal enuresis and polyuria due to impaired urinary concentration, systemic hypertension (affecting 20% of children), abdominal or back pain, and, less commonly, kidney cyst infections or gross hematuria.⁵⁸⁶ Although children with classic ADPKD typically maintain normal kidney function, hyperfiltration is observed in around 20%.⁵⁸⁷ Management of ADPKD in children is currently limited to supportive measures including blood pressure control, high water intake, salt intake restriction, maintaining a normal body mass index, and physical activity. Monitoring of the cyst burden and kidney length by ultrasonography is indicated in symptomatic children with ADPKD. ACE inhibitors or ARBs are used to manage hypertension. Tolvaptan has shown promise in slowing total kidney volume growth and kidney function decline, with clinical trials in children.⁵⁸⁸ Monitoring for extrarenal manifestations such as intracranial aneurysms is indicated in older children/late adolescence or those with strong family histories.

AUTOSOMAL RECESSIVE POLYCYSTIC KIDNEY DISEASE

Autosomal recessive polycystic kidney disease (ARPKD) is a rare cystic kidney disease that is generally diagnosed in the neonatal or infancy period, less likely during childhood, and rarely during adulthood. It is

due to mutations in *PKHD1* and *DZIP1L*, which affect fibrocystin/polyductin protein, leading to defects in tissue differentiation and cell proliferation in the kidney, liver, and pancreas.⁵⁸⁹ Severe cases of ARPKD manifest prenatally with enlarged hyperechoic kidneys, oligohydramnios, or anhydramnios, leading to the Potter sequence. Postnatal ARPKD presentation in infancy shows the presence of bilateral flank masses, enlarged hyperechoic kidneys, and bilateral cysts (usually <1 cm). At least 50% of ARPKD individuals progress to ESKF within the first decade of life. Extrarenal complications include hepatic fibrosis, resistant hypertension, and biliary abnormalities predisposing to cholangitis; therefore affected children require close follow-up by gastroenterologists with regular endoscopy to detect varices and surveillance for hepatic fibrosis. Some children will require liver transplantation.

INHERITED CILIOPATHIES

BARDET-BIEDL SYNDROME

Bardet-Biedl syndrome (BBS) is a rare autosomal recessive disorder that manifests with kidney malformations and extrarenal features, such as retinal degeneration, polydactyly, obesity, hypogonadism, and developmental delay. Pathogenic variants in *BBS1-BBS10* genes are commonly associated with BBS, with some variants also implicated in other ciliopathies.⁵⁹⁰ *BBS10* variants tend to result in more severe phenotypes. In milder cases, prenatal ultrasonography may reveal normal kidneys, kidney cysts, or urinary tract malformations, while severe forms may show enlarged, hyperechoic kidneys lacking corticomedullary differentiation. Children with BBS can sometimes exhibit an ADPKD-like phenotype because certain BBS proteins regulate ciliary trafficking of the polycystin 1 protein.⁵⁹¹ Due to the phenotypic overlap with other cystic kidney diseases, genetic analysis is crucial for diagnosis.

MECKEL SYNDROME

Meckel syndrome is an autosomal recessive ciliopathy, presenting with large polycystic kidneys, hepatic fibrosis, and occipital encephalocele. It is primarily linked to gene variants in *MKS1*, *TMEM216*, and *TMEM67*, with more than 10 loci identified. These genes encode transition zone proteins essential for ciliogenesis, and severe cases often exhibit enlarged kidneys and oligohydramnios on prenatal ultrasonography. Genetic testing is essential for diagnosis and counseling regarding pregnancy continuation and recurrence risk.

NEPHRONOPHTHISIS

Nephronophthisis (NPHP) is a group of autosomal recessive disorders characterized by progressive tubulointerstitial fibrosis, inflammation, and kidney cysts. The condition is linked to more than 20 different genes, most commonly *NPHP1*, which encode proteins localized to the cilia, basal body, or centrosomes in kidney epithelial cells.⁵⁹² These proteins interact with polycystin proteins, affecting cellular adhesion and signaling. NPHP can occur in isolation or as part of syndromic ciliopathies, such as Joubert syndrome, Senior-Loken syndrome, or

short-rib thoracic dysplasia.⁵⁹³ NPHP presents in three subtypes based on the age of onset: infantile, juvenile, and adult-onset. The juvenile subtype, associated primarily with *NPHP1* variants, is the most common, leading to ESKF around age 13. Other features include polydipsia, polyuria, anemia, and hepatic fibrosis. Adult-onset NPHP presents similarly but with a later onset of KF. Heterozygous carriers of NPHP including parents are generally asymptomatic, but they may carry the genetic risk for these ciliopathies.

FANCONI SYNDROMES—EMPHASIS ON CYSTINOSIS

Etiology

Fanconi syndrome is the result of global proximal tubular dysfunction. It is characterized by the constellation of renal tubular acidosis (RTA), low-molecular-weight proteinuria, aminoaciduria, glucosuria, and hypophosphatemia (secondary to renal phosphate wasting). Causes can be subdivided into primary/genetic and secondary. [Table 71.20](#) provides a broad list of causes. Dent disease and the oculocerebral syndrome of Lowe are described in more detail in the next section discussing urolithiasis and nephrocalcinosis. Fanconi syndrome is more likely to be seen as a consequence of drug toxicity, particularly in patients receiving chemotherapy. A number of rheumatologic causes of acquired Fanconi syndrome should also be considered in the differential diagnosis. The most common genetic cause of Fanconi syndrome in childhood is cystinosis, which is focused on here.

Cystinosis is a rare autosomal recessive lysosomal storage disorder characterized by the accumulation of cystine within lysosomes due to a deficiency in the cystine transporter, cystinosin. This leads to the formation of cystine crystals and subsequent organ damage, particularly in the kidneys, eyes, and other tissues.⁵⁹⁴ The underlying etiology of cystinosis can be traced to mutations in the *CTNS* gene, located on chromosome 17p13.⁵⁹⁴ Various mutations in this gene result in different clinical subtypes of cystinosis, with infantile nephropathic cystinosis being the most severe form. Cystinosis occurs in approximately 1 in 100,000–200,000 live births and presents typically between 6 and 12 months of age with renal Fanconi syndrome.⁵⁹⁴

There are three main forms of cystinosis. The infantile form comprises 95% of cases and presents with renal Fanconi syndrome. In contrast, approximately 5% are diagnosed during late childhood or adolescence with juvenile (late onset) cystinosis.

Presentation

Children with Fanconi syndrome present within the first year of life with polyuria, polydipsia, hypokalemia, metabolic acidosis, hypophosphatemia, dehydration, and failure to thrive due to generalized proximal tubular damage.⁵⁹⁵ This is the typical presentation of infantile cystinosis. Later-onset cystinosis has a varied presentation, ranging from isolated asymptomatic proteinuria, a mild renal Fanconi syndrome, to an overt nephrotic syndrome. The latter patients generally experience milder growth impairment and

Table 71.20 Causes of Fanconi Syndrome

Genetic

Cystinosis
Dent disease
Galactosemia
Glycogen storage disease (type 1)
Hereditary fructose intolerance
Lowe syndrome
Mitochondrial diseases
Tyrosinemia
Wilson disease

Acquired—Drugs

Nucleoside reverse transcriptase inhibitors	Tenofovir, adefovir
Nucleoside analogs	Didanosine, lamivudine, stavudine
Chemotherapeutics	Ifosfamide, cisplatin, streptozocin
Anticonvulsants	Valproic acid
Antibiotics	Aminoglycosides, expired tetracyclines
Antivirals	Cidofovir
Antiparasitics	Suramin
Miscellaneous/toxins	Fumaric acid, paraquat
Acquired—Other Conditions	
Amyloidosis	
Heavy metals	For example, lead, cadmium, mercury
Membranous nephropathy	
Multiple myeloma	
Paroxysmal nocturia	
Postrenal transplantation	
Tubulointerstitial nephritis	
Vitamin D deficiency	

exhibit a slower progression toward ESKF and extrarenal complications. The adult, non-nephropathic ocular form of cystinosis manifests solely as photophobia caused by corneal cystine accumulation, with symptoms rarely appearing before adulthood, and the kidneys appear to remain unaffected.^{596,597} As the proximal tubule is the primary site of vitamin D activation, patients with Fanconi syndrome often display vitamin D–deficient rickets.⁵⁹⁸ Cystinosis is a multiorgan disease; over time patients typically develop a number of other sequelae, including hypothyroidism, photophobia, CKD, myopathy (including difficulty swallowing), diabetes mellitus, male hypogonadism, and pulmonary dysfunction.⁵⁹⁹

Diagnosis

When the clinical features of Fanconi syndrome are identified on urine analysis and blood work (i.e., hypokalemia, hypophosphatemia, non–anion gap metabolic acidosis, low-molecular-weight proteinuria,

aminoaciduria, and glucosuria), patients should be screened for cystinosis. The current gold standard of diagnosis is the detection of elevated cystine levels in white blood cells. Other diagnostic options include molecular testing of the *CTNS* gene, with mutations of the *CTNS* gene representing 95% of all disease-causing mutations.⁵⁹⁷ In about 5% of patients, pathogenic mutations are not easily discovered by *CTNS* gene sequencing, being either deeply intronic, in the promoter region, or not commonly encountered large deletion or duplications. In patients older than 16 years, diagnosis may be made via detection of the characteristic corneal cystine crystals by slit lamp examination.^{597,600}

In children presenting with Fanconi syndrome presenting in the first year of life, the differential diagnosis should also include tyrosinemia, hereditary fructose intolerance, galactosemia, glycogen storage disease (type 1) mitochondrial disorders, Wilson disease, oculocerebral syndrome of Lowe, and Dent disease.⁵⁹⁷

Treatment

Treatment of Fanconi syndrome should be aimed at correcting metabolic abnormalities arising from tubular transport defects. This should include the administration of alkali, typically as potassium or dicitrate, volume replacement, administration of vitamin D, phosphate replacement, nutritional support, and adequate substitution of needed hormones, such as growth hormone.⁵⁹⁷

For patients with cystinosis, cysteamine therapy is given to deplete lysosomal cystine in all body cells and tissues. Cysteamine therapy has been shown to slow the progression to ESKF by 6 to 10 years and prevent or postpone the development of some extrarenal complications.^{597,601,602} Cysteamine therapy should be commenced as soon as possible. Cysteamine eye drops should also be administered to prevent the onset or progression of ocular disease.⁶⁰³ Monitoring the effectiveness of oral cystine-depleting therapy is via the leukocyte cystine level. In the case of renal failure, renal transplantation is the treatment of choice and results in resolution of the ESKF, but not the extrarenal manifestations, thus cysteamine therapy should be continued to prevent nonrenal disease.⁶⁰⁴ Cysteamine dosing is based on BSA in adults, with a maximum of 1.95 g/m² per day. Intracellular leukocyte cystine levels are monitored to optimize the dose and assess compliance. Adverse effects including bone pain, bruiselike lesions on the elbows, and myalgia may occur at higher doses. Adherence to this therapy is challenging, however, as it requires intake of multiple tablets 6 to 12 hourly and has unpleasant side effects such as halitosis. Other reversible side effects such as allergic rash, seizures, and gastrointestinal discomfort are mitigated by gradual dose titration, and copper supplementation may help prevent toxicity in patients with renal Fanconi syndrome. The combination of cysteamine therapy and multisystemic supportive treatment requires patients to take up to 60 pills per day; young children often require a gastric tube to tolerate the medications and achieve the necessary caloric intake.⁶⁰³

DISORDERS OF RENAL TUBULAR SODIUM CHLORIDE TRANSPORT

BARTTER SYNDROME

Etiology

Bartter syndrome is a hereditary renal tubular disorder caused by defective sodium reabsorption in the thick ascending limb (TAL) of the loop of Henle. This results in a hypochloremic, hypokalemic, metabolic alkalosis, and both hyperreninemia and hyperaldosteronism secondary to volume contraction. There are various clinical presentations determined by specific gene mutations, described in Table 71.21.

Bartter syndrome is classified into six types according to gene affected. Antenatal Bartter syndrome is the result of a mutation in either the *SLC12A1* gene (type 1) encoding the furosemide-sensitive sodium–potassium–chloride cotransporter, NKCC2, or KCNJ1 (type 2), the gene encoding

the renal outer medullary potassium channel, ROMK.⁶⁰⁵⁻⁶⁰⁷ Classical Bartter syndrome is caused by mutations in the *CLCKNB* gene (type 3) encoding the chloride voltage-gated channel CLCKb.⁶⁰⁸ Bartter syndrome with sensorineural deafness is the result of mutations in either BSND (type 4), which encodes the accessory protein Barttin required for CLCKb (and CLCKa) activity or disease-causing mutations in both CLCKa and CLCKb.⁶⁰⁹⁻⁶¹¹ Both CLCKb and CLCKa (type 6) are expressed in the inner ear, and their activity is required for hearing. Activating mutations in the calcium-sensing receptor (CaSR, type 5), which is expressed in the basolateral membrane of the TAL, inhibit sodium chloride reabsorption from this segment, causing Bartter-like syndrome.^{612,613} The phenotype of these patients also includes low plasma calcium and inappropriately low parathyroid hormone (PTH) levels. Polyhydramnios and transient antenatal Bartter syndrome have been ascribed to mutations in the *MAGED2* gene.⁶¹⁴

Table 71.21 Gene Defects and Description of Disorders of Tubular Salt Handling

Name	Gene Affected ^a	Clinical Description
Primary Disorders of the Thick Ascending Limb		
Bartter syndrome	<i>SLC12A</i>	Polyhydramnios, premature delivery, hypokalemic metabolic alkalosis
	<i>KCNJ1</i>	Polyhydramnios, premature delivery, hypokalemic metabolic alkalosis, transient postnatal hyperkalemia
	<i>CLCKNB</i>	Childhood/adolescent presentation, hypokalemia, metabolic alkalosis
	<i>BSND</i> ^b	Polyhydramnios, premature delivery, hypokalemic metabolic alkalosis, and sensorineural hearing loss
	<i>CASR</i>	Metabolic alkalosis, hypokalemia, hypercalciuria, hypocalcemia with low parathyroid hormone
Primary Disorders of the Distal Convulated Tubule		
Gitelman syndrome	<i>SLC12A3</i>	Metabolic alkalosis, hypokalemia, hypomagnesemia
EAST (epilepsy ataxia sensorineural-deafness tubulopathy) syndrome	<i>KCNJ10</i>	Hypokalemic metabolic alkalosis, epilepsy, ataxia, sensorineural deafness
Pseudohypoaldosteronism type 2	<i>WNK1</i> , <i>WNK4</i> , <i>CUL3</i> , and <i>KLH3</i>	Hyperkalemia, metabolic acidosis, high plasma aldosterone
Primary Disorders of the Collecting Duct		
Pseudohypoaldosteronism type 1	<i>SCNN1A</i> , <i>SCNN1B</i> , <i>SCNN1G</i>	Hyperkalemia, metabolic acidosis, high plasma aldosterone, autosomal recessive inheritance
	<i>NR3C2</i>	Hyperkalemia, metabolic acidosis, high plasma aldosterone, autosomal dominant inheritance
Liddle syndrome	<i>SCNN1A</i> , <i>SCNN1B</i> , <i>SCNN1G</i> (activating)	Hypertension, hypokalemia, suppressed renin and aldosterone

^aUnless stated otherwise genetic alterations are inactivating.

^bMutations in both *CLCKNB* and *CLCKNA* genes cause a similar phenotype.

Presentation

Antenatal Bartter syndrome is the most common form and manifests in utero with polyhydramnios that commonly results in preterm delivery. Infants typically present shortly after birth with polyuria, episodes of severe dehydration, recurrent vomiting, and failure to thrive. Dehydration may be severe enough to cause neonatal AKI. Significant hypercalciuria and nephrocalcinosis are often present. *KCNJ1* mutations are associated with hyperkalemia in the first few weeks after birth, which then changes to the classic hypokalemia typical of this syndrome.⁶¹⁵ This phenotype likely reflects the developmental expression pattern of potassium channels. Specifically, the BK big potassium channel of the collecting duct is not expressed until approximately 6 weeks after birth. As such, children with *KCNJ1* mutations cannot effectively excrete potassium until then and therefore become hyperkalemic until after this channel is expressed.⁶¹⁶

Children with antenatal Bartter syndrome and deafness typically present in a similar fashion to antenatal Bartter syndrome, with severe polyhydramnios, preterm delivery, polyuria, severe volume contraction, and failure to thrive.^{610,611} Hyperkalemia is not a feature. However, auditory testing distinguishes these subtypes as sensorineural hearing is severely affected. Patients with activating mutations in the *CaSR* can present as previously or secondary to alterations in divalent cation handling, specifically with cramps or seizures. These patients display hypocalcemia, hypercalciuria, hypomagnesemia, and inappropriately low plasma PTH levels.^{612,613,617,618}

Classical Bartter syndrome manifests later in infancy or during childhood. Presenting symptoms are varied but may include muscle weakness and cramps, fatigue, constipation, recurrent vomiting, failure to thrive, polyuria, and/or significant volume depletion. Hypercalciuria and nephrocalcinosis are not always present. *CLCKNB* is also expressed in the basolateral membrane of the distal convoluted tubule (DCT), where it contributes to magnesium reabsorption. Thus in some cases, mild hypomagnesemia has been reported.⁶¹⁹

Diagnosis

Diagnosis is based on analysis of specific genes based on the clinical presentation. A history of significant antenatal polyhydramnios, postnatal polyuria, and typical features of hypokalemic, hypochloremic metabolic acidosis should prompt hearing testing. If this is normal, then examination of *SLC12A1* should be considered. An immediate postnatal history of hyperkalemia shifting to hypokalemia several weeks later should prompt investigation of *KCNJ1*. If abnormalities in sensorineural hearing are detected, examination of the *BSND* gene should ensue, and if normal, then *CLCKB* and/or *CLCKA*. Later presentation without a significant antenatal history should lead one to suspect isolated abnormalities in the *CLCKB* gene. Finally, if serum calcium is low in the presence of low PTH, one needs to consider abnormalities in the *CaSR* resulting in its inappropriate activation.

Treatment

There is no cure for Bartter syndrome. Treatment should focus on replacing electrolyte losses and mitigating the effects of hyperaldosteronism and increased prostaglandin

release.⁶²⁰ The European Rare Kidney Disease Reference Network Working Group for Tubular Disorders recommends sodium chloride, potassium chloride, and magnesium supplementation throughout the day, with sodium not recommended in patients with nephrogenic diabetes insipidus and potassium not required to be completely normalized.³⁴² The use of gastric acid inhibitors together with nonselective cyclooxygenase inhibitors is recommended, with conversion to H₂ blockers and other antacids if proton pump inhibitor-associated hypomagnesemia occurs. Potassium-sparing diuretics, ACE inhibitors, thiazides, and ARBs are not recommended. Administration of an NSAID (1–4 mg/kg/day of indomethacin) can help to attenuate the effects of increased prostaglandin release and raise the serum potassium level. With attention to electrolyte replacement and administration of indomethacin, growth velocity can be recovered and maintained.³⁴³ Certain substances should be avoided due to their effect on electrolyte balance, such as food containing glycyrrhizic acid (licorice root), excessive alcoholic beverages (aggravate potassium loss), fruit juices, or bicarbonate-rich drinks (worsen metabolic alkalosis).³⁴⁴ Long term, approximately one-quarter of patients will develop a degree of renal impairment.⁶²¹

GITELMAN SYNDROME

Gitelman syndrome is an autosomal recessive condition caused by defects in the sodium chloride cotransporter (encoded by *SLC12A3*) resulting in a failure to reabsorb sodium and chloride from the DCT. It has an estimated incidence of 1 in 40,000 and is most commonly diagnosed in later childhood or early adulthood, rarely in the neonatal period. Patients display hypochloremic, hypokalemic metabolic alkalosis. In contrast to Bartter syndrome-affected children, patients manifest with hypocalciuria and hypomagnesemia.⁶²²⁻⁶²⁴ Patients presenting in adulthood may also have extrarenal manifestations such as chondrocalcinosis, glucose intolerance or insulin resistance, and hypertension.⁶²⁴ A prolonged QTc on ECG has been reported in approximately 50% of patients.⁶²⁵

Molecular diagnostics should be considered in patients with typical electrolyte abnormalities, including hypomagnesemia. Treatment requires replacement of electrolytes, in particular potassium and magnesium, which may cause gastrointestinal upset at high doses.⁶¹⁴ Previously, Gitelman syndrome was considered a benign or mild condition, but reports have highlighted the symptomatic variability with long-term complications including growth retardation, chondrocalcinosis, tetany, rhabdomyolysis, seizures, CKD, secondary hypertension, and ventricular arrhythmias.⁶²⁶

EAST SYNDROME

EAST syndrome is a rare multiorgan disease, which includes renal salt loss due to a failure of sodium and chloride reabsorption from the DCT.⁶²⁷ The renal phenotype is thus similar to Gitelman syndrome. Patients display hypokalemic metabolic alkalosis and hypomagnesemia due to renal wasting of this divalent cation. In contrast to Gitelman syndrome, patients with EAST syndrome also display epilepsy, ataxia, sensorineural deafness, and tubulopathy. This condition is caused by homozygous or compound

heterozygous loss-of-function mutations in the *KCNJ10* gene, which encodes the Kir4.1 potassium channel that is expressed in the basolateral membrane of the DCT, as well as in the brain and inner ear (see Table 71.21).^{627,628} This syndrome has also been referred to as the SeSAME syndrome, an acronym for seizures, sensorineural deafness, ataxia, mental retardation, and electrolyte imbalance.⁶²⁹ Treatment of the renal manifestations includes replacement of electrolytes as in Gitelman syndrome.

PSEUDOHYPOALDOSTERONISM

Pseudohypoaldosteronism (PHA) is a collection of rare heterogeneous syndromes characterized primarily by mineralocorticoid resistance, resulting in reduced potassium and proton excretion from the collecting duct.⁶³⁰ The clinical constellation of hyperkalemia, metabolic acidosis, and increased plasma aldosterone levels is universally present. In some subtypes, there is also significant salt wasting. PHA has been subdivided into three subtypes, which are described in the following sections (see Table 71.21).

PHA Type I

PHA type I is characterized by presentation in infancy with life-threatening hyperkalemic metabolic acidosis and severe salt-wasting tubulopathy leading to dehydration and failure to thrive, along with high levels of plasma renin and aldosterone.⁶³¹ It can be divided into the autosomal recessive or dominant form. Autosomal recessive PHA type I is caused by inactivating mutations in sodium channel, non-voltage-gated 1 α gene (*SCNN1A*), sodium channel, non-voltage-gated 1 β gene (*SCNN1B*) and sodium channel, non-voltage-gated 1 γ (*SCNN1G*), which encode the three α , β , and γ subunits of ENaC subunits, respectively.⁶³² As ENaC is also expressed in sweat glands, the lung, and the colon, AR PHA I manifests as a multiorgan disease, including the skin, respiratory, and renal dysfunction. In contrast, autosomal dominant PHA type I is due to mutations in *NR3C2*, the gene encoding the mineralocorticoid receptor, and is limited to renal resistance to mineralocorticoids and typically displays a less severe phenotype. Metabolic acidosis may be absent, but salt wasting is a persistent feature. Treatment of both varieties of PHA type I requires attention to sodium replacement enabling volume expansion. Potassium binders are also frequently necessary.^{633,634}

PHA Type II

In contrast to PHA type I, patients with PHA type II, which has also been called Gordon syndrome, familial hyperkalemia and hypertension, and chloride shunt syndrome, do not display a salt-losing phenotype. Rather, patients with this disorder display hyperkalemia and hypertension.⁶³⁵ This heterogeneous disorder is the result of increased sodium chloride reabsorption from the DCT, leading to volume expansion and suppressed renin levels. It is autosomal dominant and is the result of alterations in a variety of genes including *WNK1*, *WNK2*, *CUL3*, and *KLHL3*.⁶³⁶⁻⁶³⁸ Given that this disorder consists of increased sodium chloride transport from the DCT, simply blocking this pathway with a thiazide diuretic generally ameliorates the symptoms of this disease.⁶³⁶

PHA Type 3

PHA Type 3, or secondary pseudohypoaldosteronism, is a rare condition primarily affecting young children, often

associated with UTIs and CAKUT. Epidemiologic data are scarce, though a study in Ireland estimated an incidence of 1 per 13,200 live births, with possible underdiagnosis in milder cases.⁶³⁹ PHA type 3 typically presents in the pediatric population, especially males in the first months of life, with nonspecific symptoms. Studies indicate PHA type 3 is often linked to UTIs, with pathogens such as *E. coli* and *Klebsiella* spp. being prevalent. PHA type 3 manifests with nonspecific symptoms such as poor feeding, vomiting, or dehydration, accompanied by electrolyte imbalances like hyponatremia, hyperkalemia, and metabolic acidosis. The condition results from impaired aldosterone activity in the renal tubules, exacerbated by infections or malformations, leading to sodium loss and potassium retention. While symptoms are often mild, some patients may experience significant electrolyte imbalances or AKI. Differential diagnosis should include congenital adrenal hyperplasia (CAH) and other causes of hypoaldosteronism, with hormonal studies possibly needed. Treatment focuses on correcting electrolyte imbalances and addressing the underlying cause, often involving surgical intervention for CAKUT.

LIDDLE SYNDROME

Liddle syndrome describes the inappropriate increase of sodium and chloride transport from the collecting duct, which results in volume expansion and hypertension in the presence of suppressed renin and aldosterone levels. This constellation of clinical features is due to autosomal dominant activating mutations in one of the genes encoding the ENaC subunits α , β , and γ : *SCNN1A*, *SCNN1B*, or *SCNN1G*, respectively (Table 71.21).⁶⁴⁰⁻⁶⁴³ As in Gordon syndrome, therapy is directed at inhibiting the primary mechanism and involves treatment with the ENaC inhibitor amiloride. Although required lifelong, this therapy completely prevents the manifestations of this disease.

RENAL TUBULAR ACIDOSIS

A non-anion gap metabolic acidosis is consistent with the presence of renal tubular acidosis (RTA). However, especially in young children, this is more commonly associated with diarrhea, which should be excluded with a careful history. RTAs are classified according to the nephron segment affected.⁶⁴⁴ Abnormalities in proton excretion from the distal nephron are classified as distal or type I RTA. A failure to reabsorb bicarbonate from the proximal tubule is referred to as a proximal or type II RTA. Type IV, which is also called hyperkalemic RTA, results from aldosterone resistance or absence. Type III, or mixed RTA, is a rare disorder that displays both a failure to reabsorb bicarbonate from the proximal tubule and the inability to secrete protons from the distal nephron.

Plasma electrolytes are useful to identify the subtype of the RTA. Types I and II typically display normal or low serum potassium levels, while type IV has high serum potassium.⁶⁴⁵ Examination of urine parameters are useful to distinguish proximal and distal RTA. In proximal RTA the ability to secrete protons distally is maintained, so urine pH is <5.5. Protons secreted into the urine are trapped by ammonia, forming ammonium (NH_4^+). While ammonium is difficult to measure in the urine, a surrogate measure is the urinary anion gap, calculated as the difference between

urine cations ($\text{Na}^+ + \text{K}^+$) and urine Cl^- . When acidotic, a patient should secrete protons into their urine, forming ammonium, and therefore display a negative urine anion gap ($\text{urine } \text{Na}^+ + \text{K}^+ - \text{Cl}^- < 0$). If this is not the case, then there is a failure to secrete protons by the distal nephron suggesting a distal RTA. If the diagnosis remains unclear, more definitive testing can be employed. The traditional means of diagnosing proximal RTA is to bicarbonate load the patient to correct the acidosis and demonstrate a significant increase in fractional excretion of bicarbonate (i.e., to $>15\%$).⁶⁴⁶ Distal RTA is definitively demonstrated by either an acid load with ammonium chloride (100 mg/kg), which is poorly tolerated or by administering furosemide and fludrocortisone and monitoring urine pH over the next 8 hours.^{647,648} Patients with distal RTA are not able to acidify their urine below pH 5.3. Urinary acidification is discussed in further detail in Chapter 9.

Type I: Distal Renal Tubular Acidosis

The most common form of RTA is type I, or distal RTA. Clinically, patients with this disorder display hypercalciuria, hypocitraturia, nephrocalcinosis, and nephrolithiasis and may develop renal cysts later in life.^{649,650} Untreated disease often results in failure to thrive, bone demineralization, and rickets.⁶⁵¹ The etiology of this disorder is either primary (i.e., genetic) or acquired secondarily from drug toxicity or a rheumatologic condition. Genetic causes of distal RTA include mutations in the genes encoding machinery required for the secretion of protons from α intercalated cells of the collecting duct, specifically the plasma membrane H^+ -ATPase (i.e., ATP6V1B1 or ATP6V0A4), or the basolateral chloride–bicarbonate exchanger AE1 (gene name *SLC4A1*; Table 71.22).⁶⁵²⁻⁶⁵⁹ Novel genes have been found including *KCC4*, *SLC26A7*, *WNK1*, and *CLDN16*.⁶⁶⁰ Clinical phenotype is altered by

the gene affected. Mutations in *SLC4A1* produce a milder phenotype, which might not manifest until the teenage years or adulthood, and can be inherited in either an autosomal dominant or autosomal recessive fashion.⁶⁶¹ By contrast, patients with mutations in the subunits of the H^+ -ATPase typically have more severe disease, which often presents in infancy and is commonly associated with deafness.

Type II: Proximal Renal Tubular Acidosis

Proximal RTA rarely occurs in isolation and is much more frequently seen as a part of the Fanconi syndrome (i.e., with hypophosphatemia, glycosuria, aminoaciduria, and low-molecular-weight proteinuria). Common causes of Fanconi syndrome in childhood have been described earlier and in Table 71.20. Patients with isolated RTA rarely have renal calculi or nephrocalcinosis, potentially due to the increased urinary citrate excretion that typically accompanies this disorder.⁶⁵⁰ They also tend not to display abnormalities in bone mineral density or rickets. Isolated proximal RTA is caused by mutations in the basolateral sodium–bicarbonate exchanger NBCe1 (gene name *SLC4A4*), which is essential to the reclamation of filtered bicarbonate from the proximal tubule (see Table 71.22).⁶⁶²⁻⁶⁶⁵ Patients with mutations in this gene also display ocular abnormalities and intellectual disability.

Type III: Renal Tubular Acidosis

A failure to reabsorb bicarbonate from the proximal tubule and acidify the urine is known as a mixed, or type III RTA. This rare autosomal recessive disorder is a consequence of mutations in carbonic anhydrase II (CAII, gene name *CA2*), which is required for both processes (see Table 71.22).^{666,667} Type III RTA is a complex syndrome; in addition to RTA, patients also demonstrate osteopetrosis, cerebral

Table 71.22 Renal Tubular Acidosis Characteristics and Single-Gene Defects Causing Renal Tubular Acidosis

Gene	Inheritance	Associated Clinical Features
Type I—Distal Renal Tubular Acidosis		
ATP6V1B1	AR	Nephrocalcinosis/nephrolithiasis, sensorineural hearing loss
ATP6V0A4	AR	Nephrocalcinosis/nephrolithiasis, late-onset sensorineural hearing loss
ATP6V1C2	AR	Nephrocalcinosis/nephrolithiasis
FOXI1	AR	Severe early onset sensorineural hearing loss
WDR72	AR	Severe enamel hypoplasia
SLC4A1	AD (rarely AR)	Nephrocalcinosis, osteomalacia with AR inheritance can be associated with hemolytic anemia
Type II—Proximal Renal Tubular Acidosis		
SLC4A4	AR	Cataracts, glaucoma, band keratopathy
KCNK5		Cataracts, glaucoma, band keratopathy
Type III—Mixed Renal Tubular Acidosis		
CA2		Osteopetrosis (Guibaud-Vainsel syndrome)

Causes of type IV, hyperkalemic metabolic acidosis, are described in Table 6.
AD, Autosomal dominant; AR, autosomal recessive.
From Guo W, Ji P, Xie Y. Genetic diagnosis and treatment of inherited renal tubular acidosis. *Kidney Dis (Basel)*. 2023;9(5):371–383.

calcifications, and intellectual disability.⁶⁶⁸ Additional features that have been associated with the disorder include facial bone overgrowth resulting in facial dysmorphism, conductive hearing loss, and blindness.

Type IV, Hyperkalemic Renal Tubular Acidosis

Patients with type IV RTA display resistance to, or an absence of, aldosterone effect on the distal nephron. Patients typically display a milder degree of acidosis and hyperkalemia, which is proportionate to the degree of aldosterone deficiency. This disorder can result in drug administration such as NSAIDs, ACE inhibitors, CNIs, potassium-sparing diuretics, and obstructive uropathy or occur due to a single-gene defect such as occurs in PHA, which is described earlier.

Treatment of RTA involves supplementation with alkali to normalize plasma pH. In the hypokalemic forms of the disease, this is best supplied as potassium citrate.⁶⁶⁹ The amount of supplementation required differs depending on the cause. Generally distal RTA requires much less supplementation (typically 1–2 mEq/kg/day of alkali) than proximal RTA (which can require upward of 10 mEq/kg/day of alkali). In patients with distal RTA, treatment should also include minimizing the risk of stone formation and renal calcification, which can lead to renal insufficiency.⁶⁷⁰ To this end, thiazide diuretics and increased fluid intake should be considered.

NEPHROGENIC DIABETES INSIPIDUS

ETIOLOGY

A failure to respond to the action of arginine vasopressin (AVP) results in nephrogenic diabetes insipidus (NDI).⁶⁷¹ AVP is released in response to increased plasma osmolarity and results in the phosphorylation and insertion of aquaporin-2 into the apical membrane of principal cells of the collecting duct, allowing water to be reabsorbed.⁶⁷² NDI can be either congenital (i.e., genetic) or acquired. The majority (90%) of congenital diabetes insipidus is the result of mutations in the AVP receptor, V2R, encoded by the gene *AVPR2*.⁶⁷³ The other roughly 10% are due to mutations in aquaporin-2, encoded by *AQP2*.⁶⁷⁴ *AVPR2* mutations are inherited in an X-linked recessive fashion, whereas *AQP2* mutations occur in an autosomal recessive manner. Acquired NDI can be the result of renal injury (i.e., AKI, obstructive uropathy, renal dysplasia, tubulointerstitial nephritis, cystic kidney diseases, or nephronophthisis) or drug toxicity (i.e., lithium, ifosfamide, foscarnet, and amphotericin B). For further discussion on urinary concentrating and diluting mechanisms see [Chapter 10](#).

PRESENTATION

Congenital forms of NDI tend to be more severe and present in infancy (median age at diagnosis of 0.6 years old) with polyuria, hypernatremic dehydration, fever, irritability, constipation, and failure to thrive.^{675,676} Infants who are breastfed have less severe features, likely due to their reduced solute load. Untreated patients can present later in childhood with enuresis and mental retardation.⁶⁷⁷ Overall the symptoms tend to improve with age.

DIAGNOSIS

The diagnosis of NDI requires the demonstration that urine cannot be concentrated in the presence of desmopressin (1-deamino-8-d-arginine vasopressin, ddAVP). This is demonstrated by performing a ddAVP challenge and monitoring urine concentration. ddAVP can be administered intranasally, orally, intramuscularly, or intravenously. Administration of ddAVP by the intramuscular or intranasal route requires longer monitoring than oral or IV administration (6 vs. 4 hours). A diagnosis of NDI is established by the failure of the urine osmolarity to increase above 200 mOsm in this time frame. The development of a urine osmolarity >200 mOsm but <800 mOsm in children (and <500 mOsm in infants) is consistent with a partial NDI. Activation of V2R also mediates peripheral vasodilation; thus patients with mutations of *AQP2* display tachycardia and reduced blood pressure in response to IV ddAVP, in contrast to children with mutations of V2R.

TREATMENT

Early diagnosis and treatment of NDI is important avoid complications arising from severe and frequent hypernatremia. Adequate fluid intake should be ensured (at least 150 mL/kg/day), which becomes easier as children age and can respond to thirst. A thiazide diuretic (i.e., hydrochlorothiazide 2 mg/kg/day) can be employed to increase proximal sodium and consequently water reabsorption and decreased urine output. Amiloride (0.2 mg/kg/day) can also be employed to reduce urine output, via a similar mechanism. Indomethacin (1–3 mg/kg/day) has also been demonstrated to be useful in reducing urine volume, perhaps by reducing GFR, which has a role when children are being toilet trained. Typically, a combination of these drugs is employed initially to reduce urine output. The combination of amiloride, a potassium-sparing diuretic, and hydrochlorothiazide is beneficial to maintain normal plasma potassium levels.

New targeted therapeutic molecules have been explored for the treatment of NDI, but none have yet proven effective or reached clinical use. Vaptans such as tolvaptan are small-molecule agonists that bind V2R. Many mutations in V2R cause protein misfolding and intracellular retention. These membrane-permeant drugs rescue the receptor to the cell surface, thereby ameliorating the disease.⁶⁷⁸ Further, as the majority of patients have functional aquaporin-2 water channels, different attempts have been made to stimulate their phosphorylation and trafficking to the cell surface by activating the pathway downstream of V2R.⁶⁷⁹ Most promising has been the administration of phosphodiesterase inhibitors, such as sildenafil or agonists of other G-protein coupled receptors expressed in the principal cell, such as the prostanoid receptor butaprost.^{114,680} Additionally, statins have been proposed to alter the principal cell membrane cholesterol content in a way that promotes the recruitment of aquaporin-2 to the apical plasma membrane.^{681,682} The role of prostaglandin receptor agonists is currently under investigation. cAMP is a key messenger in urinary concentration, and prostaglandin E2 has demonstrated ability in increasing

cAMP in the principal cell independent of AVPR2 and thus reducing urine output in animal models. However, administering prostaglandins for NDI treatment appears to contradict the established clinical effectiveness of inhibiting prostaglandin synthesis and further data are required to address this.^{680,683} Fluconazole has also shown potential for AQP2 modulation in mouse models.⁶⁸⁴ For the remaining *AVPR2/AQP2* gene mutations with complete loss of function, gene therapy may be the only possible therapeutic strategy.⁶⁸⁵

UROLITHIASIS AND NEPHROCALCINOSIS IN CHILDREN

EPIDEMIOLOGY

The incidence of urolithiasis in children is increasing, accounting for a greater number of emergency department visits and hospitalizations.^{686,687} The incidence in the pediatric population is rare, with an estimated incidence of approximately 5 to 36 per 100,000 children, but there appears to be a steady increase over the past 10 to 20 years.⁶⁸⁸ The reason for this is unclear but is likely associated with the increasing incidence of childhood obesity, increased consumption of processed foods contributing to increased sodium chloride consumption, as well as the result of a greater number of surviving premature infants who are treated with medications causing urolithiasis.^{686,687} In contrast to adults, emerging evidence suggests that girls are more

commonly affected, especially during adolescence.⁶⁸⁹ Nephrolithiasis is discussed in more detail in [Chapter 40](#).

RISK FACTORS

HYPERCALCIURIA

Because of the difficulty in collecting 24-hour urine in younger children, hypercalciuria is often diagnosed by a spot urine collection normalized to the urine creatinine. Urinary calcium excretion decreases with age therefore normal values change with age ([Table 71.23](#)).^{690,691}

The most common etiology of hypercalciuria in childhood is idiopathic. However, the genetics of this “unknown” cause of hypercalciuria is becoming clearer ([Table 71.24](#)). Multiple drugs induce hypercalciuria including loop diuretics, vitamin D (in high doses), and corticosteroids. Finally, there are multiple genetic syndromes, which are associated with hypercalciuria. They are discussed in the next section on the genetics of pediatric urolithiasis and nephrocalcinosis.

Hypercalciuria is likely the most frequent metabolic risk factor in children, with frequencies between 50% and 60%.⁶⁹² Depending on the study, the incidence of hypocitraturia is between 15% and 50%. It has been suggested that the younger the patient with renal stones, the higher the probability of finding hypocitraturia as the underlying metabolic abnormality present.⁶⁹¹ The etiology of reduced urinary citrate excretion is unclear, except in specific genetic conditions such as distal RTA.⁶⁹³ A low urinary citrate excretion can be detecting using the calcium-to-citrate ratio or the citrate-to-creatinine ratio in spot urine

Table 71.23 Normal Pediatric Values for Urinary Excretion of Solutes

Solute	Ratio; Solute/Creatinine		24-Hour Urinary Excretion
Calcium	mol/mol	g/g	
<1 year	<2.2	<0.8	<0.1 mmol/kg
1-3 years	<1.5	<0.53	<4 mg/kg
3-5 years	<1.1	<0.4	
5-7 years	<0.8	<0.3	
>7 years	<0.6	<0.2	
Citrate	mol/mol	g/g	
0-5 years	0.12-0.25	0.20-0.42	>0.8 mmol/1.73 m ²
>5 years	0.08-0.15	0.14-0.25	>0.14 g/1.73 m ²
Oxalate	mmol/mol	mg/g	
<1 year	15-260	12-207	<0.5 mmol/1.73 m ²
1-5 years	11-120	9-96	<45 mg/1.73 m ²
5-12 years	60-150	47-119	
>12 years	2-80	2-64	
Uric Acid	mol/mol	g/g	
<1 year	<1.5	<2.2	>1 year; <815 mg/1.73 m ²
1-3 years	<1.3	<1.9	
3-5 years	<1.0	<1.5	
5-10 years	<0.6	<0.9	
>10 years	<0.4	<0.6	
Cystine	mmol/mol	mg/g	
<1 month	<85	<180	<10 years: <55 μmol/1.73 m ²
1-6 months	<53	<112	>10 years: <200 μmol/1.73 m ²
>6 months	<18	<38	>18 years: <250 μmol/1.73 m ²

Table 71.24 Genetic Conditions Displaying Hypercalciuria

Condition Name	Gene(s)	Clinical Features
Dent disease	<i>CLCN5</i>	Nephrocalcinosis, low-molecular-weight proteinuria, X-linked inheritance
Oculocerebral syndrome of Lowe	<i>OCRL</i>	Congenital cataracts, mental retardation, hypotonia, rickets, partial or complete Fanconi syndrome
Bartter syndrome	<i>SLC112A1, KCNJ1, CLCKNB, BSND</i>	Hypokalemic metabolic alkalosis, nephrocalcinosis
Autosomal dominant hypocalcemia and hypercalciuria	<i>CASR</i>	Hypocalcemia, suppressed parathyroid hormone, hypokalemic metabolic alkalosis
Familial hypomagnesemia hypercalciuria and nephrocalcinosis	<i>CLDN16, CLDN19</i>	Hypomagnesemia, hypercalciuria, nephrocalcinosis, progressive renal impairment, macular colobomas, myopia horizontal nystagmus (CCLDN19 mutations only)
Distal renal tubular acidosis	<i>ATP6V0A4, ATP6V1B1, SLC4A1</i>	Nephrocalcinosis, kidney stones, renal tubular acidosis
Williams syndrome	???	Cardiovascular disease, triangular facies, connective tissue abnormalities, intellectual disabilities

Gene	Inheritance	Accession no.	Disease
<i>ACTN4*</i>	AD	NM_004924	Familial and sporadic SRNS (usually adult)
<i>ADCK4*</i>	AR	NM_024876	SRNS
<i>ALG1</i>	AR	NM_019109	Congenital disorder of glycosylation
<i>ANKFY1</i>	AR	NM_001330063.2	Pediatric SRNS
<i>ANLN</i>	AD	NM_018685	FSGS (mainly adult)
<i>ARHGAP24</i>	AD	NM_001025616	FSGS
<i>ARHGDIA</i>	AR	NM_001185078	CNS
<i>AVIL</i>	AR	NM_006576.3	SRNS
<i>CD151</i>	AR	NM_004357	NS, pretibial bullous skin lesions, neurosensory deafness, bilateral lacrimal duct stenosis, nail dystrophy, and thalassemia minor
<i>CD2AP</i>	AD/AR	NM_012120	FSGS/SRNS
<i>CFH</i>	AR	NM_000186	MPGN type II +NS
<i>CLCN5</i>	XR	NM_001127898.4	Dent's disease ± FSGS ± hypercalciuria and nephrolithiasis
<i>COL4A3*</i>	AR	NM_000091	Alport disease/FSGS
<i>COL4A4</i>	AR	NM_000092	Alport disease/FSGS
<i>COL4A5*</i>	XR	NM_000495	Alport disease/FSGS
<i>C0Q2</i>	AR	NM_015697	Mitochondrial disease/isolated nephropathy
<i>C0Q6</i>	AR	NM_182476	NS ± sensorineural deafness; DMS
<i>CRB2*</i>	AR	NM_173689	SRNS
<i>CUBN</i>	AR	NM_001081	Intermittent nephrotic-range proteinuria ± with epilepsy
<i>DGKE*</i>	AR	NM_003647	Hemolytic uremic syndrome, SRNS
<i>DLC1</i>	AR	NM_182643.3	Childhood and adult SSNS and SRNS
<i>E2F3</i>	AD	NM_001949	FSGS + mental retardation (whole gene deletion)
<i>EMP2</i>	AR	NM_001424	Childhood-onset SRNS and SSNS
<i>FAT1</i>	AR	NM_005245.4	Combination of SRNS, tubular ectasia, hematuria, and facultative
<i>FN1</i>	AD?	NM_212482.3	Fibronectin glomerulopathy
<i>GAPVD1</i>	AR	NM_001282680.3	Early onset NS
<i>INF2</i>	AD	NM_022489	Familial and sporadic SRNS, FSGS-associated Charcot-Marie-Tooth neuropathy
<i>ITGA3</i>	AR	NM_002204	Congenital interstitial lung disease, nephrotic syndrome, and mild epidermolysis bullosa
<i>ITGB4</i>	AR	NM_000213	Epidermolysis bullosa and pyloric atresia + FSGS
<i>ITSN1</i>	AR	NM_003024.3	CNS/SRNS/SSNS (with MCD/FSGS on biopsy)
<i>ITSN2</i>	AR	NM_019595.4	SSNS/SDNS (with MCD/MPGN on biopsy)
<i>KANK1</i>	AR	NM_015158	SSNS
<i>KANK2</i>	AR	NM_015493	SSNS/SDNS ± hematuria
<i>KANK4</i>	AR	NM_181712	SRNS + hematuria

Continued on following page

Table 71.24 Genetic Conditions Displaying Hypercalciuria (Continued)

Gene	Inheritance	Accession no.	Disease
KIRREL1	AR	NM_018240.7	SRNS
LAGE3	AR	NM_006014.4	NS with primary microcephaly
LAMA5	AR	NM_005560.6	Childhood NS
LAMB2*	AR	NM_002292	Pierson syndrome
LCAT	AR	NM_000229.2	Norum disease
LMNA	AD	NM_170707	Familial partial lipodystrophy + FSGS
LMX1B*	AD	NM_002316	Nail-patella syndrome; also FSGS without extrarenal involvement
MAFB	AD	NM_005461.5	FSGS with Duane retraction syndrome
MAGI2	AR	NM_012301.4	NS ± neurologic impairment
MMACHC	AR	NM_015506.3	Cobalamin C deficiency, TMA, and nephrotic syndrome
MYO1E*	AR	NM_004998	Familial SRNS
NEU1	AR	NM_000434.4	Nephrosialidosis (sialidosis type II + childhood NS)
NPHP4	AR	NM_015102.5	Nephronophthisis with FSGS and nephrotic-range proteinuria
NPHS1*	AR	NM_004646	CNS/SRNS
NPHS2*	AR	NM_014625	CNS, SRNS
NUP85	AR	NM_024844.5	SRNS
NUP93*	AR	NM_014669	Childhood SRNS
NUPI07*	AR	NM_020401	Childhood SRNS
NUP160	AR	NM_015231.2	SRNS
NUP205	AR	NM_015135	Childhood SRNS
NXF5	XR	NM_032946	FSGS with cosegregating heart block disorder
OCRL*	XR	NM_000276	Dent disease-2, Lowe syndrome, ± FSGS, ± nephrotic-range proteinuria
OSGEP	AR	NM_017807.4	NS with primary microcephaly
PAX2	AD	NM_003987	Adult-onset FSGS without extrarenal manifestations
PDSS2	AR	NM_020381	Leigh syndrome
PLCe1	AR	NM_016341	CNS/SRNS
PMM2	AR	NM_000303	Congenital disorder of glycosylation
PODXL*	AD	NM_005397	FSGS
PTPRO	AR	NM_030667	NS
SCARB2	AR	NM_005506	Action myoclonus-renal failure syndrome ± hearing loss
SGPL1	AR	NM_003901.4	Primary adrenal insufficiency and SRNS
SMARCA1	AR	NM_014140	Schimke immuno-osseous dysplasia
SYNPO	AD	NM_007286	Sporadic FSGS (promoter mutations)
TBC1D8B	XR	NM_017752.3	Early onset SRNS with FSGS
TNS2	AR	NM_170754.3	SSNS/SDNS (with MCD/FSGS/DMS on biopsy)
TP53RK	AR	NM_033550.4	NS with primary microcephaly
TPRKB	AR	NM_001330389.1	NS with primary microcephaly
TRPC6*	AD	NM_004621	Familial and sporadic SRNS (mainly adult)
TTC21B	AR	NM_024753	FSGS with tubulointerstitial involvement
WDR73	AR	NM_032856	Galloway-Mowat syndrome (microcephaly and SRNS)
WT1*	AD	NM_024426	Sporadic SRNS (children: may be associated with abnormal genitalia); Denys-Drash and Frasier syndrome
XPO5	AR	NM_020750	Childhood SRNS
ZMPSTE24	AR	NM_005857	Mandibuloacral dysplasia with FSGS
MYH9	AD/assoc.	NM_002473	MYH9-related disease; Epstein and Fechtner syndromes
APOLI*	G1, G2 risk alleles	NM_003661	Increased susceptibility to FSGS and ESRD in African Americans, Hispanic Americans, and individuals of African descent

in small children, which identifies children who might benefit from therapy.⁶⁹⁴

HYPERURICOSURIA

A small fraction of children with nephrolithiasis (approximately 5%) will display hyperuricosuria. Like urinary calcium excretion, urinary urate excretion is

highest in infants and decreases with age (Table 71.23). The concentration of uric acid in the urine of infants can be so high that it precipitates as red-tinged crystals in the diapers and can be confused with blood. Uric acid stones may be seen in children with tumor lysis syndrome, although the frequency of this has been greatly reduced with the introduction of urate oxidases such as rasburicase. Other

causes include high dietary purine intake and hemolysis.⁶⁹⁵ Treatment in adults includes diet and xanthine oxidase inhibitors, though no consensus exists for children.⁶⁹⁶ There are also a number of rare genetic disorders, which result in increased urinary uric acid secretion, outlined as follows.

HYPEROXALURIA

The incidence of hyperoxaluria has been reported to be as high as 40%.⁶⁹⁷ The etiology of hyperoxaluria is most frequently idiopathic, although a number of primary and secondary causes should be considered. Secondary causes are much more frequent than primary genetic disorders (PH I, II, and III), which are discussed later. Increased ingestion of oxalate or oxalate precursors such as vitamin C can lead to hyperoxaluria, as can pyridoxine deficiency or ingestion of methanol. Alternatively, hyperabsorption of oxalate from the intestine, which can be due to a variety of diseases that lead to fat malabsorption including inflammatory bowel disease, postbowel resection, pancreatic insufficiency, or biliary disease, can cause hyperoxaluria. Therapies aimed at improving the underlying disorder, decreasing oxalate ingestion, increasing free water intake, and adding calcium administration, help to treat this form of hyperoxaluria. Although studies in adults have shown that dietary oxalate restriction reduces urinary oxalate excretion, restriction of dietary oxalate is not generally recommended for children due to weak evidence and a lack of studies in the pediatric population.^{698,699} However, it is suggested to limit the intake of high oxalate foods, such as spinach, rhubarb, chocolate, and nuts.⁷⁰⁰

ANATOMIC ABNORMALITIES AND INFECTIONS

A non-metabolic risk factor for kidney stone formation is urinary stasis or abnormal urinary flow. Children with congenital abnormalities of the kidney and urinary tract are more likely to develop kidney stones. UTIs with urea-splitting organisms such as *Proteus* spp. can alkalinize the urine and increase ammonium production. This in turn favors the formation of struvite stones (magnesium ammonium phosphate).⁷⁰¹ Patients with neurogenic bladders who have undergone surgical procedures including ileal augmentation are at increased risk of struvite stone formation.⁷⁰² Patients with hypercalciuria are at increased risk of developing a UTI independent of urinary tract abnormalities although the reason for this is unclear.^{703,704}

PRESENTATION

Infants and younger children often do not present with typical flank pain radiating to the groin, as is the classic presentation in adults.^{689,705} Furthermore, a significant fraction of children may not even display hematuria. Instead, young children may present with nonspecific abdominal pain, dysuria, frequency, urgency, emesis, diarrhea, or a UTI.⁷⁰⁶ Approximately 56% of children younger than the age of 5 present with flank or abdominal pain or hematuria.⁷⁰⁷ Another study in the United Kingdom reported 36% of pediatric patients presented with UTIs, 32% presented with pain, and 13% had painful gross hematuria, whereas 13% of patients were diagnosed as incidental findings in

asymptomatic patients.⁶⁹² Adolescents more commonly present similarly to adults with renal colic.

DIAGNOSIS

A high index of suspicion is key in younger children. The radiologic test of preference is a renal ultrasound; however, smaller stones and stones in the calyces, papillae, or ureter may be missed by this technique.⁷⁰⁸ A retrospective study found, however, that stones missed on ultrasound were 2.6 mm on average, with all but 3 calculi ≤ 3 mm in size, suggesting that majority of missed calculi are of low clinical consequence.⁷⁰⁹ Large radiopaque stones are visible on plain films of the abdomen; however, helical noncontrast computed tomography (CT) examination is the most sensitive test and should be reserved for when ultrasound is negative and there is a high index of suspicion.⁷¹⁰ Helical CT has the added benefit of providing further information with regard to anatomy and obstruction.

Examining urine for the presence of crystals may be useful. Further clues can be provided by urine examination for hypercalciuria, hyperoxaluria, cystinuria, and hyperuricosuria. Serum calcium, phosphorus, uric acid, and electrolytes should be measured and can help identify the underlying cause. Family history, diet, and fluid intake are important.

TREATMENT

The initial goal of treatment is to facilitate stone passage or removal, while achieving symptom control, followed by minimizing risks of long-term recurrence as children are prone to repeat stone formation, with rates shown to be as high as 50%.⁸³⁴ In many cases, renal and ureteral stones pass spontaneously in 32% to 50% and 41% to 63% of children, respectively.^{711,712} Renal calculi >7 mm are unlikely to pass, and elective surgical removal should be considered.⁷⁰⁵ Other indications for surgical removal include single kidney, prolonged obstruction, significant pain, or infection. Surgical options include extracorporeal shock wave lithotripsy (for stones >2 cm), percutaneous nephrolithotomy, and ureteroscopy. It may be necessary to place a ureteral stent in some patients to dilate the ureter before extraction can be accomplished. For stones impacted in a ureter, medical expulsive therapy can be considered. This includes hydration and use of either an α -adrenergic blocker (e.g., tamsulosin) or a calcium channel blocker. While evidence of the effectiveness of these therapies exists for adults, there is only anecdotal evidence available for children.^{712,713}

Dietary advice should otherwise be provided to decrease the risk of recurrence, which is significant in infants and children.⁷¹⁴ Fluid intake should be increased to approximately 2 L/m² to both dilute precipitating solutes and help with the passage of stones in situ. Hypercalciuric patients (and indeed most patients) benefit from a normal sodium diet (i.e., at the minimum <2000 mg sodium chloride per day). Restriction of calcium should be avoided, as this can be paradoxically associated with hypercalciuria due to the dissolution of calcium from bone.⁷¹⁵ Instead, attention should be paid to ensuring adequate calcium intake. High-fructose-containing beverages should be avoided.⁷¹⁶ In

contrast to adults, a low-protein diet is not recommended due to potential adverse consequences on growth.

Medical therapy should target underlying metabolic risk factors. Patients with hyperuricemia typically benefit from xanthine oxidase inhibition. Patients with hypocitraturia should receive potassium citrate. It is not recommended to alkalinize the urine of all stone formers as this simply changes the stones from calcium oxalate to calcium phosphate. Patients with persistent hypercalciuria despite dietary changes may also benefit from administration of a thiazide (i.e., 1–2 mg/kg per day of hydrochlorothiazide).⁷¹⁷

GENETICS OF PEDIATRIC UROLITHIASIS AND NEPHROCALCINOSIS

HYPERCALCIURIA-INDUCING CONDITIONS

Williams Syndrome

Williams syndrome is a rare autosomal dominant disorder characterized by a constellation of cardiovascular disease (typically peripheral pulmonary stenosis but can include supravalvular aortic stenosis or hypertension), distinctive triangular facies, connective tissue abnormalities, intellectual disability (typically not severe), unique personality features, early puberty, and in up to 50% of cases hypercalcemia in infancy, hypercalciuria, and nephrocalcinosis.⁷¹⁸ The etiology of this disorder is a 7q11.23 contiguous deletion of the Williams-Beuren syndrome critical region encompassing the elastin gene (Table 71.24). It is unclear why deletion of this region results in hypercalcemia, hypercalciuria, and nephrocalcinosis. Hypercalcemia tends to resolve by the age of 4 but may recur, and higher median serum calcium levels compared with controls are seen across all age groups.⁷¹⁹ The hypercalciuria responds to typical treatments including increased fluid intake, a reduced sodium diet, and thiazide diuretics, and it improves with a lactose-free diet.⁷²⁰

Dent Disease/Oculocerebral Syndrome of Lowe

Dent disease is typically an X-linked recessive disorder characterized by hypercalciuria, nephrocalcinosis, low-molecular-weight proteinuria, renal insufficiency, focal global sclerosis, and progressive CKD.⁷²¹ The majority of affected males exhibit hypercalciuria, which can be particularly high in young children. In contrast, in teenagers and adults, hypercalciuria usually resembles the levels seen in individuals who routinely form calcium stones.⁷²¹ Dent disease can also be associated with the complete Fanconi syndrome. It is most commonly caused by a mutation in *CLCN5*, which encodes a chloride-proton exchanger expressed at high levels in the kidney, with the highest levels in the proximal tubule and TAL (see Table 71.24).^{722,723} The exact molecular pathophysiology is unclear, especially with regard to the pathogenesis of hypercalciuria and nephrocalcinosis. However, approximately 60% of cases are caused by mutations in the *CLCN5* gene, which encodes a Cl[−]/H⁺ antiporter, primarily localized to early endosomes of the proximal tubule.⁷²⁴

A fraction (15%–20%) of patients with the Dent phenotype do not have mutations in *CLCN5* but instead display variations in the *OCRL* gene, which encodes the oculocerebral syndrome of Lowe protein 1.^{724,725} This protein is an inositol polyphosphate 5-phosphatase, which

regulates endomembrane lysosomal sorting. More typically, individuals with mutations in *OCRL* have congenital cataracts, intellectual disability, muscle hypotonia, and rickets, as well as evidence of partial or complete Fanconi syndrome.^{726,727} Renal insufficiency may occur in more severely affected individuals, which are predominantly boys as this is also an X-linked disorder.

Bartter Syndromes

Disorders of TAL sodium and chloride reabsorption are often accompanied by hypercalciuria and nephrocalcinosis (see Table 71.24). This association is the direct consequence of transcellular sodium chloride transport being necessary for passive paracellular calcium (and magnesium) reabsorption from this nephron segment. Hypercalciuria and nephrocalcinosis tend to be more severe in neonatal Bartter syndrome patients with mutations in *SLC12A1* and *KCNJ1* than in patients with classical Bartter syndrome due to mutations in *CLCKNB*.⁷²⁸ Additionally, types 1–4 of Bartter syndrome are inherited according to an autosomal recessive pattern, while type 5, which is transient, is X linked and is discussed in the next section.⁷²⁹

Autosomal Dominant Hypocalcemia with Hypercalciuria

Children with acquired or de novo activating mutations in either the calcium-sensing receptor gene (*CaSR*) or the G protein coupled to CaSR display a phenotype characterized by hypokalemia, hypercalciuria, and hypoparathyroidism (see Table 71.24).⁷³⁰ Interestingly, there is at least one other gene responsible for this disease that has yet to be identified, hypothesized to be a gain-of-function *AP2S1* mutation that may cause an additional form of ADH.⁷³¹ The CaSR resides both in the parathyroid gland and in the nephron, including the basolateral membrane of the TAL. In the TAL, activation of the CaSR not only prevents paracellular calcium reabsorption by increasing claudin-14 expression in this segment but also inhibits sodium and chloride reabsorption. Hence patients with these mutations can also demonstrate a Bartter phenotype. More than 400 CASR mutations have been reported, and 95% of ADH1 cases are caused by missense substitutions, whereas 5% result from in-frame or frameshift insertion/deletion mutations.⁷³²

Treatment of these defects is slightly different than classical Bartter syndrome as administration of active vitamin D is required to elevate serum calcium levels, and a thiazide diuretic is required to reduce urinary calcium excretion. This can lead to hypokalemia, which can be remedied by the addition of a potassium-sparing diuretic. Studies suggest that treatment is not required in asymptomatic patients.⁷³²

Familial Hypomagnesemia Hypercalciuria and Nephrocalcinosis

The clinical constellation of hypomagnesemia (secondary to renal wasting), hypercalciuria, and nephrocalcinosis is due to a mutation of either claudin-16 or claudin-19 (see Table 71.24).⁷³³⁻⁷³⁵ Both these tight junction proteins are expressed in the TAL of the loop of Henle, where they are essential to forming a paracellular divalent cation permeable pore. Consequently, disruption in the structure of one of these pore-forming claudins prevents the

reabsorption of calcium and magnesium from the TAL, leading to the clinical phenotype. The majority of patients tend to progress to renal impairment with upward of 50% requiring KRT.⁷³⁶ Clinically, mutations in claudin-19 can be distinguished from claudin-16 alterations by the presence of ocular abnormalities including macular colobomas, myopia, and horizontal nystagmus.⁷³⁵ Claudin-19 is also expressed to a high level in the eye consistent with this phenotype. Interestingly, milder mutations in claudin-16 have been proposed to cause isolated hypercalciuria in the presence of only mild hypomagnesemia or without any alteration in plasma magnesium.⁷³⁷ At present, there is no targeted therapy available. The standard approach involves providing supportive care through hydration, dietary adjustments, and the administration of magnesium salts, thiazides, and citrate, with the aim of slowing the progression of kidney failure. Ultimately, a kidney transplant stands as the sole curative measure for kidney failure.⁷³⁸

Distal Renal Tubular Acidosis

As discussed earlier, distal RTA is often accompanied by hypercalciuria and nephrocalcinosis, perhaps due to hypocitraturia.⁶⁵⁰ Nephrocalcinosis likely contributes to renal dysfunction in these patients, so treatment of the hypercalciuria is recommended to prevent renal insufficiency. Patients with incomplete distal RTA due to mutations in *SLC4a1* may also display hypercalciuria and kidney stones.⁷³⁹

Hereditary Hypophosphatemic Rickets with Hypercalciuria

HHRH presents with rickets, short stature, and increased renal phosphate excretion despite hypophosphatemia and hypercalciuria. It is the result of compound-heterozygous or homozygous loss-of-function mutations in the type 2, sodium phosphate transporter isoform C (NPT2c), encoded by the gene *SLC34a3*.⁷⁴⁰ This transporter contributes significant phosphate reabsorption from the proximal tubule and, when not functioning, leads to a primary renal phosphate leak. As a consequence, serum phosphate levels are reduced, driving the suppression of PTH and increasing 1,25-dihydroxyvitamin D synthesis. Patients with HHRH have low FGF23 levels, which normally regulate phosphate levels in the blood. This leads to increased intestinal phosphate and calcium absorption, resulting in hypercalciuria. Heterozygous mutations are frequently associated with isolated hypercalciuria (IH), resulting in increased risk of kidney stones or nephrocalcinosis threefold.⁷⁴¹ Additionally, inactivating mutations in *CYP24A1*, which encodes 25-hydroxyvitamin D 24-hydroxylase, are associated with the autosomal recessive idiopathic infantile hypercalcemia.⁶¹⁷

Treatment for HHRH typically involves phosphate and vitamin D supplementation. In cases of FGF23 excess, burosumab, an anti-FGF23 monoclonal antibody, is an approved option for X-linked hypophosphatemia.⁶¹⁸ However, burosumab is ineffective for HHRH since it is FGF23 independent. Adding active vitamin D metabolites can worsen hypercalciuria and nephrolithiasis, so it should be avoided. Phosphorus supplementation should be administered for at least 6 months before considering

corrective surgery to ensure proper bone healing and mineralization.

NONCALCIUM WASTING DISORDERS

Cystinuria

This is a rare autosomal recessive cause of kidney stone formation. It is associated with 1% of all cases of urolithiasis and 4% and 5% of urolithiasis in children.⁷⁴² The primary defect is in the reabsorption of dibasic amino acids, specifically cysteine, ornithine, arginine, and lysine.⁷⁴³ These amino acids are reabsorbed from the glomerular filtrate in the proximal tubule via the amino acid transporters encoded by the *SLC3A1* or *SLC7A9* genes. Mutations in *SLC3A1* cause type A cystinuria and *SLC7A9* type B cystinuria.^{744,745} They are clinically indistinguishable and do not appear to correlate with disease severity or response to therapy. Moreover, compound heterozygotes with monoallelic mutations in both genes do not display altered urinary dibasic amino acid excretion.⁷⁴⁶

Patients may present at any age but typically present in the early teens. Severe cases may present in the first year of life. Diagnosis is suggested by family history and confirmed by isolation of pathognomonic hexagonal crystals on urinalysis.⁷⁴³ The cyanide-nitroprusside test is a rapid screening test used historically, but due to low reproducibility and sensitivity it is no longer recommended.⁷⁴² Patients with cystinuria generally display cystine excretion more than five times the upper limit for age. Patients with Fanconi syndrome and heterozygotes will have intermediate excretion values and tend not to form stones.

First-line therapy includes increasing fluid intake, urinary alkalization, and a lower-sodium diet. In cases where significant cystinuria persists despite these measures, a number of additional therapies have been proposed. α -Mercaptopyrroline (tiopronin) forms a soluble dimer with cysteine and can be administered at a dose of 10 to 15 mg/kg/day.⁷⁴⁷ This therapy reduces new stone formation and may dissolve existing calculi. d-Penicillamine is a chelating agent that binds to cysteine and homocysteine, increasing their solubility. This compound has the potential for significant side effects including nephrotoxicity and monitoring is required. Other potential therapies that bind cysteine increasing its solubility include captopril and buccillamine.⁷⁴⁸

Hyperuricosuria (Including 2,8-Dihydroxyadenine Urolithiasis)

A number of rare single-gene defects of purine metabolism result in hyperuricosuria and uric acid lithiasis. Defects in hypoxanthine-guanine phosphoribosyl-transferase (HPRT1) cause Lesch-Nyhan syndrome, an X-linked disorder characterized by dystonia, gout (including hyperuricosuria), intellectual disability, and self-mutilation.⁷⁴⁹ Patients with glycogen storage disease type 1a have mutations in the enzyme glucose-6-phosphatase.⁷⁵⁰ They demonstrate a severe phenotype typically in the first year of life, which includes growth retardation, lactic acidemia, hypoglycemia, hepatomegaly, hyperlipidemia, and hyperuricemia leading to hyperuricosuria and uric acid stone formation. Another rare enzymatic deficiency is adenine phosphoribosyl-transferase deficiency, which leads to the overproduction

of 2,8-dihydroxyadenine that is excreted in the urine and can precipitate causing calculi.⁷⁵¹ Urolithiasis is the most common manifestation of this disorder, and patients can present at any age with calculi. A urinalysis will display characteristic round, brown crystals suggestive of the disorder, which can be analyzed by enzymatic activity testing, or the diagnosis may be established by demonstration of a biallelic pathogenic variant in the *APRT* gene. A low-purine diet and ample food intake are recommended treatments for the renal manifestations of these disorders. Medical therapy can include xanthine oxidase inhibitors.

Primary Hyperoxalurias

This is a group of rare autosomal recessive disorders that display increased generation and urinary excretion of oxalate (i.e., hyperoxaluria).⁷⁵² Three different subtypes, which result from mutations in three different genes that participate in oxalate metabolism, have been described.

Hyperoxaluria Type 1. Type 1 hyperoxaluria accounts for the majority (70%–80%) of primary hyperoxaluria cases.⁷⁵³ Many patients with type 1 hyperoxaluria will develop progressive severe nephrolithiasis and nephrocalcinosis leading to renal failure. However, there is significant phenotypic variation, with some patients developing renal failure in infancy and others not developing their first kidney stone until adulthood. The estimated prevalence of primary hyperoxaluria is <3:1 000 000, with type 1 hyperoxaluria accounting for ~80% of patients with hyperoxaluria.⁷⁵⁴ Symptoms typically appear during childhood and can lead to ESKF over a wide range of ages, from infancy to the sixth decade of life, with incidences of pediatric ESKF between 1% and 10%, depending on the country.⁷⁵⁴ Mutations in alanine-glyoxalate aminotransferase are responsible for hyperoxaluria type 1.⁷⁵⁵ The variability of disease severity is due to mutations that permit residual enzyme activity. The spectrum of mutations includes mostly point mutations leading to missense, nonsense, a number of splicing mistakes, and minor deletions/insertions.⁷⁵⁶ Importantly, residual alanine-glyoxalate aminotransferase enzyme activity in some cases can be stimulated by the cofactor pyridoxine and therefore may respond to treatment.⁷⁵⁵ Patients with responsive mutations tend to display a more benign course. In the most severe cases, infants can progress to renal failure within the first year of life; other cases might not be diagnosed until presentation of ESKF.

Systemic oxalosis occurs when plasma oxalate levels rise with decreasing GFR, due to reduced oxalate clearance. Systemic oxalosis results in the precipitation of calcium oxalate crystals in multiple tissues including the skeleton, heart, blood vessels, joints, retina, and skin.

Genetic analysis is helpful for therapeutic decision making and prognostication. The earlier in the course of the disease therapy is instituted, the better the reported outcome and the longer renal function can be preserved.⁷⁵⁷ Therapy involves increasing fluid intake (to at least 2.5 L/m²), avoidance of high oxalate-containing foods (i.e., chocolate, tea, rhubarb, and spinach), and administration of potassium citrate to both ensure adequate urinary citrate excretion and alkalinize the urine. Both of these urinary alterations tend to prevent

calcium oxalate precipitation. A trial of high-dose pyridoxine (5–10 mg/kg/day) should be administered and urine oxalate followed to identify responders, as approximately 30% of PH I patients have some degree of sensitivity to pyridoxine.⁷⁵³

Lumasiran and nedosiran are liver-directed RNA interference (RNAi) drugs that have emerged as a newer therapy resulting in sustained reduction in urinary and plasma oxalate in infants and young children with PH1.⁷⁵⁸ More follow-up data are needed regarding the use of these drugs in reducing stone disease and preventing kidney failure.

Stiripentol is an oral medication targeting lactate dehydrogenase A, used to treat Dravet syndrome, a genetic epileptic encephalopathy. Patients with Dravet syndrome generally have lower urinary oxalate excretion compared with healthy individuals. Stiripentol has been used in a patient with PH1 and led to a substantial reduction in urine oxalate levels.⁷⁵⁹

Clinical indications for early onset of dialysis are high plasma oxalate levels despite oxalate-lowering therapy (RNAi therapy or pyridoxine) and signs of systemic oxalosis.⁷⁰⁰ Once kidney failure ensues, even intensive dialysis (i.e., up to 40 hours of hemodialysis per week) is often ineffective at preventing the consequences of systemic oxalosis. Isolated kidney transplantation is not recommended in most patients unless lumasiran therapy is available, as it does not correct the primary metabolic problem of alanine-glyoxalate aminotransferase expression in the liver. In select patients who respond to pyridoxine with normalization or near normalization of urinary oxalate excretion, isolated kidney transplantation may be beneficial. Liver transplantation remains the only cure for PH1. A substantial body of evidence demonstrates that liver transplantation can reverse hyperoxaluria and prevent the further development of oxalate-related disease in patients with PH1.⁷⁰⁰ Isolated liver transplantation should be considered when GFR is >40 mL/min/1.73 m².⁷⁶⁰ For patients with PH and kidney failure, combined liver and kidney transplantation results in better kidney graft survival than isolated kidney transplantation (87% vs. 14% at 15 years).⁷⁰⁰ Data on kidney transplantation alone with lumasiran therapy is encouraging, although this therapy is extremely costly.⁷⁶¹

Hyperoxaluria Type 2. Primary hyperoxaluria type 2 results from mutations in the glyoxylate reductase/hydroxypyruvate reductase (*GRHPR*) gene, which leads to increased amounts of glyoxylate and hydroxypyruvate available for conversion to both oxalate and L-glyceric acid, which are excreted by the kidney.^{753,762} Type 2 hyperoxaluria is typically less severe than type 1. It traditionally presents in adolescence with repeated calcium oxalate stone formation. A minority of patients develop chronic renal impairment. Diagnosis is suggested by the demonstration of significantly elevated urinary oxalate excretion and the presence of increased urinary L-glycerate. Definitive diagnosis is made via mutation analysis of the *GRHPR* gene or by demonstration of reduced liver GRHPR enzyme activity. Pyridoxine is not beneficial. Liver transplantation is not beneficial. Patients with this disease may instead benefit from isolated renal transplantation.

Hyperoxaluria Type 3. Primary hyperoxaluria type 3 results from mutations in the 4-hydroxy-2-oxoglutarate aldolase (*HOGA1*) gene.^{763,764} This enzyme is localized to the mitochondria in both the liver and kidney and catalyzes 4-hydroxy-2-oxoglutarate to glyoxylate and pyruvate. Of the three subtypes, this is the most mild clinically and is associated with recurrent calcium oxalate stone formation. Diagnosis is suspected by increased urinary oxalate excretion and can be confirmed by genetic analysis of the *HOGA1* gene. Nephrocalcinosis and renal impairment are rarely seen.

More discussion on the genetics of tubulopathies and urolithiasis can be found in [Chapter 44](#).

PEDIATRIC ASPECTS OF CHRONIC KIDNEY DISEASE

GROWTH, NUTRITION, AND DEVELOPMENT

Childhood CKD impairs statural growth in a manner related to the age of CKD onset. Approximately 50% of children requiring KRT before their 13th birthday have a final height below the normal range.⁷⁶⁵ Results from the Chronic Kidney Disease in Children Study (CKiD, a U.S. registry of children with a GFR <75 mL/min/1.73 m²) demonstrated that height is below the normal range (median height standard deviation score [HtSDS] −0.55, fall in HtSDS of 0.12–0.16 for each decline of 10 mL/min/1.73 m²) in children with predialysis CKD as well.⁷⁶⁶ One-third of all postnatal growth occurs during the first 2 years of life. Birth with CKD and inhibition of growth during this critical 2-year period can have major effects on the total amount of growth achieved.⁷⁶⁷ The approach to pediatric CKD is discussed in further detail in [Chapter 72](#).

The etiology of growth failure in children with CKD is multifactorial including disturbances in growth hormone and insulin-like growth factor-I, electrolyte abnormalities, nutritional deficiency, metabolic acidosis, uremia, anemia, inflammation, vitamin D deficiency, hyperparathyroidism, and hypogonadism.⁷⁶⁸ Additionally, many medications such as corticosteroids and CNIs disrupt linear growth. Corticosteroids cause reduced pulsatile secretion of GH, impaired sensitivity of the GH receptor to IGF-1, and decreased production of IGF-1.⁷⁶⁸ Management of pediatric growth and development is detailed in the next section but generally involves close monitoring of growth and anthropometric data, assurance of adequate caloric intake, correction of acid-base/electrolyte abnormalities, treatment of renal osteodystrophy, increased physical activity, and treatment with recombinant human growth hormone if required.⁷⁶⁹

PREVENTION AND TREATMENT OF GROWTH FAILURE IN CHRONIC KIDNEY DISEASE

In infants with CKD, the most important measures for avoiding growth failure, defined as reduced adult height (< −2 SD), are the provision of adequate energy intake, correction of metabolic acidosis, and maintenance of fluid and electrolyte balance.⁷⁷⁰ Supplementary feedings via nasogastric tube or gastrostomy are frequently required to achieve these targets. Caloric intake should be targeted

to provide 80% to 100% of the regular daily allowance for healthy children. Increasing caloric intake above 100% of the recommended daily allowance does not induce further catch-up growth but rather results in obesity. Protein intake should be at least 100% of the dietary reference intake but should not exceed 140% of the dietary reference intake in patients with CKD stage 2 or 3 and 120% of the reference intake in children with CKD stage 4 or 5. Excessive protein intake should be avoided to limit phosphorus and acid load in patients with advanced CKD.⁷⁷¹

Metabolic acidosis should be treated by oral alkaline supplementation aiming for serum bicarbonate levels equal to or above 22 mEq/L.⁷⁶⁹ In addition, the supplementation of water and electrolytes is essential for children with polyuria and/or salt-losing nephropathies, as occurs in tubulopathies and renal dysplasia. Early and consistent provision of supplementary nutrients, fluids, and electrolytes, with delivery ensured by enteral feeding via a nasogastric tube or percutaneous gastrostomy whenever necessary, markedly improves growth and development of young children with CKD.^{767,772} In the postinfantile phase of childhood, nutrition, fluid, and electrolyte balance continue to be permissive factors for adequate longitudinal growth, but catch-up growth can rarely be achieved by dietary and supplementary measures alone.

If growth failure is imminent and/or has already occurred, despite adequate provision of nutrients, salts, and fluid, treatment with recombinant GH (rGH) may be indicated. The presence of an open epiphysis on the radiograph of the left wrist indicates potential for growth, which is a prerequisite for the indication of rGH treatment in children presenting with persistent short stature.⁷⁷³ The efficacy and safety of rGH therapy in children with CKD have been demonstrated in numerous short- and long-term trials and have been comprehensively reviewed.⁷⁷⁴

CARDIOVASCULAR COMORBIDITY IN PEDIATRIC CHRONIC KIDNEY DISEASE

CKD-associated cardiovascular disease manifests relatively early in the course of CKD, leading to significant morbidity and mortality.^{775,776} Among individuals younger than 19 years of age with ESKF, cardiac disease is responsible for 22 deaths/1000 patient-years, accounting for 16% of all deaths in whites and 26% in African Americans.⁷⁷⁵ This represents up to a 1000-fold increase in risk compared with that in the general population.⁷⁷⁷

HYPERTENSION

The prevalence of hypertension is 40% to 50% in children with CKD stages 2 to 4 and approaches 70% by the time ESKF has developed.⁷⁷⁸ High blood pressure is associated with target organ damage during childhood and is closely related to the rate of progression of renal failure. Current management guidelines recommend regular blood pressure screening for all children with CKD.⁷⁷⁹ In addition to clinic measurements, ambulatory blood pressure monitoring should be performed at least once a year and within 1 to 2 months of modification of therapy in patients receiving antihypertensive medication. Children with CKD should be considered at increased risk for cardiovascular

complications, and therapeutic interventions should be initiated when blood pressure exceeds the 90th percentile for age and height. The KDIGO guidelines for blood pressure management in CKD recommend that a 24-hour mean arterial pressure by ambulatory blood pressure monitoring (ABPM) should be lowered to ≤ 50 th percentile for age, sex, and height.⁷⁸⁰ When ABPM is not available, manual auscultatory office BP obtained in a standardized setting should target an SBP < 90 th percentile for age, sex, and height of normal children. An ABPM should be measured once per year, and office BP should be measured at least every 3 to 6 months at all appointments. Lifestyle changes such as increased physical activity, avoidance of salty foods, and reduction of weight in obese patients are also recommended for children with CKD but are rarely effective as sole measures.^{781,782} ACE inhibitors and ARBs should be the first choice of antihypertensive medications and, if given at appropriate dosages (e.g., ramipril 1.5–6 mg/m²/day or candesartan 0.2–0.4 mg/kg/day).^{783,784} If the blood pressure-lowering effect is insufficient, a loop diuretic should be added (e.g., furosemide, 2–4 mg/kg/day), followed by a calcium channel antagonist (e.g., amlodipine, 0.2 mg/kg/day). The timing of drug dosing should be adapted to achieve optimal blood pressure control throughout 24 hours as verified by ambulatory blood pressure monitoring. Uncontrolled and/or unrecognized hypertension is common. Even in children taking antihypertensive medication, up to 35% have BP above the 90th percentile. In children with elevated systolic or diastolic BP, almost 40% were prescribed no antihypertensive medication. Hypertension is discussed in more detail in [Chapters 46 and 48](#).

ANEMIA

Data derived from the CKiD study indicate that 26% of children with nonglomerular and 43% of children with glomerular CKD had hemoglobin levels less than the fifth percentile.⁷⁸⁵ In children, growth, neurocognitive development, school attendance, and exercise capacity are important clinical outcomes related to anemia. The prevalence of anemia in pediatric CKD patients has been reported as approximately 30% in CKD stages 1 and 2, 66% in CKD stage 3, and 93% with CKD stages 4 and 5.⁷⁸⁵ Anemia is associated with an increased risk of morbidity and mortality in both pediatric ESKF and nondialysis CKD patients, as well as progression of CKD.⁷⁸⁶

Erythropoiesis-stimulating agents (ESAs) are central to the treatment of anemia. While the efficacy of ESAs is clear in childhood CKD, the desired therapeutic target of blood hemoglobin has not been defined in children via randomized trials.⁷⁸⁷ Studies in adults with CKD have raised important concerns regarding increased cardiovascular morbidity and mortality in patients with hemoglobin > 120 g/L.⁷⁸⁸ The effects of partially correcting severe anemia versus fully normalizing hemoglobin levels through ESA therapy on quality of life have been studied extensively. Most research shows that complete normalization of hemoglobin does not result in additional quality of life improvements. More importantly, fully normalizing hemoglobin increases the risk of cardiovascular and thrombotic events.⁷⁸⁹

INTERMEDIATE CARDIOVASCULAR ENDPOINTS IN CHILDREN WITH CHRONIC KIDNEY DISEASE

LVH and increased carotid intima-media thickness have been demonstrated in hypertensive children with advanced CKD, in whom alterations of mineral metabolism and volume overload are important superimposed risk factors, but also in children with early essential hypertension and in children with masked hypertension. LVH is the most common identifiable cardiac alteration in CKD and the most important indicator of cardiovascular risk in adult and pediatric patients with ESKF.⁷⁹⁰⁻⁷⁹³ LVH is present in approximately 15% to 30% of children with stage 2 to 4 CKD, in 50% to 75% of children undergoing dialysis, and in 50% to 67% of renal allograft recipients.⁷⁹⁴⁻⁸⁰⁰ In addition to the morphologic change of LVH, subclinical impairment of left ventricular systolic function is found in about 20%, 32%, and 37% of children in CKD 1, 2, and 3, respectively.⁸⁰¹

CKD-associated bone mineral disorder also contributes to the development of calcifying uremic vasculopathy.⁸⁰² Additionally, children with predialysis CKD show early changes in arterial gene transcription consistent with calcium accumulation and matrix remodeling.⁸⁰³ Coronary artery calcifications are found in individual adolescent patients undergoing dialysis and in more than 90% of young adults with childhood-onset CKD.⁸⁰⁴ Early signs of vasculopathy, such as an increased intima-media thickness and stiffness of the carotid artery, can be detected as early as the second decade of life.⁸⁰⁵ Both morphologic and functional alterations are progressive over time and are most marked in adolescents undergoing dialysis, but they also may be seen in children with moderate CKD and appear to improve partially after successful renal transplantation. Intima-media thickness and stiffness of the carotid artery correlate with the degree of hyperparathyroidism, serum calcium-phosphorus ion product, and cumulative dose of calcium-containing phosphate binders.

PROGRESSION OF CHRONIC KIDNEY DISEASE IN CHILDREN

Progression of pediatric CKD is mainly determined by the age and degree of renal failure at the time of first disease manifestation. Two-thirds of children with CAKUT progress to ESKF during the first 2 decades of life.¹⁸ The physiologic early increase in GFR is typically extended to the first 3 to 4 years of life in children with CKD, which may reflect adaptive hypertrophy of the reduced number of functioning nephrons.¹⁷⁶ In approximately 50% of children, this early “improvement” phase is followed by a period of stable or slowly deteriorating renal function, which usually lasts for 5 to 8 years. Around the onset of puberty, the GFR loss tends to accelerate, typically leading to ESKF in late adolescence or early adulthood. The reasons for this nonlinear progression of renal failure are incompletely understood and may include an insufficient capacity of the nephrons to adapt to the rapidly growing metabolic needs during the pubertal growth spurt, adverse renal effects of puberty-related increases in sex steroid production, and/or an accelerated sclerotic degeneration of a diminishing number of remnant hyperfiltering nephrons. In children with severe renal hypoplasia, the early increase in GFR may be blunted

and more rapid progression to ESKF may occur. About 20% of patients with renal hypoplasia maintain a stable GFR, even beyond puberty. Follow-up studies have shown that in patients with mild bilateral renal hypoplasia, progressive deterioration of GFR frequently ensues in the third decade of life.

RISK FACTORS FOR PROGRESSION OF KIDNEY DISEASE

Having a blood pressure >90 percentile reduced the time to needing KRT or reaching 50% decline in GFR by 67% in glomerular disease and 38% in nonglomerular disease.⁸⁰⁶ The randomized, prospective ESCAPE trial (Effect of Strict Blood Pressure Control and ACE Inhibition on the Progression of Chronic Renal Failure in Pediatric Patients) demonstrated that in children with CKD due to various disorders, intensified antihypertensive treatment targeting blood pressure in the low-normal range (<50th percentile for age) was associated with improved long-term renal survival.⁸⁰⁷ Proteinuria predicts renal disease progression in children with renal hypodysplasia and, even in children with normal kidney function, persistent nephrotic-range proteinuria is a risk factor for progressive renal injury. Other reported risk factors associated with CKD progression include low birth weight or prematurity, uric acid disturbances, lead or heavy metals, hyperlipidemia, metabolic acidosis, oxidative stress, and disorders of bone and mineral metabolism.⁷⁸⁶ Discussion of kidney replacement therapy in children is outlined in [Chapter 72](#).

CONCLUSION

While many aspects of kidney disease in children may resemble those in adults, it is crucial to recognize that pediatric kidney diseases often have distinct causes, presentations, and treatment considerations that differ from adult cases. For example, congenital anomalies, genetic conditions, and growth-related factors play a much larger role in pediatric kidney diseases. Additionally, children may respond differently to medications, and long-term management strategies must consider their developmental and nutritional needs. The unique physiology of growing children, combined with developmental factors, demands tailored approaches to diagnosis and management. Understanding these differences is essential for effective care and long-term outcomes in pediatric patients.

Author comments regarding drug dosing in children: The authors have taken the utmost care with the medication dosages given in this chapter. Nevertheless, errors cannot be ruled out. Changes may also result from new findings. We would therefore like to point out that the responsibility for the dosages of medication used always lies with the prescribing doctor.

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complete set of references.

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